

Wood Anatomy of Brassicales: New Information, New Evolutionary Concepts

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Abstract Wood anatomical data for the 19 families of Brassicales are presented, based on light microscopy and scanning electron microscopy (SEM), arranged according to recent molecular phylogenetic evidence. Because of large species numbers and diversity in ecology and growth form, Brassicales are an ideal case study group for understanding wood evolution. Features newly reported include vested pits in Cleomaceae, Koeberliniaceae, Pentadiplandraceae, Salvadoraceae, and Setchellanthaceae. Vesturing of primary xylem helices is shown for *Raphanus* (first report in angiosperms). Fiber dimorphism is newly reported in some genera of the crown group (Capparaceae + Cleomaceae + Brassicaceae). The fiber-tracheid is probably the ancestral imperforate tracheary element type for Brassicales, and from it, libriform fibers, living fibers (including septate fibers), and tracheids have likely been derived. The Baileyan concept of unidirectional evolution from tracheids to libriform fibers must have many exceptions in angiosperms, and tracheids are not uniform. Tracheids occur in Emblingiaceae, Koeberliniaceae, Pentadiplandraceae, Stixaceae, and Tropaeolaceae. Synapomorphies can be identified, as in the Akaniaceae—Tropaeolaceae clade (rays of two sizes, living fibers, scalariform perforation remnants) and the Moringaceae–Caricaceae clade (ground tissue of wood composed of thin-walled fibers or similar parenchymatous cells). Wood of Brassicales is mostly not paedomorphic, although paedomorphic characters suggesting secondary woodiness occur within the families Brassicaceae (abundance of upright ray cells, raylessness), Caricaceae, Cleomaceae, and Moringaceae. Brassicales are probably ancestrally woody, and wood of Sapindales and Malvales has a number of key character states (plesiomorphies) like those in Brassicales, as would be predicted by current molecular phylogenies. Surveys of large taxonomic groupings, such as Brassicales, tend to yield more examples of homoplasies and apomorphies that can be interpreted in terms of adaptation and functional interlinkage (e.g., ray evolution paralleling imperforate tracheary element evolution). In turn, these features can be interpreted in terms of ecology (e.g., xeric habitats) and growth forms (e.g., tree succulents). The assemblages of wood character information in a reasonably well known order of angiosperms permits hypotheses about wood evolution in angiosperms as a whole. Some of the more important hypotheses presented include: (1), that evolution of wood (and other) characters is always progressive, with gene overlays (silencing, modification, etc.) and simultaneous changes in multiple features, so that ancestral conditions are never truly



re-attained. (2). Not all characters are of equal value in water economy of any given plant; some (presence of tracheids) may supersede others, and xeromorphic characters can be arranged relative to each other in tiers, although various taxonomic groups have different rosters of conductive safety features. (3). Heterochrony (protracted juvenilism, accelerated adulthood) is extensively represented in angiosperms, and acts as an overlay that is a source of diversity that angiosperms have drawn on since their inception (probably as minimally woody plants). (4). There may be no “purely taxonomic” characters, because genes of an organism relate primarily to changes, ancient and new, that are of adaptive significance, although we may not be able to detect selective value, past or present. Although many families of Brassicales are small and represent occupancy of specialized or extreme habitats (*Batis*, *Koeberlinia*, *Moringa*), active speciation in Brassicaceae and Capparaceae is related to tolerance of drought and cold with mechanisms such as vested pits, narrow vessels, and abbreviation in life cycle length.

Keywords Ecological wood anatomy · Fiber dimorphism · Irreversibility · Raylessness · Wood xeromorphy · Vestures

Introduction

Comparative wood anatomy offers new opportunities in the era of molecular-based phylogeny. The presence of molecular trees with high degrees of statistical probability means that we can see how, and how rapidly, wood evolves with respect to particular environmental factors. For example, can woodiness change over short periods of time? Can degree of woodiness fluctuate in both woodier and less woody directions in some groups but not in others? Does wood anatomy change rapidly and opportunistically, or is it one of the more conservative features of plant evolution? Which wood features can change more rapidly, and in what ways? Which features are active in countering dry or freezing conditions?

Some orders and families of angiosperms are much better than others for answering such questions. The size of Brassicaceae (3710 species) and Capparaceae (480) and their wide range of ecological distribution provide a much better material for understanding wood evolution than an assemblage of species with stereotyped habitat preferences. The families of Brassicales with fewer species are, in a different way, informative because they represent families that are superficially so different that they were earlier not included in the order tell us how by securing special ecological niches, a clade can survive and be transformed into growth forms as diverse as annuals (Limnanthaceae) or succulent trees (Caricaceae).

At a histological level, Brassicales offer many characters the phylogeny of which can be elucidated by distribution within the order as now conceived. Imperforate tracheary element types and vesturing presence or absence (Fig. 1, columns at right) are two of these. Tracheids in the sense of Bailey & Tupper (1918), Bailey (1936), Carlquist (1961, 1988), IAWA Committee on Nomenclature (1964) and Sano et al. (2011), cells capable of conduction, occur in five families of Brassicales. Implicit in the concepts of Bailey (1944) and the tabular data on bordered pits by Metcalfe & Chalk

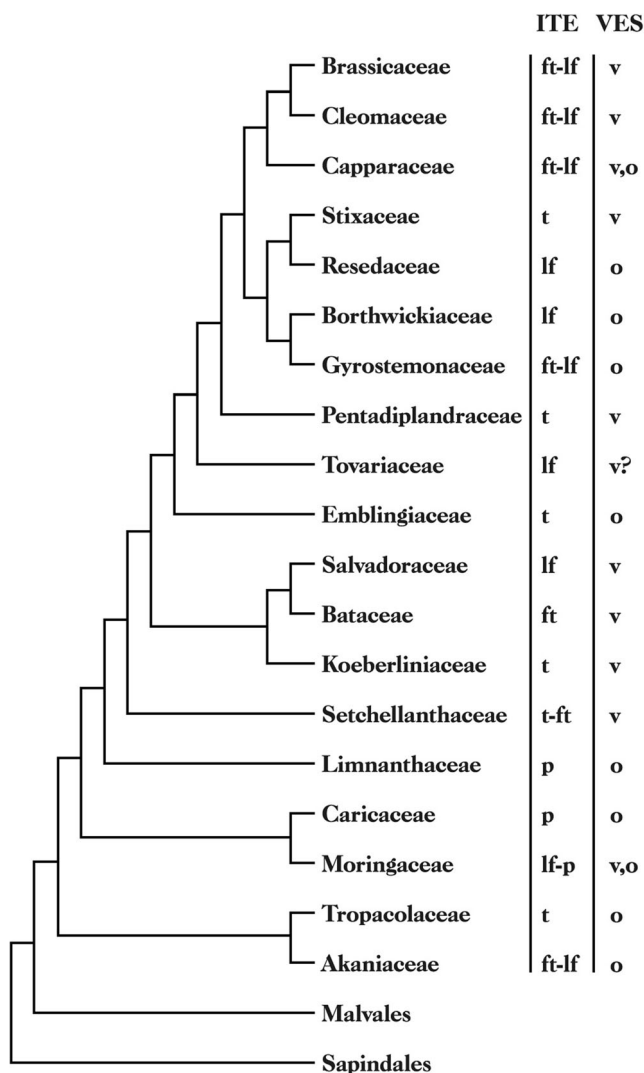


Fig. 1 Phylogeny of the 19 families of Brassicales, a combination of the trees of Hall et al., (2002, 2004) Su et al. (2012) to show all of the families. First column at right (ITE), indicates the type of imperforate tracheary elements present (ft= fiber-tracheid; lf= libriform fiber, which may also be a living fiber; p= parenchyma instead of fibrous tissue; t= tracheids). Second column at right indicates presence of vesturing (0= no vesturing observed; V= vestured lateral wall pits on vessels, as well as, in some cases, on imperforate tracheary elements

(1950, xlv) is the idea that the tracheid is the primitive (plesiomorphic) type of imperforate tracheary element in angiosperms, and that it has evolved, irreversibly, into fiber-tracheids in various clades, followed by libriform fibers. Libriform fibers are often thought to be dead at maturity, but in fact, when liquid-preserved materials are studied, they prove to have living contents in an appreciable number of genera. Many workers have relied on the present of septa in libriform fibers (which are then termed septate fibers) as evidence that libriform fibers have prolonged longevity. Brassicales is

an ideal group for demonstrating whether or not the tracheid to libriform fiber progression is irreversible, and if it is not, what ecological factors favor occurrence of tracheids in particular clades.

Vestured pits and vestured vessel walls (which may be construed so as to include “warts” on the lumen surface) occur in most Brassicales (Fig. 1) but not all. Are vestured pits apomorphic or plesiomorphic? Are they a constant feature of a species, or can they appear sporadically within a particular species? What is the relationship between vesturing and ecology? Our access to this character (or these characters) depends on access to scanning electron microscopy (SEM). Although vesturing can be detected in many instances with careful light microscopy, availability of SEM has permitted not only much greater certainty about whether vesturing is present in a particular species or not, it can be used to demonstrate that not all vestured pits are alike. Perhaps no order is more important in demonstrating the evolutionary nature and diversity of vesturing than Brassicales, and significant new observations are included in the present study. Examination of vesturing in terms of particular taxa and clades (e.g., Jansen et al. 2001) is required in order to demonstrate the ecological significance of vesturing and its probable phylogenetic status; a broad-brush analysis of vesturing on a global basis (Jansen et al. 2004) cannot provide the selective basis for this feature. Clearly interest in vesturing has peaked in recent years (e.g., Jansen et al. 2001), but much remains to be done, both with SEM and transmission electron microscopy (TEM).

Other wood anatomical features well represented in Brassicales that invite study and interpretation include presence of axial parenchyma as a background cell type. Kribs (1937) gives us the impression that all axial parenchyma is produced by modifications of more plesiomorphic types (e.g., diffuse parenchyma evolving into paratracheal, etc.), but is this an accurate view of axial parenchyma evolution? Brassicales offer some excellent examples of such diverse types as pervasive (axial parenchyma forming the ground tissue of secondary xylem) and bands that occur in latewood, especially as a stem of a “woody herb” senesces (e.g., the *Castilleja* example: Carlquist, 2015a).

Distinctive successive cambia appear in the group of genera recognized here as Stixaceae (Carlquist et al., 2013), and structural modes referable to successive cambia occur in Capparaceae (Adamson, 1936; Metcalfe & Chalk, 1950) and Brassicaceae (Metcalfe & Chalk, 1950). Interxylary phloem also occurs conspicuously in Brassicaceae (this paper) and Salvadoraceae (Carlquist, 2002). The characteristics of these cambial variants and their probable ecophysiological significance must ultimately be explored with reference to all occurrences, not just those in Brassicales (Carlquist, 2007). However, each instance in which these occur contributes a vital fragment to our understanding of these special conditions, too often neglected by wood anatomists and plant physiologists.

Fiber dimorphism proves to be another neglected phenomenon (Carlquist 1958, 2014). The occurrence of fiber dimorphism in angiosperm woods seems a logical evolutionary development, but, like successive cambia, it characterizes a small number of families and needs further exploration.

Helical sculpture in vessels is quite varied in expression (Carlquist, 1988). The occurrences in Brassicales are intriguing because helical sculpture is less characteristic of genera in the order than it is in some other angiosperm orders (e.g., Asterales). The association of helical sculpture with xeric ecological conditions is notable (Carlquist, 1966), but this feature can be found in genera that are not at all xeric in preferences, but

are subject to winter freezing (*Acer*). Water is minimally available to the plant in either case, but can we differentiate among the two modes of occurrence? Helical sculpture also occurs in various forms, ranging from helical thickenings on the lumen side of a vessel wall to grooves that interconnect pit apertures (“coalescent pit apertures”) or widen the inner pit apertures of pits.

Most of the abovementioned features have been shown to have some degree of association with xeric conditions. The habitats of Brassicales are mostly xeric in some way or to some degree, and in this respect, the order is ideal as a template for identifying which features are most indicative for wood xeromorphy, and which seem to be most effective in prevention of cavitation or in restoring the integrity of water columns once embolisms have formed. Thus, Brassicales can play a key role in understanding of wood xeromorphy as a whole. Of orders of comparable size and ecological distribution, only Asterales and Caryophyllales have offered such excellent materials for the understanding of wood xeromorphy (Carlquist 1966, 2010; Mauseth, 1993).

In the past, many workers have arranged anatomical information about angiosperm woods in a systematic fashion (e.g., Solereder 1885, 1908; Metcalfe and Chalk, 1950). This method is ideal for retrieval of anatomical information, but has the drawback that it might lead us to think of wood characters as purely “taxonomic” in significance, whereas ecology and growth forms are the factors that explain occurrences of wood characters. To be sure, ecological, physiological, and growth form correlations may be difficult to demonstrate, but the adaptive approach is to be preferred to considering that any given feature is “of taxonomic value.” This concept does not negate the fact that wood characters have distinctive systematic distributions. The primary selective factor in wood anatomy may be the balance between conductive efficiency and conductive safety (Carlquist, 2012)—a balance that is attained in many different ways. Mechanical strength inevitably also is a prime factor in wood patterns, one that must be analyzed separately from hydraulic capabilities, although there can be overlap.

There has been a tendency to relate the study of wood anatomy to woodier species, based on the economic value of those species. In fact, non-woody species, such as the majority of Brassicales—have secondary xylem, which deserves study equally. One could even say that one cannot understand woody species if one does not also understand the secondary xylem of non-woody or less woody species (Carlquist, 2009a), because early angiosperms probably had secondary xylem but would be classified as “non-woody” if we were to see them in comparison today’s angiosperms.

Brassicales as currently recognized includes 19 families (Fig. 1). One could register an objection to inclusion of so many families. In fact, some coalescence has occurred. For example, Akaniaceae and Bretschneideraceae have been merged. One could even add a twentieth family, because the fossil genus *Dressiantha* belongs to Brassicales but appears in a clade that includes no other families of the order (Gandolfo et al., 1998). The history of Brassicales (sometimes called Capparales) is succinctly offered by Rodman et al. (1998) and by Hall et al., (2002, 2004) and need not be repeated here. There is understandably little impetus to condense Brassicales into one or two families when about half of the families currently included were, in phylogenies proposed before global molecular-based phylogenies emerged, placed in entirely different orders: Tropaeolaceae and Limnanthaceae in Geraniales. Caricaceae was placed in Violales, perhaps close to Passifloraceae. *Emblingia* was variously claimed to be related to

Goodeniaceae, Polygalaceae, or Sapindaceae. Now that these families (and others) have been newly integrated into Brassicales, should we reconsider whether there should be condensation into fewer families? Actually, Brassicales, like Caryophyllales (Carlquist, 2010) proves to be an order in which further aggregation of families would result in umbrella families that cannot be described because of character sets that would be so heterogeneous that the resulting umbrella families would have unifying features too few, too vague, and too rife with exceptions.

For reasons just suggested, descriptions of wood characters of the 19 families of Brassicales are presented here. Without such descriptions, conclusions about wood in relation to phylogeny, ecophysiology, and habit would not be documented. The descriptions of wood of Brassicaceae and Capparaceae could be amplified so as to include variations as yet undescribed, of course, but the generalizations offered seem adequate for the purposes of the present paper.

The topology of the tree offered in Fig. 1 is likely to be modified somewhat as more molecular data becomes available. I have tried to place the families according to the most recent trees, but no molecular tree presently offered includes all 19 families. The tree in Fig. 1 is essentially an attempt to integrate the phylogeny of Su et al. (2012) with that of Hall et al. (2004). The familial unit Stixaceae Doweld has been hesitantly adopted because of the amazing similarities in wood anatomy between wood of *Forchhammeria* and *Stixis*. It includes *Forchhammeria*, *Neothorelia*, *Stixis*, and *Tirania*, genera once included in Capparaceae (Pax & Hoffmann, 1936). The tree offered in Fig. 1, although based on those of Hall et al. (2004) and Su et al. (2012), is not the same as those or other trees because all trees offered to date do not include all 19 families of Brassicales. The wood descriptions and illustrations of the present paper are organized according to the tree of Fig. 1, reading from bottom to top. The descriptions of wood of the families are offered because if available elsewhere, they are scattered through the literature, and are not organized consistently to each other. In addition, the recent alterations in content of some of the families has now been altered (e.g., Capparaceae now excludes a number of genera, notably those of Cleomaceae, included by Pax & Hoffmann, 1936), so that older descriptions of wood of Capparaceae are no longer valid. The descriptions serve as a convenient data source when discussing the systematic distribution and evolutionary changes of any of the wood characters of Brassicales.

Materials and Methods

This paper is based on a series of my monographs of brassicalean families, cited at the beginnings of each of the family descriptions, plus monographs by others that cover more than one or two species. Much new original data is presented here, as are original illustrations (particularly SEM images). The bibliography of Gregory (1994) contains references to non-monographic sources of information on small numbers of species in larger families and genera. Documentation of collections is offered in the monographs cited, as well as in the captions of figures. Authors of binomials are given in the captions; for species not illustrated, binomial authors are provided in the running text. Species numbers and geographical distributions are from Stevens (2015). Terminology is in accord with Carlquist (1988, 2001), which in turn is essentially that of IAWA

Committee on Nomenclature (1964), plus modifications necessitated by insights from physiology (Sano et al., 2011) and ontogeny (Carlquist, 2007).

Most studies were based on dried wood samples, but liquid-preserved materials proved valuable where soft tissues were concerned in families such as Caricaceae, Moringaceae, Salvadoraceae (interxylary phloem) and Stixaceae (successive cambia in *Forchhammeria*). Both sliding microtome sectioning and paraffin sectioning following softening with ethylene diamine (Carlquist, 1982) were employed. For SEM studies, sections made by hand with a single-edged razor blade proved entirely satisfactory. These hand sections were made from material boiled in water and stored in 50 % ethanol. The sections were rinsed in distilled water and then flattened by drying between clean glass slides under pressure. The thickness of such sections, which would be disadvantageous in light microscopy, was valuable because three-dimensional images of cells, rather than slices of cells, could be obtained. Some permanent slides made with sliding microtome sectioning and a Canada balsam mounting medium were used for light microscopy, but replicates of some of these slides, especially those from earlier years, were soaked in xylene and the sections recovered. Once cleansed with several changes of xylene and dried between clean glass slides, these sections were entirely satisfactory for SEM work. The majority of the SEM work was done with a Hitachi S2600N.

Quantitative data included here as indicators of relative mesomorphy or xeromorphy of particular woods, and as a way of documenting presence of protracted juvenilism (paedomorphosis). More extensive measurements can be found in the papers cited for each family. Quantitative data (vessel diameter, ray height, etc.) vary with portion of plant, size of sample, position within a growth ring, etc., and without knowing these factors, quantitative data reveal only generalized information. The present paper contains original qualitative, as well as observations in such resources as Solereder (1908) and Metcalfe & Chalk (1950). Wood terminology follows Carlquist (1988, 2001). The sequence of characters discussed in each family begins with transactional plan of wood and storying, if any, as seen in low power transverse and tangential sections respectively. The sequence of features that follow are vessel elements, then imperforate tracheary elements, axial parenchyma, rays, and crystals. An attempt has been made to cover the diversity within each of these categories, but in the larger families, the coverage is necessarily incomplete.

Results

1. Akaniaceae (Carlquist 1996; Heimsch, 1942).

Bretschneidera (Fig. 2).

Wood plan. Growth rings moderately well developed (Fig. 2a), latewood vessels about half the diameter of earlywood vessels.

Storying. Wood non-storied (Fig. 2b).

Vessels. Vessels in small groups, groups larger in latewood (Fig. 2a). Perforation plates predominantly simple; variously scalariform plates occasional (Fig. 2e). Lateral wall pitting (both vessel to vessel and vessel to septate fiber) alternate or opposite. Vessel to ray pitting scalariform, opposite, or transitional. Vessel pits non-vestured.

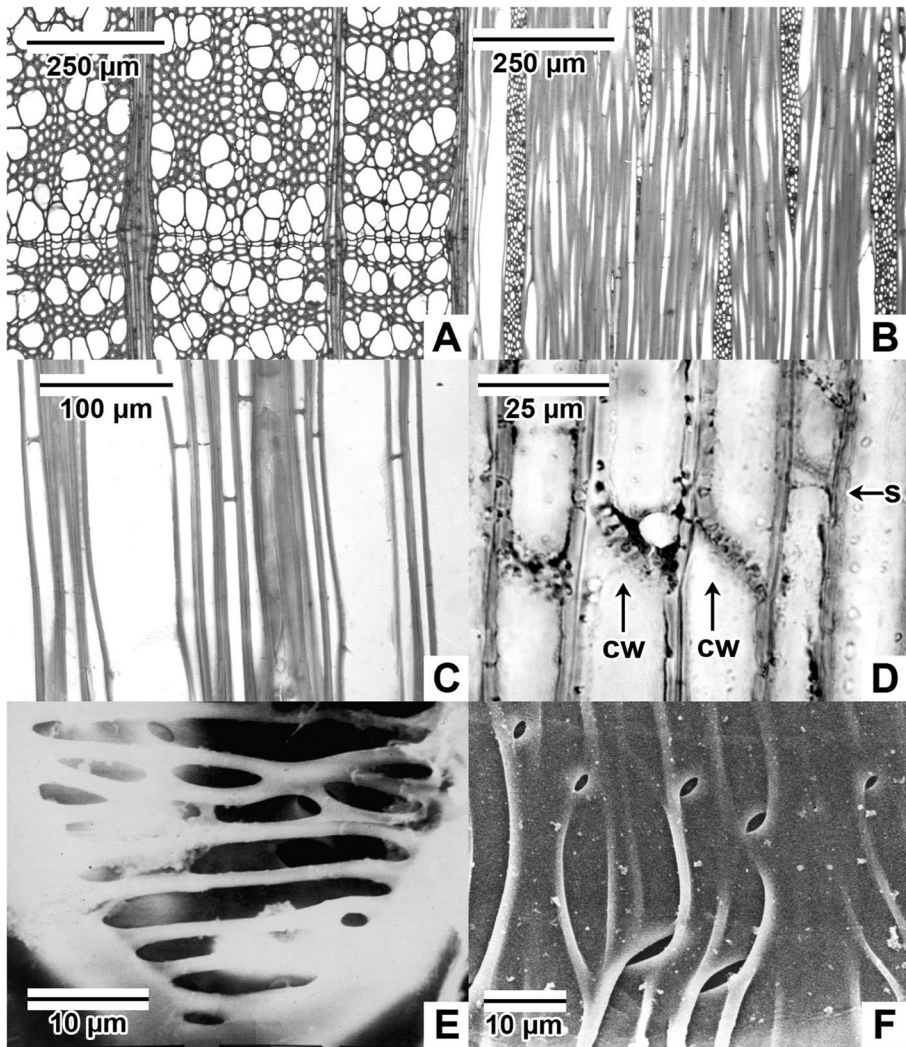


Fig. 2 Wood features of *Bretschneidera sinensis* Hemsl., MADw-22016. **a.** Transverse section: end of growth ring, just below center. **b.** Tangential section; portions of five multiseriate rays. **c–f.** Radial sections. **c.** Septa are present (top) in several fibers. **d.** cross-walls (cw) in three axial parenchyma strands; bordered pits, seen in sectional view, are present; one septum (s) in a fiber at right is present. **e–f.** SEM micrographs. **e.** Portion of a rare scalariform perforation plate. **f.** Inner vessel surface, showing helical thickenings

Helical thickenings on the vessel lumen surface, sometimes anastomosing, often fading into the vessel wall (Fig. 2f).

Imperforate tracheary elements. All imperforate tracheary elements are moderately thin-walled septate fibers, with small simple or vestigially bordered pits (Fig. 2c and d).

Axial parenchyma. Axial parenchyma is scanty paratracheal (vasicentric) and terminal. Axial parenchyma in strands of two to seven cells. The cross-walls between the cells of the strands have bordered pits (Fig. 2d).

Rays. Multiseriate rays (Fig. 2b) relatively wide (5.8 cells at widest point). Uniseriate rays also present. The ray type is Heterogeneous Type II of Kribs. Tip cells and sheathing cells of the multiseriate rays upright, the multiseriate ray cells otherwise procumbent.

Crystals not observed.

Akania (Fig. 3).

Wood plan. Growth rings very inconspicuous (Fig. 3a).

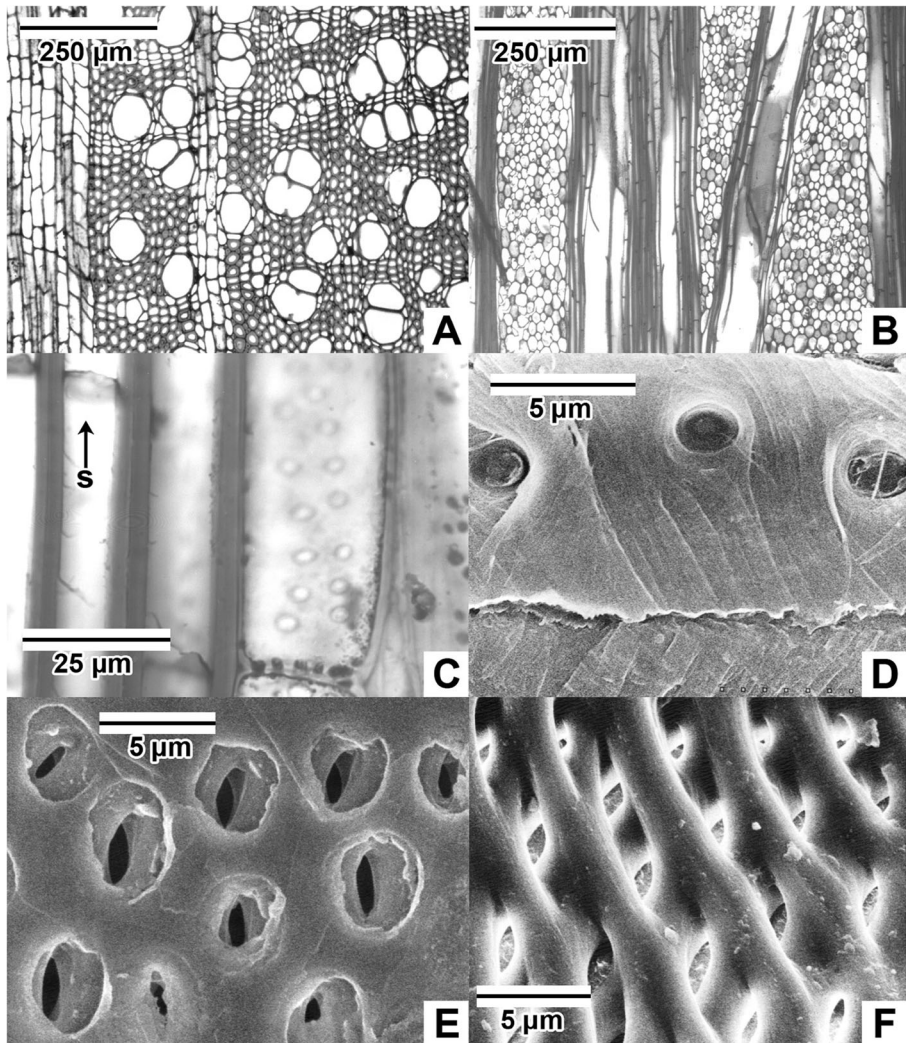


Fig. 3 Wood of *Akania bidwillii* (R. Hogg) Mabb., Forestry Commission of New South Wales SFCw-D10096. **a.** Transverse section; paratracheal parenchyma adjacent to vessels below the magnification scale. **b.** Tangential section; portions of three wide multiseriate rays present. **c–f.** Radial sections. **c.** Septate fiber, upper left (s = septum). Axial parenchyma strand, to right of center; pits on lateral wall, in face view, with small borders; cross wall (near bottom) with bordered pits in sectional view. **d–f.** SEM micrographs. **d.** Vestigially bordered pits on imperforate tracheary element. **e.** Outer surface of vessel; pits alternate, non-bordered. **f.** Inner surface of vessel; helical thickenings present

Storying. Wood not storied (Fig. 3b).

Vessels. Vessels solitary or in small groups (Fig. 3a). Perforation plates mostly simple. A few multiperforate perforations (modified scalariform, with variously shaped perforations) present. Lateral wall pits of vessels circular (Fig. 3e). Pits non-vestured. Vessel-to-vessel pits alternate (Fig. 3e), vessel-to-axial parenchyma and vessel-to-ray pitting scalariform to transitional. Lateral walls of vessels with varied helical sculpturing, grooves interconnecting pit apertures or helical thickenings present (Fig. 3f).

Imperforate tracheary elements. Septate fibers present (Fig. 3c). Pits of the septate fibers are simple or with narrow borders (Fig. 3d). Septate fibers rather thick walled.

Axial parenchyma in strands of four to eight (mostly five) cells. Pits on the cross-walls of the axial parenchyma strand often with perceptible borders. Axial parenchyma vasicentric scanty plus bands in latewood.

Rays. Rays Heterogeneous Type IIA of Kribs (1935), but rays of two distinct sizes; the wider rays average 7.1 cells across at widest point as seen in a tangential section (Fig. 3b). Multiseriate rays have an upright cell at upper and lower tip, and upright sheathing cells, but otherwise are composed of procumbent cells. Uniseriate rays are composed of upright cells.

Crystals. Occasional solitary rhomboidal crystals in ray cells.

Comments: The number of resemblances between wood of *Akania* and that of *Bretschneidera* is so compelling that one is amazed that these two genera had been relegated to widely-separated parts of the angiosperm tree (Carlquist, 1996). One also is amazed that two woods so similar could occur in such a disjunct pattern. To be sure, not all of the similarities may be synapomorphies. We have as yet no picture of the woods ancestral to Brassicales, although Akaniaceae gives us a good guideline of what to look for in Sapindales, Malvales, and related clades.

The list of similarities between the two genera includes: vessels solitary or in small groups; helical sculpture on walls of vessels; vessel to vessel pitting alternate; vessel to axial parenchyma and vessel to ray pitting scalariform to transitional; pits non-vestured; scalariform perforation plates or modifications of them scarce, but present throughout the secondary xylem; imperforate tracheary elements are septate fibers with simple or vestigially bordered pits; axial parenchyma scanty vasicentric and terminal; bordered pits present on cross-walls of the strands of axial parenchyma; multiseriate rays wide, with tip and sheath cells upright but ray cells otherwise procumbent; uniseriate rays composed of upright cells.

The few differences (septate fiber walls thicker in *Akania*; growth rings more apparent in *Bretschneidera*; crystals not observed in *Bretschneidera*) are those one might expect of congeneric species. The gross morphology of the flowers of the two genera, however, shows some conspicuous (but perhaps unimportant) differences.

2. Tropaeolaceae (Carlquist & Donald, 1996). (Fig. 4)

The information on wood anatomy of Tropaeolaceae is restricted to root and stem material of a single species, the commonly-cultivated *Tropaeolum majus*. One should keep in mind that this species is a scandent annual herb.

Wood plan (Fig. 4a). Growth rings absent. Vessels prominently dimorphic, the smaller ones grading into vasicentric tracheids.

Storying. Wood non-storied (Fig. 4b).

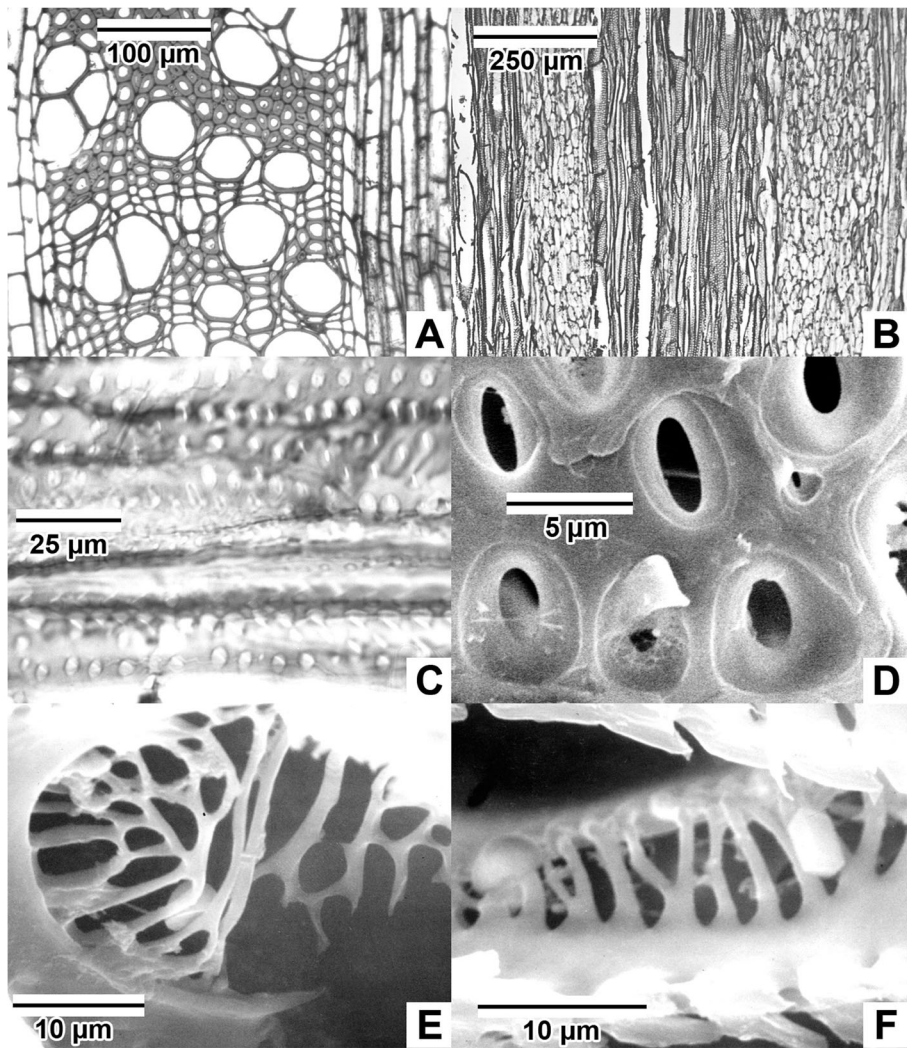


Fig. 4 Wood of *Tropaeolum majus* L. (Tropaeolaceae), cultivated in Santa Barbara, CA. **a.** Transverse section; thick-walled cells are mostly narrow vessels or vasicentric tracheids; thin-walled cells adjacent to vessel (center) are axial parenchyma. **b.** Tangential section; portion of two multiseriate rays composed of upright cells. **c–f.** Radial sections. **c.** Portions of narrow vessels and vasicentric tracheids; pits bordered. **d–f.** SEM micrographs. **d.** Non-vestured pits seen from outer surface of vessel. **e.** Irregular perforation plate. **f.** Narrow nearly scalariform perforation plate

Vessels. Wide vessels (ca. 80 µm in diameter) solitary or in pairs; narrow vessels (ca. 28 µm in diameter) present (together with vasicentric tracheids) in large numbers. Perforation plates mostly simple, but occasionally scalariform or scalariform-like, both in wide and narrow vessels (Fig. 4e and f). Lateral wall pitting of vessels and vasicentric tracheids covered with alternate bordered pits (Fig. 4c). Pits not vestured (Fig 4d).

Imperforate tracheary elements. In transection (Fig. 4a), wood of *Tropaeolum* appears to have fibers as a background cell type, but these

cells, when seen in longitudinal section and in macerations, prove to be both narrow vessel elements and vasicentric tracheids. Libriform fibers are also present. The libriform fibers are alive at maturity and contain globular starch grains.

Axial parenchyma. Axial parenchyma is sparse, vasicentric (Fig. 4a), with primary walls. Axial parenchyma in strands of two cells.

Rays. Multiseriate only in the stem, more than 1 cm in average height. Roots with a few uniseriates in addition to the multiseriates. Ray cells mostly upright, with a few square to procumbent cells in central portions of rays.

Crystals. No crystals observed.

Comments: Molecular data places Tropaeolaceae adjacent to the two genera of Akaniaceae (Gadek et al., 1992; Rodman et al., 1998; Hall et al., 2004; Soltis et al., 2011). The wood of Tropaeolaceae can, in this context, be seen to be essentially an herbaceous version of wood of Akaniaceae, modified for the vining habit (vessels dimorphic: Carlquist, 1985a).

3. Moringaceae (Olson, 2001, 2002, 2007; Olson & Carlquist, 2001). (Fig. 5)

Our knowledge of wood anatomy of Moringaceae is definitive, thanks to the field work of Mark Olson, especially in northeastern Africa. The resulting wood monograph (Olson & Carlquist, 2001), groups the species of the single genus, *Moringa*, according to habit, which also closely follows the phylogeny of the genus (Olson, 2001): bottle trees, sarcorhizal trees (thick roots with soft wood), slender trees (e.g., the widely-cultivated *M. oleifera*), and tuberous shrubs (stems of a finite duration, borne atop a large underground tuber). The species illustrated here (Fig. 5) include bottle trees (*M. drouhardii*, *M. hildebrandtii*, *M. stenopetala*) and the multi-stemmed tree *M. oleifera* (Fig. 5f).

Wood plan: Olson & Carlquist (2001) recognize four groupings of species. The anatomical differences can be seen most clearly in transverse sections.

- (a) Bottle trees. Wood background of stems composed of thin-walled wide libriform fibers (Fig. 5a) with occasional bands of narrow fibers (Fig. 5b). Degeneration of some wide fibers in young stems (Fig. 5a) can result in formation of air spaces. Wide bands of paratracheal axial parenchyma yield, often abruptly, to libriform fibers. Wood of roots is similar, with wider bands of axial parenchyma, narrower bands of libriform fibers.
- (b) Sarcorhizal trees. Stem wood of *M. arborea* Verdc. as in the bottle trees, wood of *M. ruspoliana* Engl. with wide bands of wide water-storing libriform fibers rather than wide bands of axial parenchyma. Axial parenchyma vasicentric, a single layer thick, in *M. ruspoliana*. Roots have wider bands of irregular diameter axial parenchyma cells and occasional bands of libriform fibers.
- (c) Slender trees. Stem background tissue consists of a preponderance of libriform fibers, with variable amounts of earlywood axial parenchyma in the wet season. Roots have alternating bands of libriform fibers and paratracheal axial parenchyma bands (much as in stems of the bottle trees).
- (d) Tuberous shrubs. Stems slender, like those of the slender trees, with a similar preponderance of libriform fibers. Root secondary xylem background tissue is

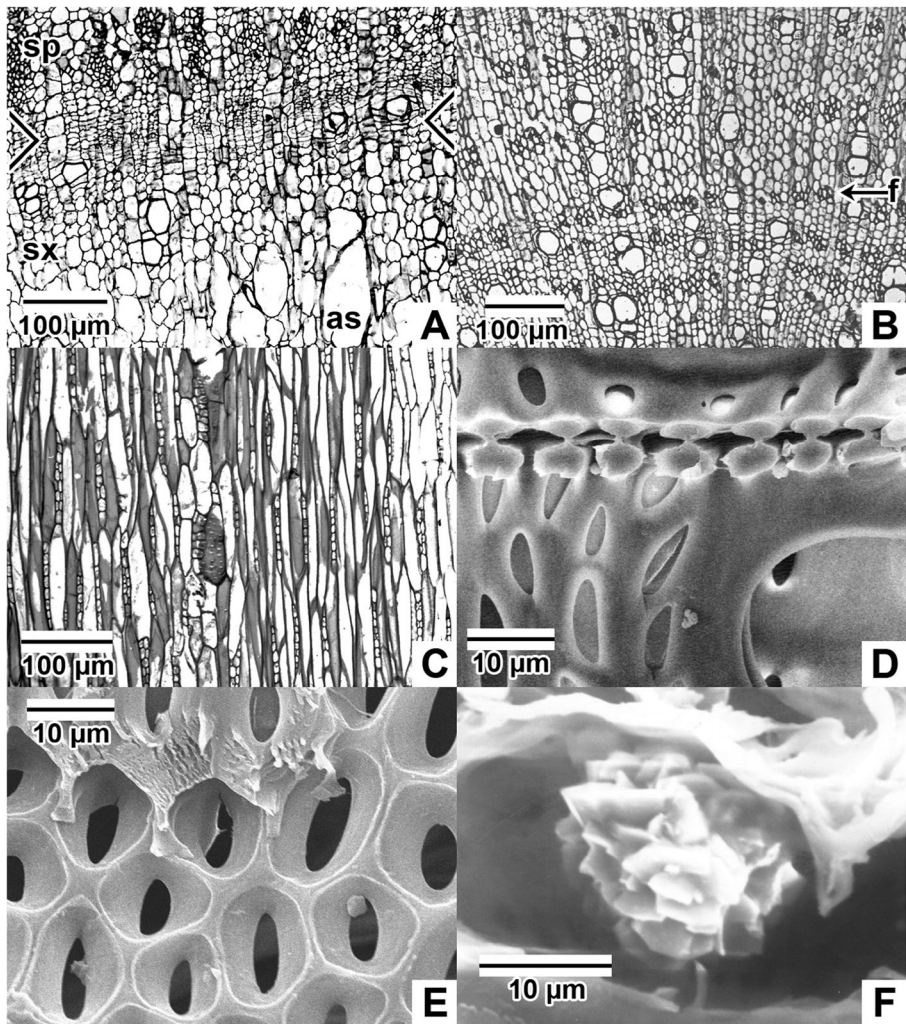


Fig. 5 Wood of *Moringa* (Moringaceae). **a–b.** *M. hildebrandtii*, Arid Lands Greenhouses, transverse sections from young plants. **a.** Section including cambium (pointers), secondary phloem (sp) above; air spaces (as) below in parenchymatous secondary xylem (sx). **b.** Section with band of fibers (f) in thin-walled secondary xylem. **c.** *M. stemopetala* (Bak. F.) Cufod., Olson 675 (UNAM), tangential section; storing prominent in rays and fibers. **d–f.** Radial section portions, SEM micrographs. **d–e.** *M. drouhardii* Jum, Arid Lands Greenhouses. **d.** Perforation plate portion and intervessel pitting in sectional view. **e.** Outer surface of vessel: non-vestured alternate pits. **f.** *M. oleifera* Lam., cultivated in Honolulu, HI. Druse in a ray cell

composed almost entirely of thin-walled axial parenchyma, rich in starch, as in stems of Caricaceae.

Storing. Larger, less juvenile stems have storied libriform fibers and axial parenchyma (Fig. 5c). Rays are storied only in older stems that have the most pronounced storing of axial elements.

Vessels. Vessel elements are circular in transverse section (Fig. 5a and b), solitary or in small groupings, widest in diameter in the bottle trees (148 μm), intermediate in slender trees (129 μm) and narrowest in stems of tuberous shrubs (40–80 μm mean diameter). Vessel diameter is wider in roots than in stems in any given species (Olson & Carlquist, 2001). Perforation plates all simple (Fig. 5d). Lateral wall pitting of vessels alternate, from circular to oval (Fig. 5d and e), somewhat laterally elongate to pseudoscalariform. Shallow helical grooves present in vessels of *M. riva* stems. Vessel-to-vessel pitting in my material of *M. oleifera* non-vestured (Fig. 5d and e), but vestured pits reported and figured for that species by Jansen et al. (2001). Jansen's report of vesturing is enclosed in parentheses, his symbol for indicating that not all collections or portions show vesturing in a species. When present, vestured pits in vessels are like those shown here for *Setchellanthus* (Fig. 8e), with vestures present only at the edges of the pit aperture. Vessel density greatest in stems of tuberous shrubs (32 per mm^2), intermediate in sarcorrhizal trees (20), and slender trees (13), and least in bottle trees (7).

Imperforate tracheary elements. These can all be categorized as libriform fibers, but range from rectangular with blunt ends to fusiform, with some intermediate fibers cuboidal but with fusiform tips at either end (Fig. 71–75 in Carlquist & Olson, 2001). The acicular libriform fibers are more common in the slender trees. Wider libriform fibers are thought to be related to water storage. Criteria for distinguishing between axial parenchyma and libriform fibers include subdivision in axial parenchyma cells, slit-like pits on libriform fibers (as opposed to small circular pits on parenchyma walls), and greater density of pits on walls of axial parenchyma cells.

Axial parenchyma. Some axial parenchyma cells may be non-subdivided, but their co-occurrence with subdivided (strands of) parenchyma, as well as the above criteria permits discrimination between the two cell types. Axial parenchyma strands can be up to four cells in length; the proportions of undivided and subdivided axial parenchyma vary according to species (see radial sections of wood in Olson and Carlquist, 2001).

Rays. The rays are Heterogeneous Type IIB of Kribs (1935), with the proportion of upright to procumbent cells varying according to degrees of wood juvenilism. Tall non-storyed rays composed chiefly or entirely of upright cells are indicative of pedomorphosis in wood of the genus. The opposite non-juvenilistic condition is illustrated here for *M. stenopetala* (Fig. 5c).

Crystals. Both druses (Fig. 5f) and solitary rhomboidal crystals have been reported in ray cells of *Moringa*, depending on species (Olson & Carlquist, 2001). Crystals are more common in pith and secondary phloem than in wood.

Starch storage. Starch is abundant in axial parenchyma of the tuberous species.

4. Caricaceae (Carlquist, 1998a; Fisher, 1980) (Fig. 6).

Wood plan. The secondary xylem of Caricaceae consists of vessels and bands of laticifers embedded in a tissue composed of axial parenchyma (Fig. 6a–d). The laticifers are articulated and anastomosing (Fig. 6c). The bands of laticifers sometimes cross rays rather than being confined to fascicular xylem. No fibers have been reported in the secondary xylem, but the secondary phloem is rich in fibers (accounting for the fact that some xylarium samples consist entirely or mostly of secondary phloem).

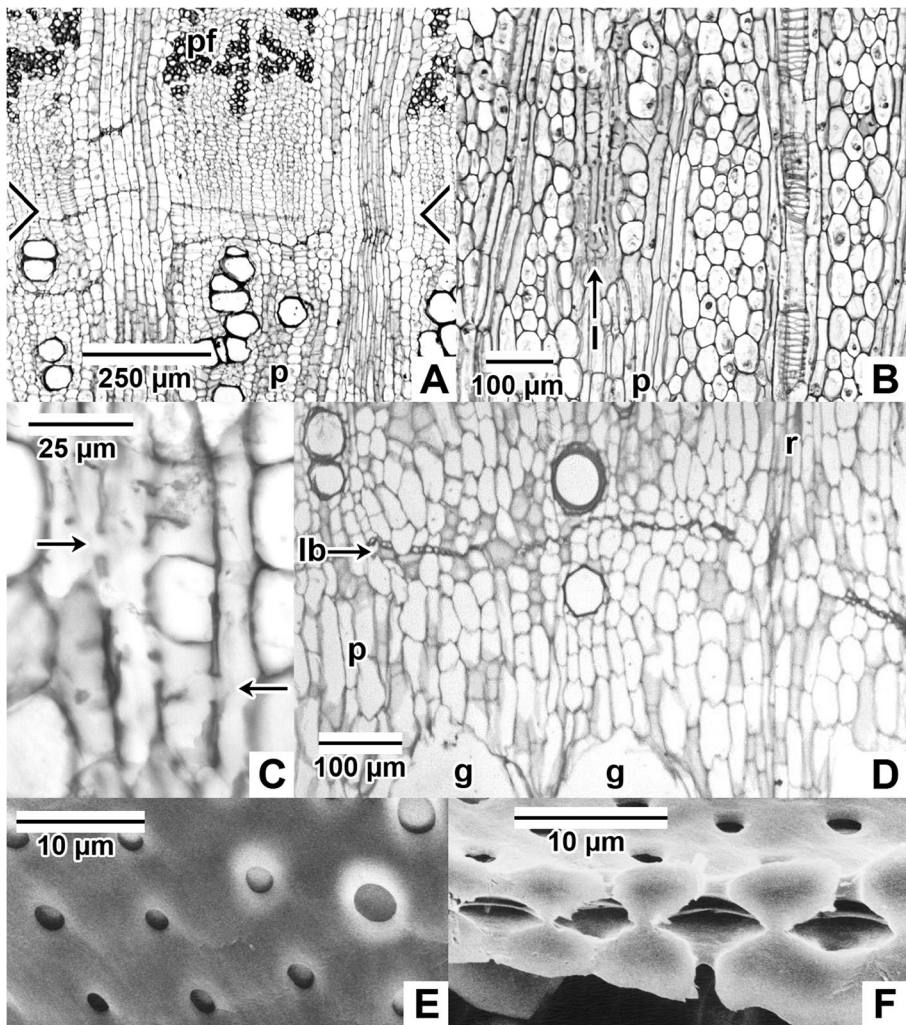


Fig. 6 Wood of Caricaceae. **a.** *Carica pentagona* Heilbron, Carlquist 8179 (SBG); transverse section of stem, cambium indicated by pointers. Secondary phloem with fibers (pf) above; parenchymatous secondary xylem (p) below. **b–c.** *C. quercifolia* Solms, cultivated at University of California, Santa Barbara. **b.** Tangential section; axial parenchyma (p) present instead of fibers (I=laticifers). **c.** Portion of a laticifer band; arrow indicates one of many pores interconnecting the cells forming the anastomosing laticifers. **d–f.** *C. papaya*, cult. Honolulu, HI. **d.** Transverse section showing laticiferous band (lb), gaps (g) in parenchyma (p) and a ray (r). **e.** **f.** SEM micrographs. **e.** Inner surface of vessel pits circular, non-vestured. **f.** Inter-vessel pits in sectional view, non-vestured

Storying. The vascular cambium is storied, and occasional axial parenchyma cells are storied (Fisher, 1980).

Vessels. Vessels are often grouped (Fig. 6a), although solitary vessels (Fig. 6d) are not uncommon. The average number of vessels per group for the family, 1.6, can be interpreted as indicating that the secondary xylem is mesomorphic, if only by means of succulence (Carlquist, 1998a). The vessel diameter and low vessel density (Fig. 6d) are also indicative of mesomorphy. Perforation plates are simple. Vessel-to-vessel lateral

wall pitting consists of circular alternate pits (Fig. 6e and f) scalariform pitting (horizontal length of pits conforming to facet width) or pseudoscalariform pitting (elongate pits extending over more than one facet) is common on vessel to axial parenchyma and vessel to ray contacts of vessels. Pits are non-vestured (Fig. 6e–f).

Axial parenchyma. Strands consist of one, two, or three cells. The axial parenchyma distribution type can be termed pervasive (Carlquist, 1988). Axial parenchyma proliferation occurs in the form of divisions that increase the stem thickness (Arnold & Baas Becking 1949). Radial and tangential cell expansion, as well as divisions, participate in this process. Some lignification occurs in a few parenchyma cells adjacent to vessels (Fisher, 1980).

Rays. Vascular rays are multiseriate; a few uniseriate rays are present (Fig. 6b, to the left of the laticifers). Radial sections show that a few upright cells are present in rays, but the majority of the ray cells are procumbent. The procumbent nature of ray cells is achieved, in part, by radial stem expansion. Increase in tangential width of multiseriate rays by radial divisions was reported by Fisher (1980).

Crystals. Crystals in ray cells of Caricaceae have been reported only in *Cylicomorpha*, where druses occur in some ray cells (Carlquist, 1998a). Druses are abundant in the cortex of *Carica* but absent in the secondary xylem (Fisher, 1980).

Starch. Starch is present in axial parenchyma, especially near vessel elements (Carlquist, 1998a) and at the margins of rays (Fisher, 1980).

Comments. The secondary xylem of Caricaceae, despite its parenchymatization and succulence, is not really juvenilistic in most features, showing that paedomorphosis and succulence do not necessarily go together. Caricaceae could be called “woody succulents,” provided that one understands that the axial and ray parenchyma cells, which have primary walls only, may function in any of several ways not studied experimentally as yet, such as photosynthate storage and retrieval.

Olson (2002) has noted that *Cylicomorpha* may be sister to the remainder of Caricaceae. In this regard, occurrence of druses in the rays of some *Moringa* species and in the rays of *Cylicomorpha* is an interesting point of resemblance between *Cylicomorpha* and Moringaceae. *Cylicomorpha* is African, as are most *Moringa* species (*M. oleifera*, native to India, is an exception). When viewed as the sister family of Moringaceae, Caricaceae appear to have developed a curious division of labor between strength in secondary phloem fibers and storage of water and photosynthates in axial parenchyma and rays, with their thin primary walls. In this regard, there may be a correlation between the large fleshy fruits of Caricaceae, compared to the follicular fruits of Moringaceae.

The presence of laticifers in Caricaceae is clearly an apomorphy; laticifers are unknown elsewhere in Brassicales. Other apomorphies and synapomorphies of Caricaceae and Moringaceae are listed by Olson (2002).

5. Limnanthaceae (Carlquist & Donald, 1996) (Fig. 7).

The information here is based on a single species of *Limnanthes*. The family is small (two genera, eight species), and is not diverse with respect to habit. All are annuals that occur in places moist during winter, but drying out during warmer months.

Wood plan: The basal stem of *Limnanthes* consists of a circle of bundles. Secondary growth is mostly limited to these areas (Fig. 7a), so that a complete

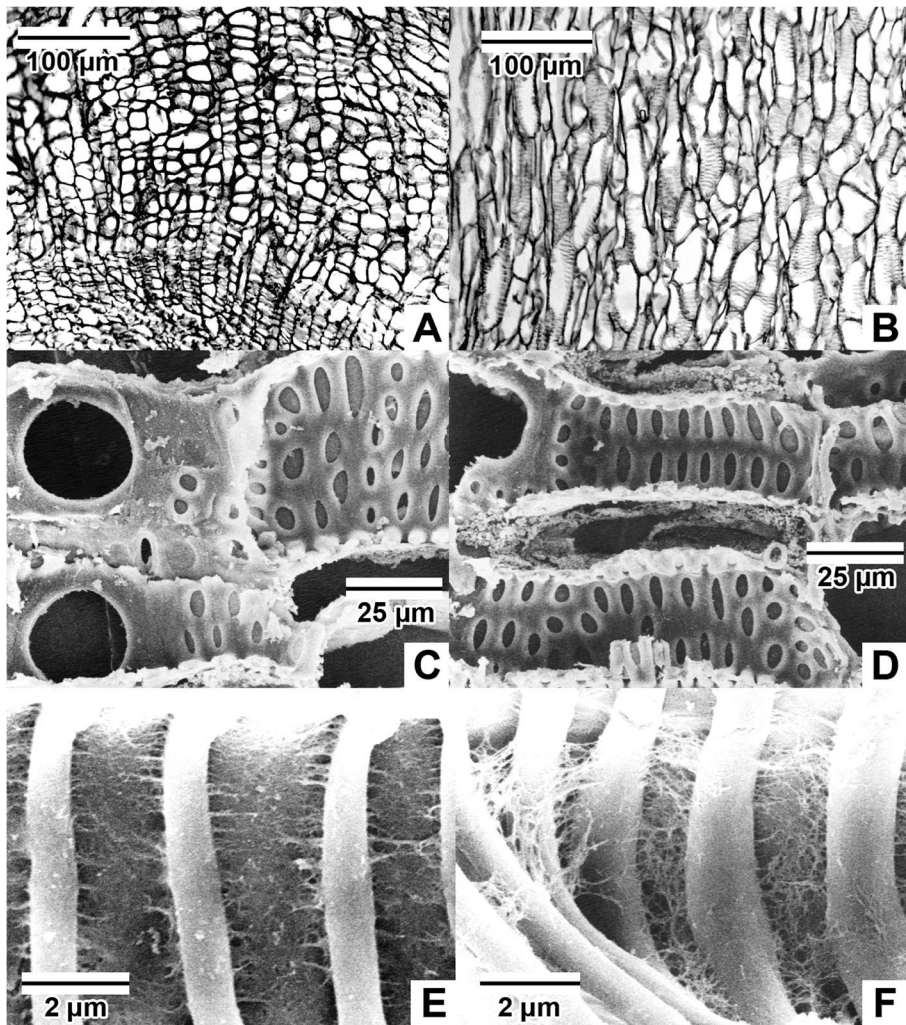


Fig. 7 Wood of *Limnanthes douglasii* R. Br. (Limnanthaceae), cultivated, Santa Barbara Botanic Garden. **a.** Transverse section, secondary phloem at bottom. Axial parenchyma is of the pervasive type. **b.** Tangential section. Axial parenchyma is difficult to distinguish from rays. **c–f.** SEM micrographs from radial sections. **c.** Perforation plates; alternate pitting. **d.** Vessel elements short, pitting mostly scalariform, non-vestured. **e–f.** Helices of protoxylem vessels seen from inside vessels. **e.** Strands of wall material extend onto helices. **f.** Strands are reticulate

cylinder does not typically form. Some of these units are tangentially wide, up to a third of the stem circumference. Zones of primary rays with little secondary growth are interpolated among these units. The wood consists wholly of vessels and parenchyma (Fig. 7a and b).

Storying. No storying has been observed (Fig. 7a and b). One would not expect storying in a species with so little secondary xylem accumulation.

Vessels. Vessel diameter increases from protoxylem into secondary xylem, then decreases again as secondary growth proceeds to cessation. Vessels are mainly in

radial groupings, Vessel elements are very short (average = 72 μm) and have simple perforation plates (Fig. 7c and d). Vessel-to-vessel pits are alternate (Fig. 7c), but vessel-to-axial parenchyma and vessel-to-ray pitting tends to be scalariform (Fig. 7d). Vessel diameter is narrow, about 28 μm in the specimen studied. Vessels are angular, square to polygonal in transection (Fig. 7a).

Axial parenchyma. Axial parenchyma is scattered among the vessels, a pattern termed intervacular by Carlquist (1988, 2001). Axial parenchyma is not subdivided into strands (Fig. 7c, lower right; Fig. 7d, an axial parenchyma cell lies between the two vessels). Axial parenchyma has primary walls only.

Rays. Uniseriate and biseriate rays observed (Fig. 7b). These are difficult to discern in tangential sections because ray cells and axial parenchyma appear similar. Ray cells have primary walls only.

Stand on primary xylem helices. On the insides of primary xylem vessels, SEM studies reveal distinctive strands of wall material, possibly pectic in nature, stretching between the helices and the primary wall (Fig. 7e and f). These are most apparent close to the gyres (Fig. 7e), but may also form a webbing superimposed on the primary wall (Fig. 7f). Reports of such structures in protoxylem of angiosperms other than *Limnanthes* are not evident to me, although protoxylem is little studied by means of SEM.

Comments. *Limnanthes* wood seems clearly related to the annual habit. However, there are many kinds of annual habits in angiosperms, and these have been little explored. The stems of annual Brassicaceae and Tropaeolaceae reveal wood plans different from that of Limnanthaceae.

6. Setchellanthaceae (Carlquist & Miller, 1999) (Fig. 8).

The single species of *Setchellanthus* is a shrub, branched from the base, up to one meter in height. It is native to arid hillsides in restricted areas of northern and central Mexico.

Wood plan. Growth rings are only minimally developed, evident in terms of fluctuation more in terms of imperforate tracheary element diameter than in vessel diameter (Fig. 8a). Because vessels are mostly solitary, and vary in diameter, precise delimitation of earlywood and latewood is difficult.

Storying. Neither axial xylem nor rays are storied (Fig. 8b).

Vessels. Vessels are circular in outline (Fig. 8a). Most vessels are solitary (vessels per group from 1.03 to 1.12), which would correlate with the presence of vasicentric tracheids (see Carlquist, 1984). Vessel density is low, 35–113 μm (Fig. 8c). Vessel diameter of several collections studied ranges from 28 to 39 μm . Vessel element length ranges from 148 to 198 μm . Vessel-to-vessel pitting is alternate, the circular pits small (2 μm in diameter) with vesturing along the edges of pit apertures (Fig. 8e). Vesturing was not reported earlier (Carlquist and Miller 1999). Helical sculpture (grooves and ridges) is present on some vessel walls (Fig. 8f). Yellowish deposits are present in some vessels (Fig. 8c).

Imperforate tracheary elements. Fiber-tracheids (Fig. 8c) and vasicentric tracheids (Fig. 8d) are present. There are occasional vestigial vestures on pits of the vasicentric tracheids. Imperforate tracheary element lengths (292–230 μm) as well as vessel element length decrease as the stem widens.

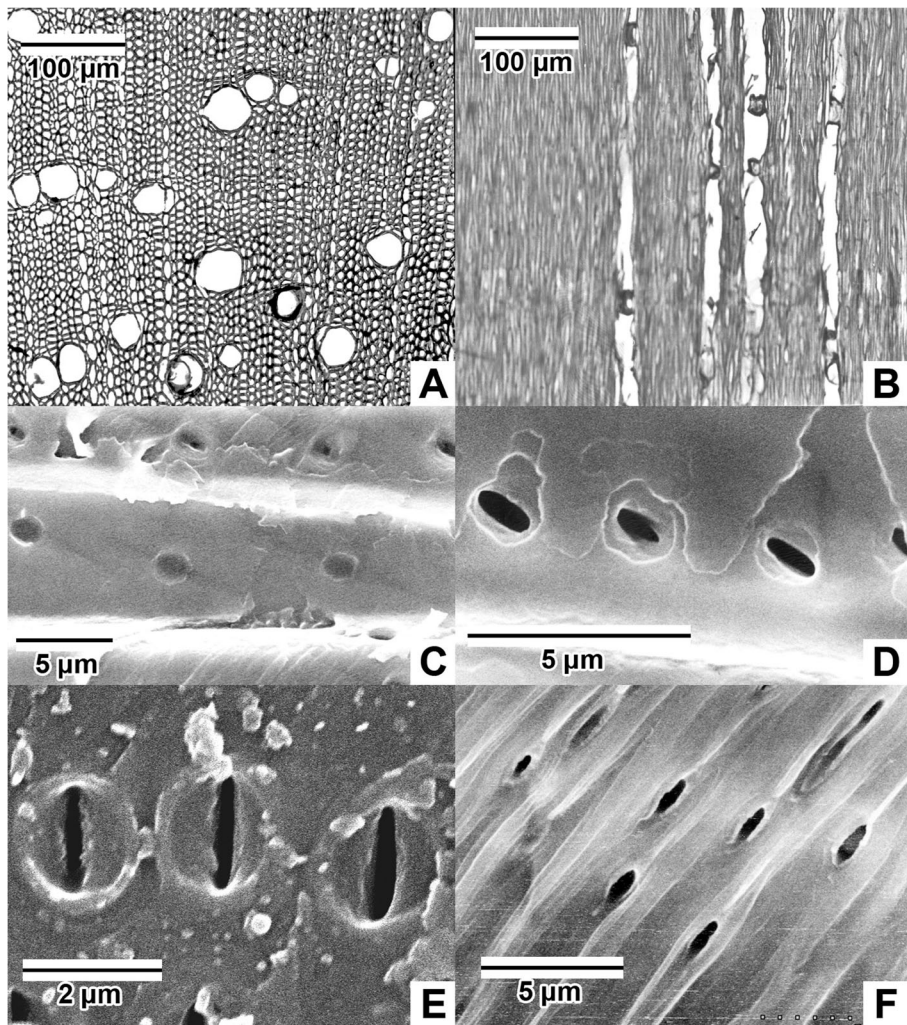


Fig. 8 Wood of *Setchellanthus caeruleus* Brandegees (Setchellanthaceae), Iltis et al. 31801 (WIS). **a.** Transverse section. Axial parenchyma is vasicentric scanty plus diffuse and is visible in color only because of counterstaining. **b.** Tangential section. Rays are uniseriate, mostly one cell high, better visible at higher magnifications. **c–f.** SEM micrographs from radial sections. **c.** Pits scattered along libriform fiber. **d.** Bordered pits on vasicentric tracheid. **e.** Three vestured pits seen from outer surface of a vessel. **f.** Helical sculpturing on inner vessel surface

Axial parenchyma. Axial parenchyma consists of single cells rather than strands. Axial parenchyma is in the form of diffuse cells, diffuse-in-aggregates, and vasicentric scanty. These were apparent only in preparations that had been counterstained, permitting axial parenchyma cells to be distinguished in transection from imperforate tracheary elements.

Rays. Rays are all uniseriate, mostly one cell in height, but occasionally two. Rays are so slender tangentially that they are not readily apparent in Fig. 8b. Rays clearly

belong to Paedomorphic Type III of Carlquist (1988), a type seen in small shrubs such as *Empetrum*. Ray cells have secondary walls.

Crystals. No crystals were observed.

Comments. The decrease in vessel element length and imperforate tracheary element length over time, as well as the exclusively upright cells of rays, are hallmarks of paedomorphosis (Carlquist, 1962). *Setchellanthus* wood is distinctive in Brassicales in having vasicentric tracheids, and these may play an important role in conductive safety in this arid-land shrub.

7. Koeberliniaceae (Gibson, 1979 and original data below) (Fig. 9).

Earlier descriptions of wood of Koeberliniaceae include *Canotia* (e.g., Metcalfe and Chalk, 1950), but *Canotia* has been excluded and placed in Celastraceae). *Koeberlinia* is monogenetic, and had been thought to consist of the single species *K. spinosa*, which occurs in summer-wet desert areas of southern California to Texas and adjacent portions of Mexico. In 2008, a second species, *K. holacantha* W. C. Holmes, K. L. Yip, & Rushing was discovered in similar areas of Bolivia.

Wood plan. Growth rings are well developed (Fig. 9a), with earlywood vessels ca. 50 μm in diameter. Latewood vessels occupy the bulk of each growth ring and are 25 μm or less in diameter. A few patches of vessel-free wood seen in the first growth ring of one specimen.

Storying. Tangential sections show storying in all of the fascicular elements but not in the rays (Fig. 9b).

Vessels. Vessels are solitary (Fig. 9a), circular in outline. Vessel elements are short (average = 77 μm). Perforation plates are simple. Vessel-to-vessel pits are circular and alternate. Vessel to ray pits are circular. Vessel pits are vested (suspected by Gibson, 1979, but not observed by Jansen et al., 2001); the vestures form a series of minute knobs along the edges of the pit aperture (Fig. 9d). Helical thickenings on inner surfaces of vessels (Fig. 9e), most conspicuously in latewood.

Imperforate tracheary elements. Tracheids with fully bordered pits form the background cell type of fascicular secondary xylem (Fig. 9c). The bordered pits are vested. Mean tracheid length = 83 μm).

Axial parenchyma. Diffuse axial parenchyma is present. It is sparse and randomly scattered. Because of the high vessel density, one could say that vasicentric scanty parenchyma is in part present, because axial parenchyma—vessel contacts are inevitable. Parenchyma mostly in strands of two cells.

Rays. Rays are Heterogeneous Type IIB of Kribs (1935). Uniseriates are relatively few. Multiseriate rays are composed mostly of procumbent cells; a few square and upright cells may be found as sheathing cells. Multiseriate rays average 320 μm in height, but are quite variable in dimensions, and up to eight cells wide at widest point. Ray cells with secondary walls, some with bordered pits. Uniseriates are composed mostly of square to upright cells.

Crystals. Druses are present in ray cells of secondary xylem near the pith (Fig. 9f), but rhomboidal crystals predominate in secondary xylem. Rhomboidal crystals are borne one per ray cell and vary greatly in size.

Comments: Presence of tracheids in wood of Koeberliniaceae is unusual among Brassicales, but probably accords with the desert habitat. Tracheids are present in wood

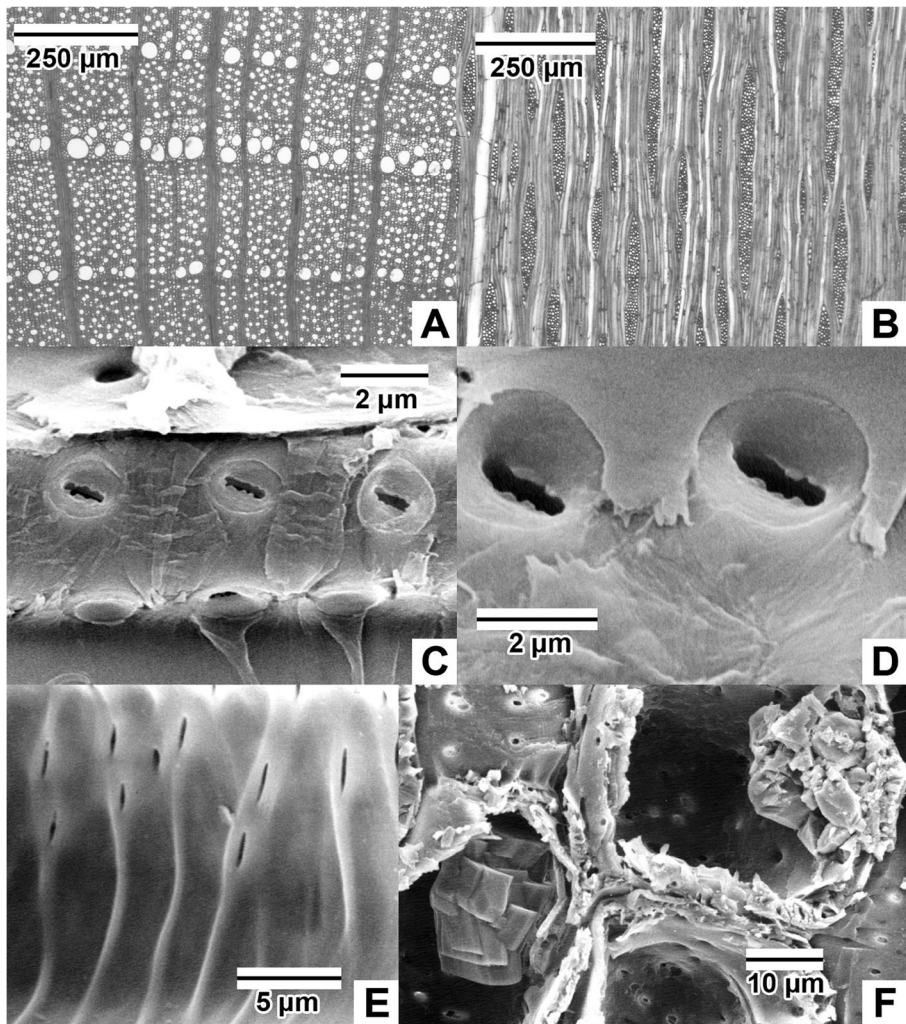


Fig. 9 Wood of *Koeberlinia spinosa* Zucc. (Koeberliniaceae), A. C. Sanders 9521 (SBG). **a.** Transverse section, Vessels solitary, earlywood vessels markedly larger in diameter. **b.** Tangential section. Multiseriate rays present, storying present but inconspicuous, **c–f.** SEM images of radial section portions. **c.** Outer surface of tracheid to show spacing of pits, which are vested. **d.** Two vessel pits, seen from outside of vessel; vesturing present along pit aperture. **e.** Inner surface of vessel; helical thickenings present. **f.** Ray cells containing intact druse (*lower left*) and druse fractured by sectioning (*upper right*)

of such desert genera as *Krameria* and *Prunus*. The presence of tracheids is associated with solitary vessels, a correlation proposed for angiosperm woods at large earlier (Carlquist, 1984).

8. Bataceae (McLaughlin, 1959; Carlquist, 1978) (Fig. 10).

Bataceae consists of two subtropical to tropical species of *Batis*, *B. argillicola* van Royen (South Pacific) and *B. maritima* (New World). Both tend to occupy muddy

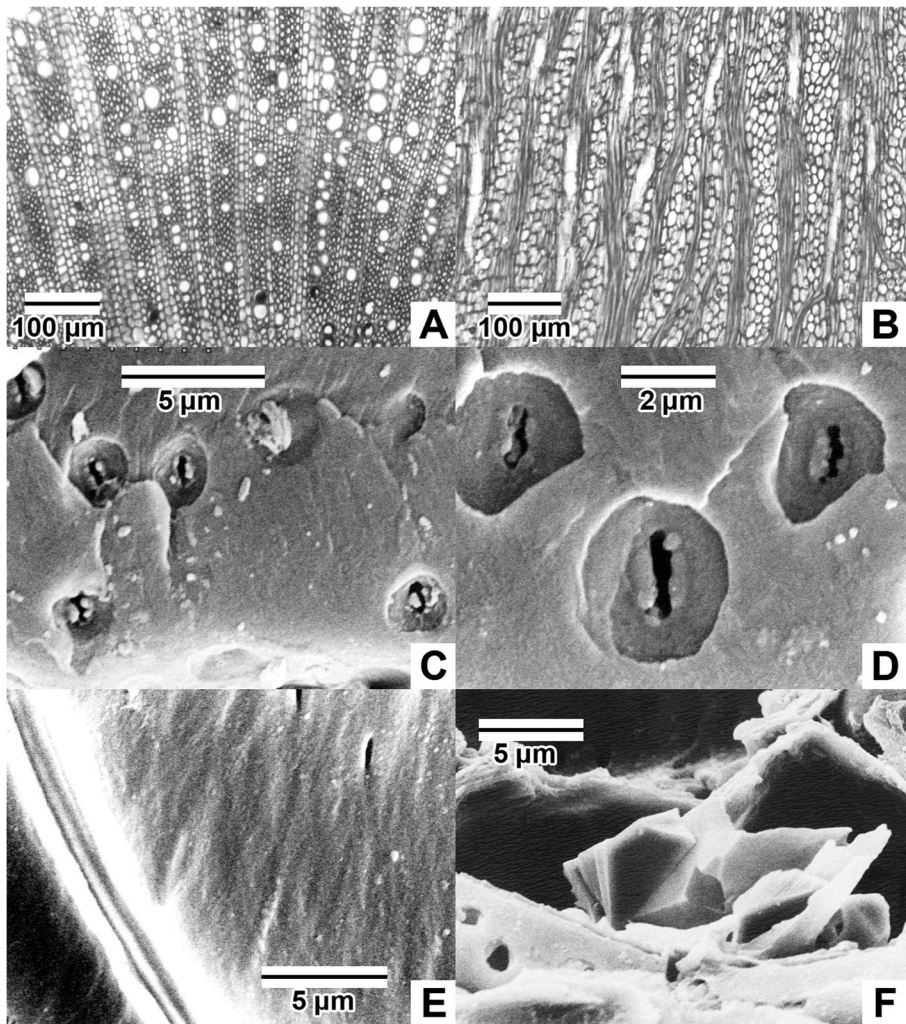


Fig. 10 Wood of *Batis maritima* L. (Bataceae), Carlquist s. n. RSA, Hanauma Bay, Oahu, HI. **a.** Transverse section. Increase in vessel diameter over time. **b.** Tangential section. Abundant multiseriate rays. **c–f.** SEM micrographs from radial section portions. **c.** Outer surface of tracheid, Note vestures. **d.** Outer surface of vessel, showing three pits with vestured pit apertures. **e.** Inner surface of vessel (perforation plate at left). Wall is striated. **f.** Portion of a druse from a ray cell

saline flats near seacoasts and to be shrubs up to a meter in height with succulent leaves. The data here are based on materials of *B. maritima*.

Wood plan. Little fluctuation in vessel or imperforate tracheary element diameter, growth rings are lacking (Fig. 10a shows increase in vessel diameter distal to the pith).

Storying. Storying is evident in larger stems, in which the storied pattern of axial parenchyma and imperforate tracheary elements agrees with the storying of vessel elements (Carlquist, 1978).

Vessels. Vessels are circular in transverse section (Fig. 10a). About a third of the vessels are solitary; the others are grouped in radial multiples. Vessel-to-vessel and

vessel-to-ray pitting consists of circular alternate pits. The vessel-to-vessel pits are vested (Fig. 10d), as reported by Jansen et al. (2001). Mean vessel diameter = 40 μm ; mean vessel element length = 104 μm ; mean number of vessels per sq. mm in transverse section, 126. Faint helical striations visible with SEM on vessels walls (Fig. 10e).

Imperforate tracheary elements. Minute borders are present on pits of imperforate tracheary elements, which are therefore termed fiber-tracheids here (Fig. 10c). These fiber-tracheid pits are vested (Fig. 10c).

Axial parenchyma. Axial parenchyma is paratracheal, one to several cells thick around vessels or vessel groups; also present in the form of short apotracheal bands. Axial parenchyma in strands of two cells near vessels, but undivided distal to the vessels.

Rays. All rays multiseriate, composed of upright and procumbent cells (Fig. 10b); procumbent cells more common in wood of wider stems. Multiseriate rays two to eight cells wide.

Crystals. Crystals were not reported earlier, but druses are present in some ray cells (Fig. 10f).

Comments. The degree of juvenilism in wood of *Batis* is not markedly pronounced, and the wood does not qualify as genuinely paedomorphic.

9. Salvadoraceae (den Outer & van Veenendal 1981; Fahn et al., 1986; Carlquist, 2002) (Figs. 11 and 12).

Salvadoraceae consist of three genera of shrubs of warm and dry and somewhat alkaline habitats.

Wood plan. Interxylary phloem strands present (Fig. 12a and e), markedly contrasting with thick-walled fibers and variously arranged axial parenchyma bands (Figs. 11 and 12a and e).

Storying. Axial parenchyma is clearly storied in accordance with the vessel elements, but storying less evident in the libriform fibers (Figs. 11b and 12b and f).

Vessels. Vessels mostly grouped (*Azima*, 4.2 vessels per group; *Dobera*, 2.0; *Salvadora*, 5.2; Figs. 11a and 12a and e). Vessels circular in transection, narrow (vessel diameter in *Azima*, 25 μm ; *Dobera*, 15 μm ; *Salvadora*, 12 μm). Vessel elements short (*Azima*, 229 μm ; *Dobera*, 134 μm ; *Salvadora*, 175 μm). Vessel-to-vessel and vessel-to-ray pits are alternate and circular to polygonal. Vested pits not reported earlier for Salvadoraceae (Jansen et al., 2001; Carlquist, 2002), but newly reported here for *Azima* (Fig. 11d). The vestures of *Azima tetracantha* are distinctively large and few per pit. In addition to the vested pits, sphaeroidal vestures are present on flanges that parallel the perforations plates in *Azima tetracantha* (Fig. 11e and f). The perforations are either bordered (Fig. 11e) or non-bordered, but the vestures, similar in size to those on the vessel pits, are not on the borders on the perforation plates themselves, but on flanges separated from the perforation borders by a groove. These vestures may be crowded (Fig. 11e) or somewhat sparse (Fig. 11f). This represents a new kind of vesturing in angiosperms. Helical striations were observed on vessel walls in *Salvadora* (Fig. 12c).

Imperforate tracheary elements. Libriform fibers are present, mostly quite thick-walled (Figs. 11a and 12a and e), and with simple pits. The libriform fibers are about four times the length of vessel elements in the various species, an evidence of intrusive growth that lessens the storying evident in vessels and axial parenchyma.

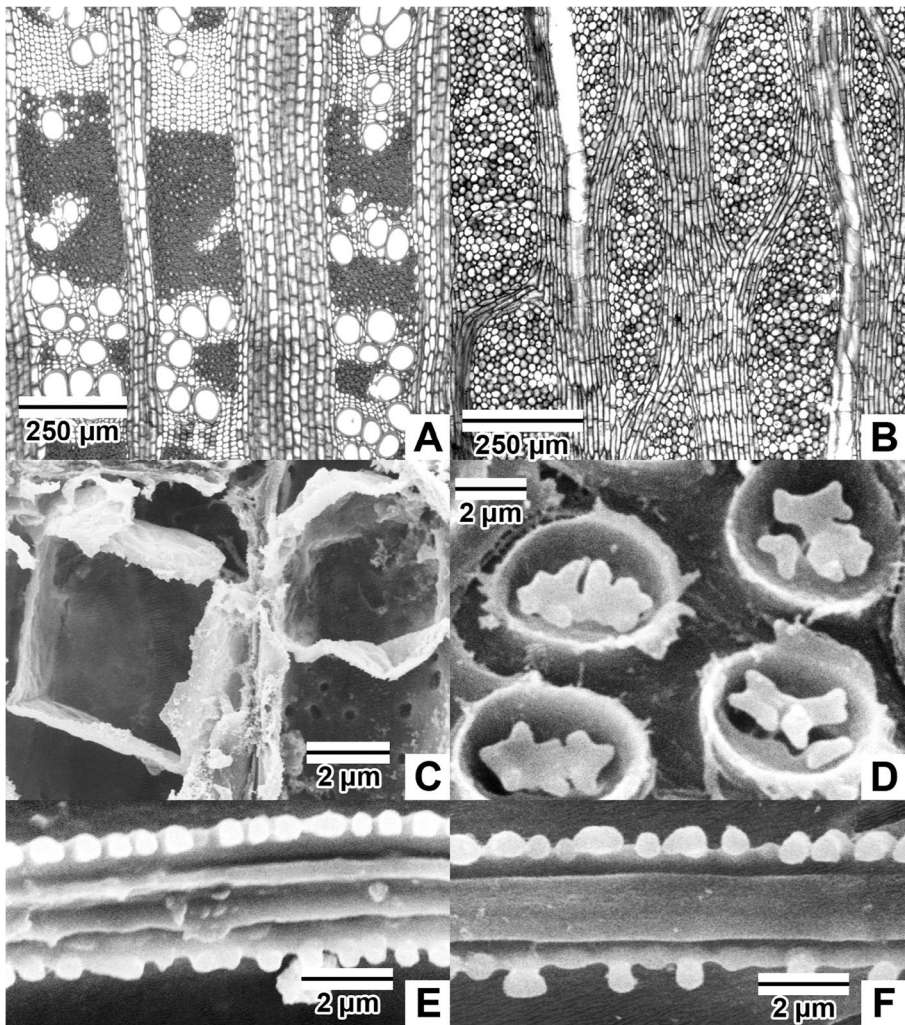


Fig. 11 Wood of *Azima tetraacantha* Lam. (Salvadoraceae), Carlquist 4733 RSA. **a.** Transverse section. Patches of thick-walled libriform fibers alternate with large axial parenchyma bands. **b.** Rays are notably wide; storying is clear in axial parenchyma. **c–f.** SEM micrographs of radial section. **c.** Two encapsulated crystals in ray cells. **d.** Four vested pits (note large, lobe-like vestures) as seen from outer surface of vessel. **e–f.** Vestures along flanges that parallel the perforation plate borders. **e.** Flanges beside bordered perforation plate. **f.** Flanges beside non-bordered (borders fused) perforation plate

Axial parenchyma. Axial parenchyma strands are composed chiefly of two cells in *Azima* (Fig. 11b), but undivided cells are more common in *Salvadora* (Fig. 12b). As seen in transection (Figs. 11a and 12a and e), axial parenchyma takes the form of tangential bands that are usually paratracheal or in contact with vessels (Fig. 11a). A few diffuse cells are present (Fig. 11a).

Rays. Vascular rays are mostly multiseriate in *Azima* (Fig. 11a) and *Dobera* (Fig. 12e), but mostly uniseriate to triseriate in *Salvadora* (Fig. 12b). Most ray cells are square to procumbent; the proportion of procumbent cells increases with increase in

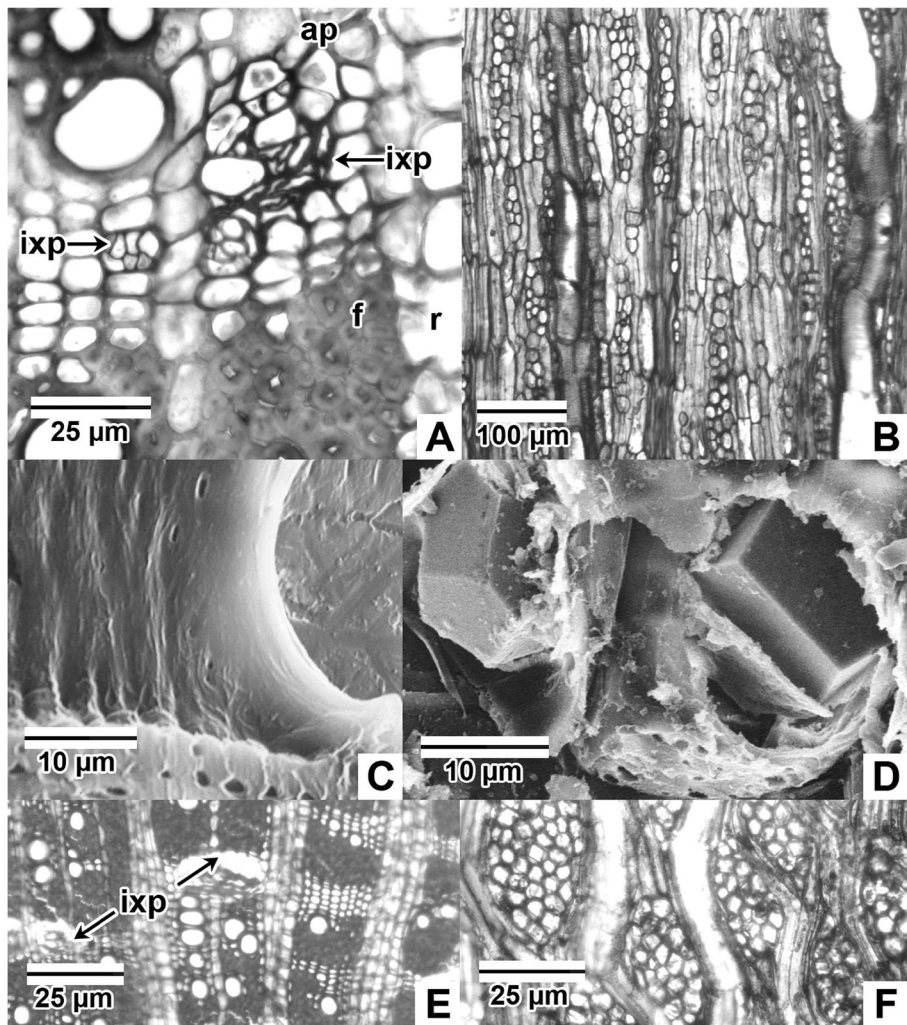


Fig. 12 Wood of Salvadoraceae. a–d. *Salvadora persica* L., cultivated at University of California, Santa Barbara., cult. no. 83160. **a.** Transverse section portion that includes vessels (upper left), axial parenchyma (ap), thick-walled fibers (f) and two strands of interxylary phloem (ixp). **b.** Tangential section. Rays are uniseriate, biseriate and triseriate; axial parenchyma is storied. **c–d.** SEM micrographs from radial sections. **c.** Inner surface of vessel; wall striate. **d.** Encapsulated crystals in ray cells, polyhedral crystal at left. **e–f.** *Dobera macalusoi* Mattei, MADw-26260. **e.** Transverse section. Sites of two interxylary phloem strands (ixp) indicated by arrows. **f.** Tangential section; rays are multiseriate, very short

stem diameter. The rays qualify as Heterogeneous Type IIB of Kribs (1937), perhaps transitional to Homogeneous Type II, in which all cells are procumbent and all rays are multiseriate.

Crystals. Crystals are very common (in almost every ray cell) in *Azima* and *Dobera*, less so in *Salvadora*. Crystals are rhomboidal to polyhedral (Figs. 11c and 12d). The crystals are encapsulated, and a variously thin layer of cell wall material can be seen

(Figs. 11c and 12d). Crystals that are smaller than those in rays may be found occasionally in axial parenchyma.

Starch. Starch grains are common both in axial parenchyma and in ray cells, and may be seen in liquid-preserved material (*Salvadora*). Starch grains may be single or borne in mutually-compressed pairs.

Comments. Interxylary phloem strands are present in all three genera (den Outer and van Veenendal 1981; Carlquist, 2002). Interxylary phloem is otherwise present in Brassicales only in Brassicaceae; the term “included phloem” is an umbrella term that fails to distinguish between Interxylary phloem and successive cambia and must be discontinued. Delayed onset of production of interxylary phloem strands was observed in *Azima tetracantha* (Carlquist, 2002), in which the first cm. of secondary xylem was observed not to have any such strands. Strands vary in size from very small (Fig. 12a, left) to moderately large (Fig. 12a, right). Interxylary phloem strands are embedded within axial parenchyma (Fig. 12a). Cambial activity within the strands leading to the formation of new secondary phloem cells (Carlquist 2013). In this way, longevity of the phloem strands is assured.

The wood of Salvadoraceae is notable for xeromorphy: narrow grouped vessels, short vessel elements, conspicuous storying, thick-walled libriform fibers, presence of Interxylary phloem strands, and large rhomboidal to polyhedral encapsulated crystals. The wood of Salvadoraceae cannot be characterized as paedomorphic.

10 Emblingiaceae. (Fig. 13).

There is a small amount of data provided by Metcalfe based on an *Emblingia* twig in Erdtman et al. (1969). The present account is new information, and is based on a somewhat larger twig. Study of a more mature wood sample of the single species is needed. *Emblingia* is a relatively small shrub native to limited sandy coastal areas of southwestern Australia. The placement of *Emblingia* as sister to core Brassicales was proposed by Hall et al. (2004). Prior to availability of molecular data, *Emblingia* was placed in various families, including Goodeniaceae, Polygalaceae, and Sapindaceae (Erdtman et al., 1969).

Wood plan. Growth rings are evident on the basis of wider vessels in earlywood (Fig. 13a).

Storying. Storying is not present in the tangential section of Fig. 13b, although that is only a three-year old stem.

Vessels. Vessels are circular in transection, and solitary (Fig. 13a). Vessel elements are caudate, and about 240 μm long in the material studied. Lateral wall pits are alternate (Fig. 13e–f), with narrow slit-like pit apertures (Fig. 13e). Pits are not vested (Fig. 13e and f). Inner surfaces of vessels are faintly striate as seen with SEM (Fig. 13e). Perforation plates are simple and bordered (Fig. 13e).

Imperforate tracheary elements. All imperforate tracheary elements are tracheids, with bordered, non-vestured pits (Fig. 12c and d). Tracheids are about 420 μm in length.

Axial parenchyma. Axial parenchyma is scanty vasicentric, with wall thickness similar to that of the tracheids. Axial parenchyma is in strands of two to four cells.

Rays. Rays are uniseriate and biseriate (Fig. 13b). Uniseriate rays look rather similar to axial parenchyma strands in tangential sections. The ray type is Paedomorphic Type I of Carlquist (1988). Ray cells have secondary walls.

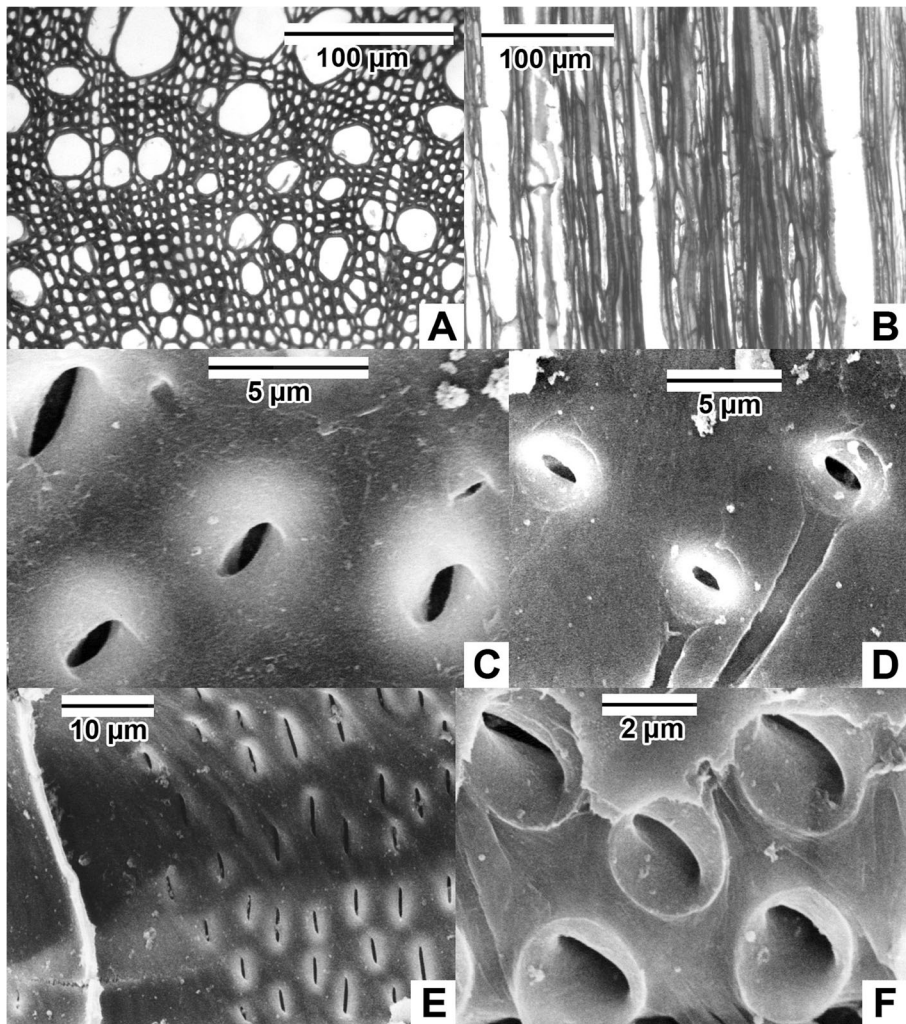


Fig. 13 Wood of *Emblingia calceoliflora* F. Muell. (Emblingiaceae), A. S. George 9756 (PERTH). **a.** Transverse section; vessels are solitary. **b.** Tangential section. Rays are juvenilistic, composed of upright cells. **c–f.** SEM micrographs from radial section portions. **c.** Tracheid inner surface. **d.** Tracheid outer surface. **e.** Inner surface of vessel, non-bordered perforation plate at left. **f.** Outer surface of vessel; pits are non-vestured

Crystals. No crystals were observed.

Comments. *Emblingia* is distinctive in having tracheids, which are associated with solitary vessels, in agreement with the scheme of Carlquist (1984). The wood of the material studied was pedomorphic in ray structure, a condition that very likely persists beyond the third year.

11. Tovariaceae (Carlquist, 1985b) (Fig. 14a–e).

There are two species of the neotropical genus *Tovaria*, the sole genus of the family. They are short-lived shrubs of unstable scree. The only study to date, based on a

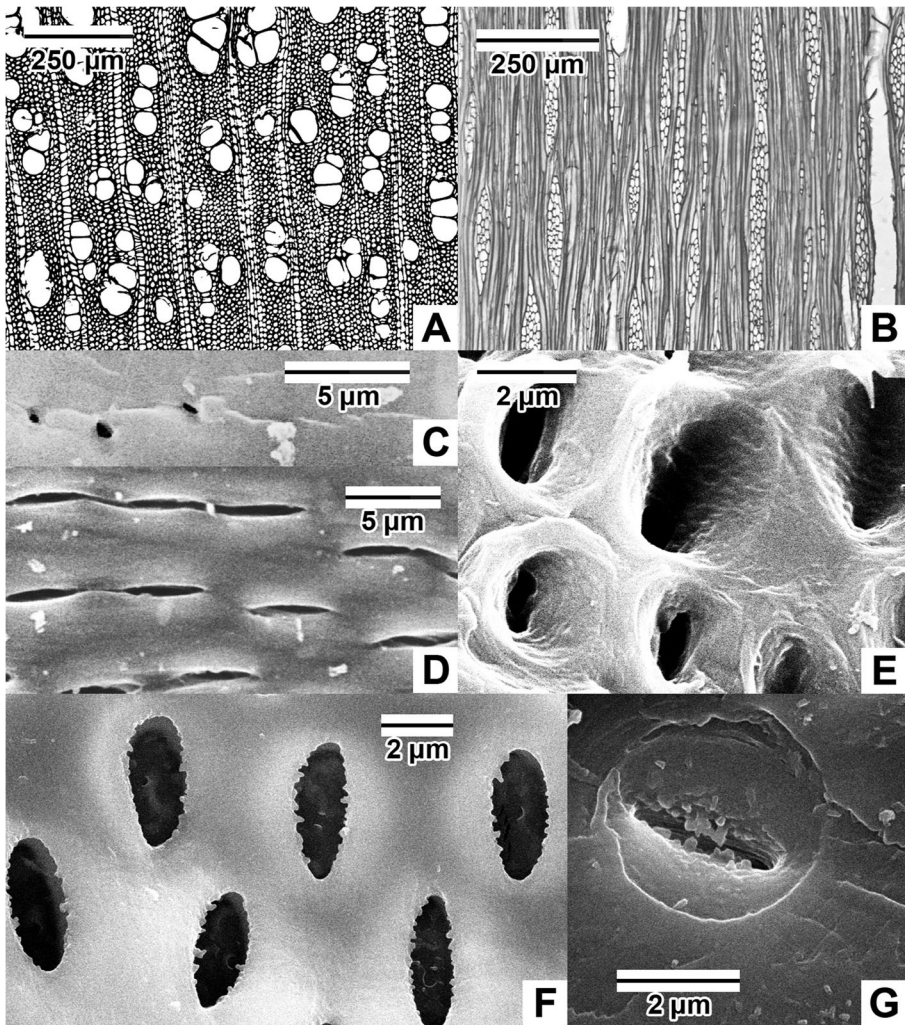


Fig. 14 Wood of Tovariaceae and Pentadiplandraceae. **a–e.** *Tovaria pendula* R. & P., Carlquist 7156 RSA. **a.** Transverse section. Vessels mostly grouped. **b.** Tangential section. Rays Heterogeneous IIB, wood non-storied. **c–e.** SEM micrographs from radial sections. **c.** Outer surface of libriform fiber. **d.** Inner surface of vessel. **e.** Outer surface of vessel, pit cavities textured but not truly vestured. **f–g.** *Pentadiplandra brazzeana* Baill., J. Hall 263 WIS. **f–g.** SEM micrographs of radial section portions. **f.** Vessel seen from inner surface; pit apertures are vestured. **g.** Pit of tracheid, seen from outer tracheid surface; vestures are present in pit cavity

collection of *T. pendula* from montane cloud forests of Peru, represents the mature wood pattern of this species.

Wood plan. The secondary xylem is uniform with no growth ring activity (Fig. 14a).

Storying. No storying is present in the tangential section (Fig. 14b).

Vessels. Vessels are grouped (1.77 vessels per group, Fig. 14a), and are circular in transverse section. Vessels are rather wide for the order (68 µm). Mean vessel element length is 306 µm). The number of vessels per square mm is 2.40. Pit apertures are narrow and slit-like (Fig. 14d). Although some texturing appears in SEM micrographs

of pit cavities (Fig. 14e), this does not seem referable to vesturing, although further studies of other specimens should be undertaken. No helical sculpturing appears on the vessel walls (Fig. 14d).

Imperforate tracheary elements. Pits are very small and circular and apparently non-bordered (Fig. 14c). Libriform fibers are thus present.

Axial parenchyma. The parenchyma type is vasicentric scanty, in strands of two or three cells.

Rays. Both uniseriate and multiseriate rays are present, the latter more common. Ray cells are mostly upright, with a few procumbent cells in central portions of multiseriate rays (Paedomorphic Type I of Carlquist, 1988). Mean height of multiseriate rays is 536 μm ; mean width at widest point is 3.8 cells. Ray cells have thin but lignified walls, like the walls of libriform fibers and vessels.

Crystals. No crystals observed.

Comments. The wood of *Tovaria*, based on stems of maximal diameter that were available, can be termed paedomorphic according to the concepts provided earlier (Carlquist, 1962).

12. Pentadiplandraceae (data original) (Fig. 14f and g).

No description of wood anatomy of *Pentadiplandra* has been provided prior to the present essay. The sole species of the family, *P. brazzeana*, is a scandent shrub native to Cameroons. The data herewith were derived from SEM studies of a small-diameter stem with about 2 mm thickness of secondary growth. This is not sufficient to provide reliable data on most characters, but the data on vestures seem decisive.

Vessels. The vessels are circular in outline and are mostly solitary. Lateral wall pits are alternate and circular. Vestures are present, and best seen from the inside of vessels (Fig. 14f), although they are also visible in pits as seen from the outer surfaces of vessels. The vestures are restricted to the edges of pit apertures.

Imperforate tracheary elements. Tracheids are present, based on two criteria: the prominent borders on pits (Fig. 14g) and the fact that vessels are not grouped (Carlquist, 1984). Vestures are present on the margins of the pit apertures (Fig. 14g). Tracheid wall thickness, about 2.5 μm .

Rays. In the limited material available, all rays were multiseriate, rays three to four cells wide at their widest point. Ray cells were all observed to be upright, which is to be expected in wood of a young stem or branch.

Crystals. Single rhomboidal crystals seen in pith cells and in ray cells close to the pith.

13. Gyrostemonaceae (Carlquist, 1978) (Fig. 15; Fig. 16a–d).

Wood plan. As seen in transverse sections, tangential bands of axial parenchyma in which vessels are embedded are conspicuous (Figs. 15a and d, 16a). These bands are mostly not annual. The boundary between latewood and earlywood is illustrated for *Tersonia brevipes* (Fig. 16a).

Storying. Storying is visible in vessel elements and axial parenchyma, but only to a limited degree in imperforate tracheary elements (Figs. 15b and e, 16b) because of the intrusive nature of the imperforate tracheary elements. Tangential sections of bark show

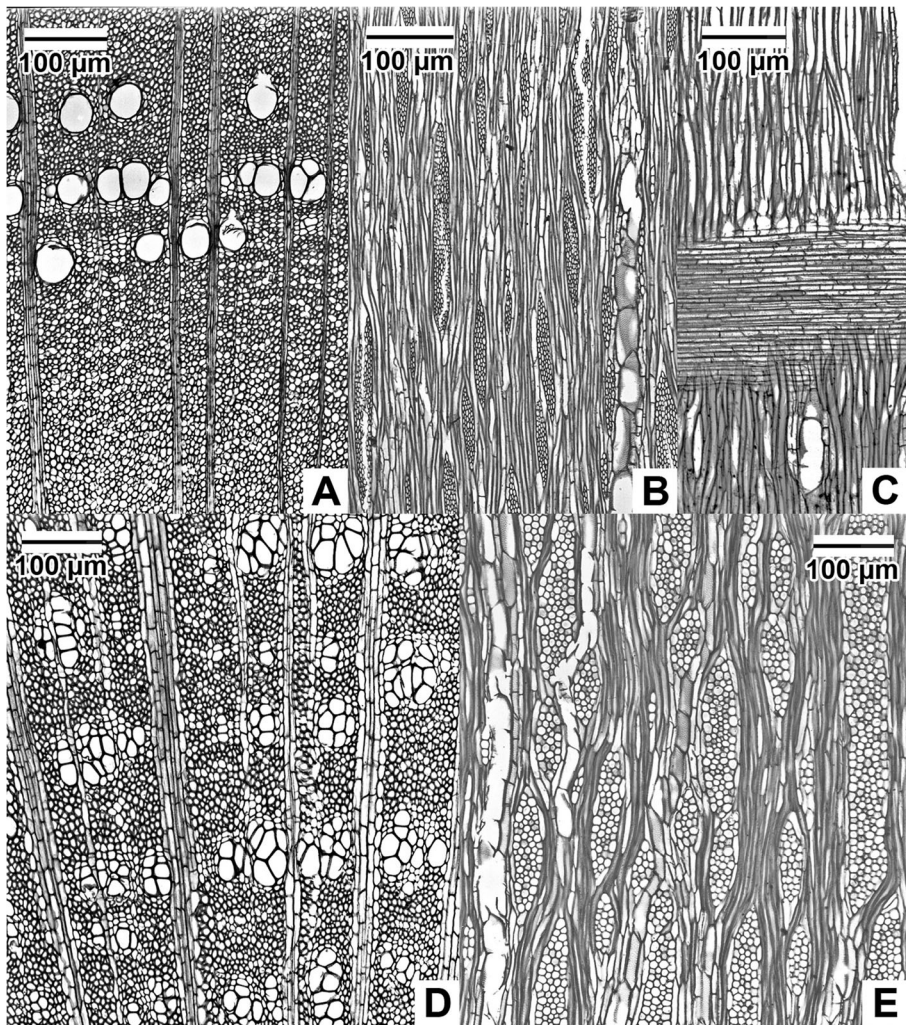


Fig. 15 Wood of Gyrostemonaceae. **a-c.** *Codonocarpus cotinifolia* F. Muell, Carlquist 5150 RSA. **a.** Transverse section. Growth ring prominent. **b.** Tangential section. Wood non-storied. **c.** Radial section. All ray cells procumbent. **d-e.** *Gyrostemon subnudus* (Nees) Baill., Carlquist 5518 RSA. **d.** Transverse section. Vessels are in large multiples, with a tendency for the groupings to be in tangential bands. **e.** Tangential section. Some storied in axial parenchyma (bottom center); rays abundant, Type Heterogeneous IIB

that fusiform cambial initials in the family are storied in samples old enough to show mature wood patterns for the genera.

Vessels. Vessels are circular in outline, less often oval in *Codonocarpus* (Fig. 15a) and *Tersonia* (Fig. 16a), but conspicuously grouped in *Gyrostemon* (Fig. 15d) and *Didymothecca*. Mean vessel diameter ranges from about 50 to 100 µm (main stem, branches, and root included). Number of vessels per sq. mm ranges from 14 (e.g., *Codonocarpus*, Fig. 15a) to almost 100 (*G. subnudus*, Fig. 15d). Mean vessel element length ranges from 148 to 321 µm in the collections of *G. ramulosus* Desf., a range as great as that of the entirety of species investigated (Carlquist, 1978). Perforation plates

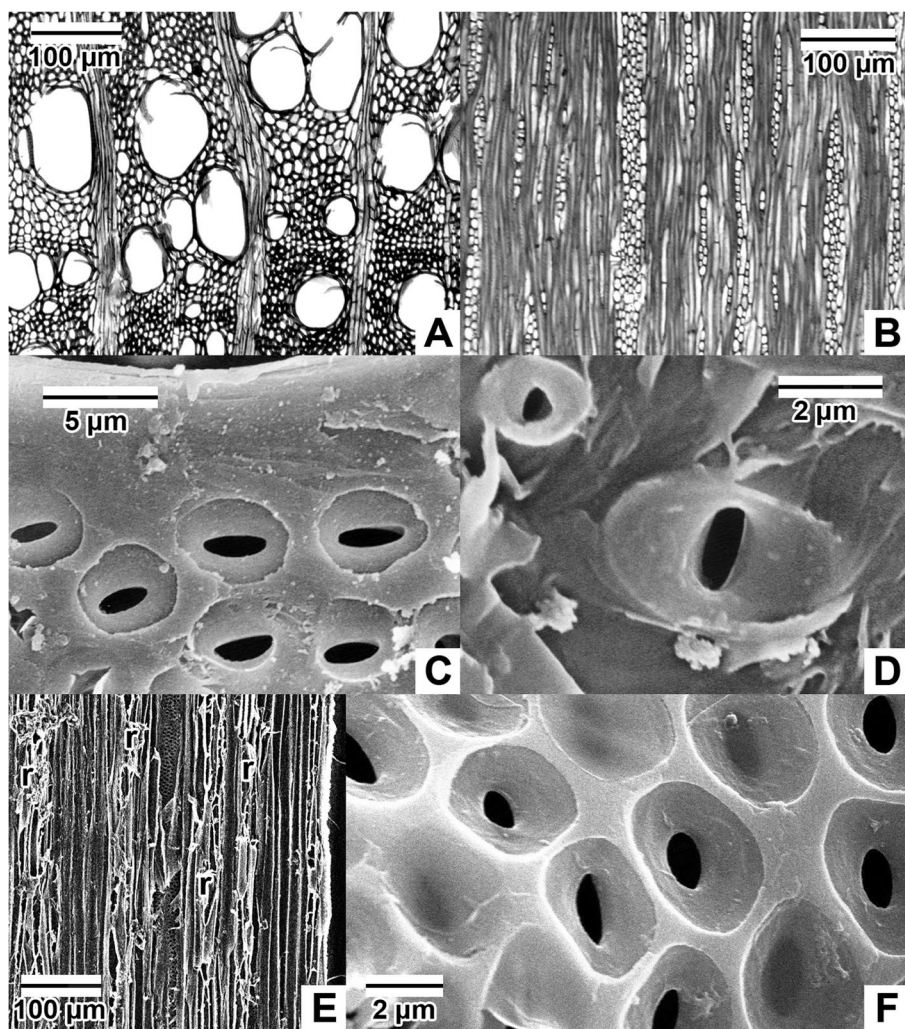


Fig. 16 Wood of Gyrostemonaceae (a–d) and Borthwickiaceae (e–f). a–d. *Tersonia brevipes* Moq., Carlquist 5395. a. Transverse section. Vessels markedly narrower in latewood (bottom) than in earlywood (above). b. Tangential section, rays Heterogeneous Type IIB. c–d. SEM images from radial section. c. *Codonocarpus cotinifolia*, Carlquist 5150 RSA. Outer surface of vessel (perforation plate at top); pits non-vestured. d. *Gyrostemon ramulosus*. Carlquist 3153 RSA. Outer surface of imperforate tracheary element, pits bordered. e–f. *Borthwickia trifoliata* W. W. Sm., H. T. Tsai 60547 A, SEM micrographs. e. Lower power tangential section; rays (r) composed of upright cells. f. Outer surface of vessel element; pits non-vestured

are simple. Lateral wall pits of vessels are circular and alternate and non-vestured (Fig. 16c).

Imperforate tracheary elements. Nature of pitting suggests that there is a range from fiber tracheids to tracheids in these cells, depending on the density of the bordered pits (Fig. 16d). The imperforate tracheary elements are mostly tracheid-like in *Tersonia*, but fiber-tracheids with much-reduced pit borders in *Codonocarpus*. Wall thickness is about 3 μm .

Axial parenchyma. Axial parenchyma is in bands associated with the largest vessels, but elsewhere is diffuse or, more commonly, diffuse-in-aggregates. Axial parenchyma characteristically occurs in strands of two cells, with relatively thin but lignified walls.

Rays. Rays are mostly multiseriate, but a few uniseriates are also seen in all species studied. Procumbent cells are present almost exclusively (Fig. 15c). Upright and square cells are present at tips of rays and occasionally as sheathing cells along the sides of rays (Fig. 15e). Rays are thus Heterogeneous Type II transitional to Homogeneous Type II. Mean multiseriate ray height ranges from 391 μm in *Codonocarpus* to 408 μm in *G. subnudus* (Fig. 15e) to 1538 μm in *Tersonia brevipes* (Fig. 16b).

Crystals. No crystals were observed.

Comments. The relatively tall rays of *Tersonia* suggest greater juvenilism, but other common indicators of protracted juvenilism (abundance of upright ray cells, absence of storying) are not present. Gyrostemonaceae are woodier than one might think on the basis of its lack of study in earlier wood anatomy literature, but *Gyrostemon* and *Codonocarpus* qualify as trees. *Tersonia* is a sprawling subshrub. Succulent leaves and stems are present in the genera other than *Codonocarpus*.

Starch is present in rays (especially in *Didymothea*), but is less common in axial parenchyma.

14. Borthwickiaceae (data original) (Fig. 16e and f).

The single species of *Borthwickia* now constitutes its own family (Su et al., 2012). There is no description of the wood anatomy of this species prior to the present essay. The data herewith were derived from a stem about 4 mm in diameter from a single herbarium specimen, and thus represent incomplete material. Hopefully, a larger-diameter stem will be collected for study so that the nature of secondary xylem can be more accurately described. *Borthwickia* is a shrub or small tree from subtropical regions of China (Su et al., 2012).

Wood plan. There is no evidence of growth rings in the material studied.

Storying. There is no storying in the specimen at hand (Fig. 16e); storying would not be expected in a relatively young stem.

Vessels. Vessels are circular, mostly solitary. Vessels have alternate circular pits. There is no evidence of vesturing in the pits (Fig. 16f).

Imperforate tracheary elements. Pits with vestigial borders were seen (with SEM). These could be termed either libriform fibers or fiber-tracheids, depending on the cell viewed.

Axial parenchyma. Axial parenchyma is vasicentric; strands of two cells were observed.

Rays. Multiseriate rays composed of upright cells are present in the young stem examined (Fig. 16e).

Crystals. No crystals were observed.

Comments. The rays in this specimen are clearly juvenile, with only upright ray cells. This may well change as a stem increases in diameter, so more material must be examined before we can say whether or not such juvenilism is present in larger stems as well.

15. Resedaceae (Carlquist, 1998b; Schweingruber, 2006) (Fig. 17).

The six genera of Resedaceae are small, with only *Reseda* containing more than 10 species (Bolte, 1936). The few Resedaceae that qualify as woody (*Caylusea*, *Ochradenus*) do not have the clearly woody texture that shrubs in Ericaceae or Rhamnaceae do. One could regard these Resedaceae as herbs the longevity of which exceeds two or three years.

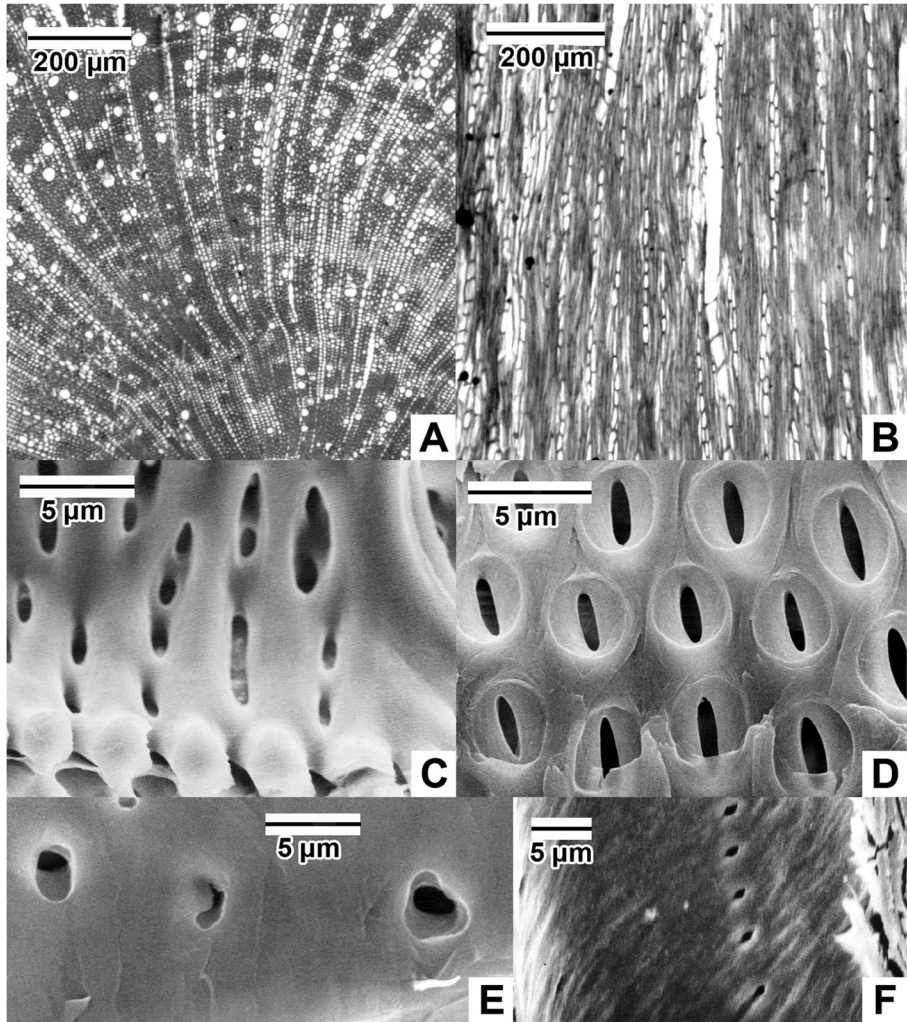


Fig. 17 Wood of Resedaceae. **a–b.** *Caylusea hexagyna* (Forssk.) M. L. Green, Podlech 42683 RSA. **a.** Transverse section; axial parenchyma in banded patterns. **b.** Tangential section; rays composed of upright cells. **c–f.** SEM images from radial sections. **c.** *Oligomeria linifolia* J. F. Macbr. ME Jones 24096 POM. Inner surface of vessels, showing grooves interconnecting pit apertures. **d–e.** *Reseda alba* L. E. Kennedy April 2, 1923 POM. **d.** Outer surface of vessel; pits non-vestured. **e.** Outers surface of fiber-tracheid. Pit borders are narrow. **f.** *Reseda luteola* L., Gander 2415 RSA. Inner surface of vessel; wall striate

Wood plan. Growth rings are present or are lacking, depending on seasonality of growth in particular species (Fig. 17a).

Storying. Wood non-storied in all species studied (e.g., Fig. 17b). This may be correlated with the relative juvenility of resedaceous woods.

Vessels. Vessels are circular in outline. Mean number of vessels per group ranges from 1.8 (*Reseda crystalline* Webb & Berth.) to 3.1, and averages 2.3 for the family as a whole. Solitary vessels are about as common as groups of two or three vessels (Fig. 17/a). Vessels are narrow, ranging from 15 μm mean diameter in *Caylusea* (Fig. 17a) to 31 μm (*R. luteola*), and 25 μm for the family as a whole. Mean number of vessels per sq. mm. ranges from 33 to 301; the lower vessel density is in *R. alba* and reflects the wood mesomorphy in this cultivated species. Vessel element length for the family has a range from 95 to 197 μm (145 μm for the family as a whole).

Perforation plates are all simple. Vessel pitting is alternate (Fig. 17c and d). Pits are not vested in my material (Fig. 17d), although Schweingruber (2006) thought there might be some vesturing in *R. suffruticosa* Loefl. On the basis of light microscopy. Grooves that widen some pit apertures horizontally and interconnect two or three pit apertures (Fig. 17c) are present in some species. Fine striations were seen with SEM on vessel walls of *R. luteola* (Fig. 17f).

Imperforate tracheary elements. These cells have been reported to be libriform fibers, but SEM study reveals small borders (Fig. 17e), so fiber-tracheids may be present at least in some species.

Axial parenchyma. Vascentric scanty axial parenchyma is present. In *Caylusea*, however, vascentric axial parenchyma is scarce, but banded and confluent patterns occur (Fig. 17a). Marginal (probably terminal) axial parenchyma occurs in *R. alba*, *R. crystallina*, and *R. lutea* L.

Rays. Rays are (narrow) multiseriate and uniseriate in about equal numbers (Fig. 17b). Ray cells are predominantly upright as seen in radial sections,

Crystals. No crystals have been observed in wood of Resedaceae, although they are present in some other plant portions (Bolle, 1936).

Comments. In comparison to Gyrostemonaceae, Resedaceae are relatively non-succulent. In the perennial or shrubby species, wood xeromorphy is evident in the narrowness of vessels and in the short vessel element length. Resedaceae have more paedomorphic character state expressions than do Gyrostemonaceae.

16. Stixaceae (Carlquist et al., 2013) (Figs. 18 and 19).

The family Stixaceae Doweld is tentatively recognized here because as shown by Su et al. (2012) and Hall et al. (2004), the family Capparaceae becomes monophyletic only with the removal of *Forchhammeria*, *Stixis*, and other genera once included in it by such authors as Pax and Hoffmann (1936). We need wood data from *Neothorelia*, *Tirania*, and more species of *Stixis*, and such data are likely to yield more diversity in this family, judging from some preliminary work (Carlquist, unpublished data). The grouping in the family Stixaceae of three Asiatic genera together with the neotropical *Forchhammeria* may seem phytogeographically less than plausible, but equally long disjunctions separate the two genera of Akaniaceae; instances of marked disjunction in other families could also be cited.

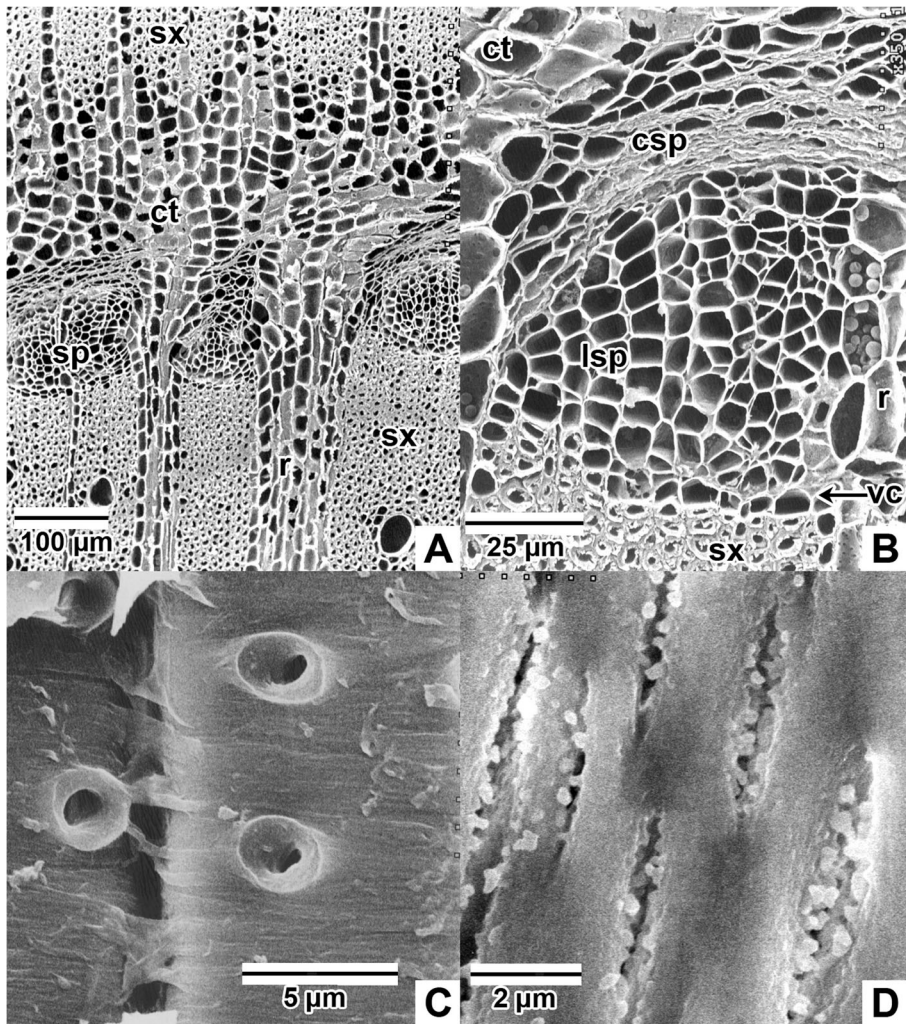


Fig. 18 Wood anatomy of *Forchhammeria* (Stixaceae), SEM micrographs. **a–c.** *F. watsonii* Rose Desert Botanical Garden (Phoenix), liquid preserved material. **a.** Portions of two vascular increments, showing conjunctive tissue (ct), rays (r), secondary phloem (sp), and secondary xylem (sx). **b.** Portion of a vascular increment, centered on secondary phloem. Labels indicate crushed secondary phloem (csp), living secondary phloem (lsp), conjunctive tissue (ct), ray containing starch (r) and secondary xylem (sx). **c.** Tracheid outer surface, showing pit borders. **d.** *F. trifoliata* Radlk. ex Millsp., Usw-29650. Inner surface of vessel; pit apertures are very wide and abundantly vested

Wood plan. Successive cambia (Fig. 18a) are present in all species of *Stixis* and *Forchhammeria* for which material is available and has been studied. The plan of successive cambia follows that seen in other angiosperms with successive cambia (Carlquist, 2007), with vascular increments separated by conjunctive tissue.

Storying. Storying is not present. Although larger stems of *Forchhammeria* and one species of *Stixis* were available, secondary xylem consists of bands each generated by a vascular cambium (which in turn is produced by a master cambium). Thus, each vascular increment can be regarded as a short segment of ontogenetic change, with

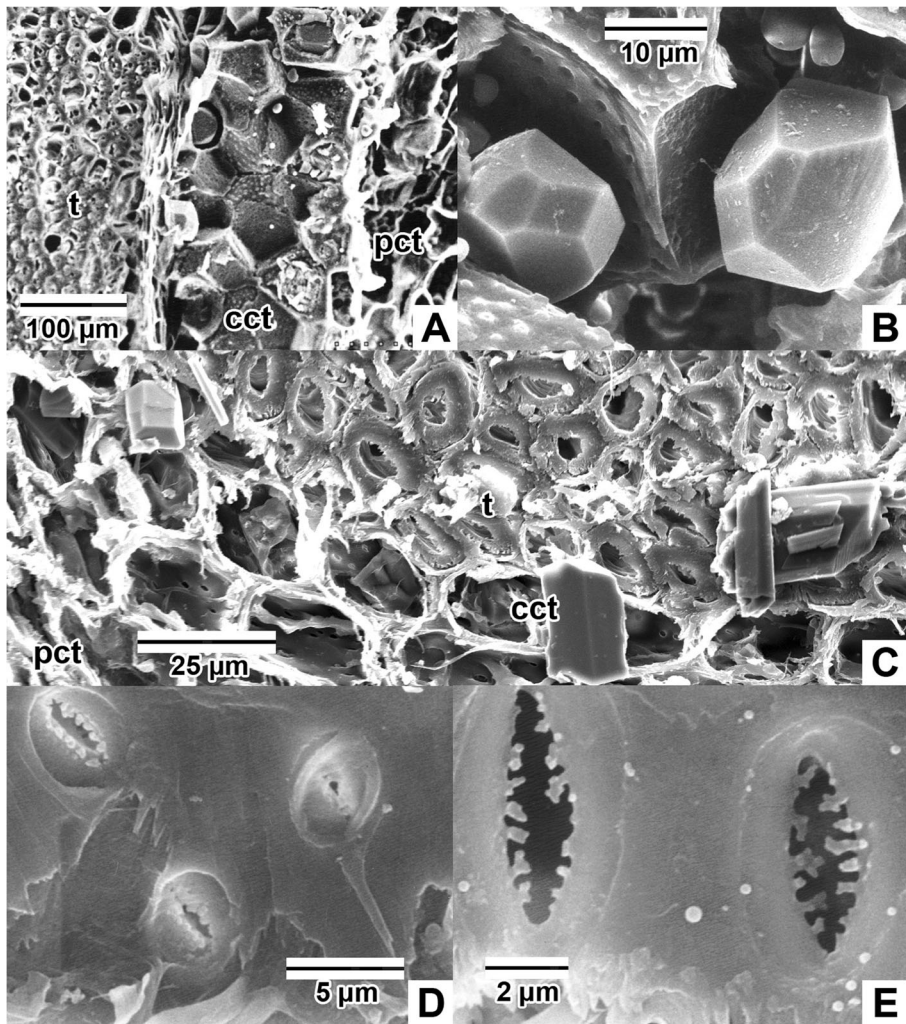


Fig. 19 Wood of Stixaceae. **a–b.** *Forchhammeria pallida* Liebm., Olson 873 UNAM. **a.** Portion of transection; abaxial (outer) face at right. **a.** Portion of transection, showing tangential band of crystal-containing sclereids in the conjunctive tissue (cct) and parenchymatous conjunctive tissue (pct) to right; tracheids of secondary xylem (t) at left. **b.** Two sclereids from the band of crystal-containing sclereids, cut open and exposing polyhedral crystals. **c–e.** *Stixis parviflora* (Griff.) Pierre, USW-28950. **c.** Transverse section of a portion of secondary xylem (t = tracheids), below which is a single discrete layer of crystal-containing conjunctive tissue (cct) and the parenchymatous conjunctive tissue (pct); some crystals dislodged from cells in the cct layer. **d.** Outer surface of tracheid, showing three vestured pits. **e.** Outer surface of vessel; two pits with vestured margins of pit apertures

more extensive ontogenetic change (needed for storing to be achieved) observable in angiosperm species with indefinite production of a single wood cylinder by a single vascular cambium.

Vessels. Vessels are solitary in species of *Forchhammeria* studied to date. Grouped vessels are advantageous in angiosperms at large unless (in accordance with the

concept (Carlquist 1984) vessels are associated with tracheids. This phenomenon demonstrates that tracheid presence is more effective than vessel grouping in promoting conductive safety. It also incidentally shows that solitary vessels would be advantageous in any wood if selection for xeromorphy is not an issue. Figures for vessel density in *Forchhammeria* are provided by Carlquist et al. (2013). Some vessel grouping was observed in *S. philippinensis* Mert., and this may related to the fact that vessels can be embedded not in a mass of tracheids, but in some places, in axial parenchyma. Vessel density is lower (16 per square mm) than for angiosperms as a whole, in accordance with the figures for successive cambial species by Carlquist (1975).

Pit apertures as seen on the inner surfaces of vessel walls are slit-like (Figs. 18d and 19e), and pit apertures adjacent in a helical direction may be coalescent. Vesturing is maximally present in vessels of *Forchhammeria trifoliata* (Fig. 18d), *Stixis parviflora* (Fig. 19e), and *S. philippinensis* (unpublished data). Vestigial vesturing was observed in other species of *Forchhammeria*. Pits cavities are circular and pits are alternate. Perforation plates are all simple.

Imperforate tracheary elements. Tracheids with conspicuous bordered pits occur in *Forchhammeria* (Fig. 18c) and *Stixis* (Fig. 19d). Pits may be vested (Fig. 19d) or non-vested (Fig. 18c) in *Forchhammeria*, but vestures in *Stixis parviflora* and *S. philippinensis*. The length of tracheids in *Forchhammeria* as a whole (1279 μm) is greater than the corresponding vessel element length (206 μm), suggesting that there is considerable intrusive growth.

Axial parenchyma. Axial parenchyma in *Forchhammeria* is diffuse, with tendencies toward vasicentric in *F. pallida* and *F. watsonii* (Fig. 18a) and diffuse-in-aggregates in *F. trifoliata*. Axial parenchyma is sparse in *Stixis parviflora*, but abundant, both paratracheal and in bands, in *S. philippinensis*. Axial parenchyma cells in *S. parviflora* have oval pits unlike the small slit-like pits one associates with libriform fibers. Strands of axial parenchyma are two to five cells in *Forchhammeria*.

Rays. Rays are predominantly multiseriate, with occasional uniseriates (*F. watsonii*) or extremely few (*F. trifoliata*, *Stixis pauciflora*). Rays are composed mostly of procumbent cells. Thus, they are transitional between Heterogeneous II and Homogeneous II of Kribs (1935). Rays are thus similar to those in Gyrostemonaceae.

Crystals. There are no reports of crystals in secondary xylem per se.

Conjunctive tissue. Conjunctive tissue may consist wholly of parenchyma, as in *F. watsonii* (Fig. 18a) or *Stixis* (Fig. 18c, bottom), or it may contain a band of sclereids, as in *F. pallida* (Fig. 19a). The sclereids of *F. pallida* contain a large polyhedral crystal each (Fig. 19b). A band of cells, some walls of which are thicker, suggestive of incomplete sclereids, occur in the last-formed conjunctive tissue in *Stixis pauciflora* (Fig. 19c). The crystals in this layer of cells are varied in shape and size.

Secondary phloem. Each vascular cambium in a species with successive cambia is capable of producing more secondary phloem (lsp, Fig. 18b) as well as more secondary xylem (sx) overtime. Bands of crushed secondary phloem (Fig. 18b, csp) are evidence of continued secondary phloem production.

Starch. Starch grains are common in phloem rays (Fig. 18b, middle right).

Comments. The inherent taxonomic interest of so many distinctive features in which *Stixis* resembles *Forchhammeria* seems clear: successive cambia, ray histology, tracheid presence, vested pits, and crystal-bearing cells in conjunctive tissue.

The vessel diameter, vessel density, and vessel element length of *Forchhammeria* and *Stixis* would not suggest wood xeromorphy. However, various botanists have noticed that in Mexico, shrubs of *Forchhammeria* have green leaves during the dry months when other vegetation in the dry scrub habitats of *Forchhammeria* show drought deciduousness in foliage. The only wood feature that would explain this is presence of tracheids and vestured pits in secondary xylem. Tracheids are present in such large numbers that one can hypothesize that they play a role not merely in preservation of water columns, but in active conduction also. Occasional portions of secondary xylem in *Forchhammeria* are free from vessels.

17. Capparaceae (Figs. 20, 21, 22 and 23).

No survey of wood anatomy of Capparaceae exists. This is understandable in view of the wide geographical distribution and varied degrees of woodiness of the family. There are not even any generic monographs of wood anatomy within the family. Descriptions of wood of assemblages of species based on floristics exist (e.g., Cozzo, 1946; Stern et al., 1963). Gregory (1994) lists 39 such references, to which can be added Fahn et al. (1986). The accounts of Solereder (1908) and Metcalfe & Chalk (1950) therefore are by default, the best overall sources of information on wood anatomy. SEM images in the present essay are all original (as are the light microscope photomicrographs).

Capparaceae are recognized here in the sense of Hall et al. (2002), excluding Cleomaceae and Brassicaceae. With that definition, each of these families becomes monophyletic. The earlier vision of the family (Pax & Hoffmann, 1936) was more inclusive, and could not be maintained with the advent of molecular data. For the history of the taxonomy and phylogeny, the account of Hall et al. (2002) is admirable and need not be repeated here.

The revised Capparaceae includes some true trees (e.g., *Boscia*, *Cadaba*) as well as shrubs branched from the base (e.g., *Capparis*). Even though Capparaceae s. s. is a woody group, there are no wood features that clearly separate the family from Cleomaceae and Brassicaceae.

Wood plan. Wood consists of an ordinary woody cylinder, with growth rings minimal in the majority of species (Figs. 20a–d and 21a and e). However, some degree of successive cambial activity has been reported in *Boscia* (Adamson, 1936), *Cadaba*, and *Maerua* (Metcalfe & Chalk, 1950).

Storying. Storying is minimally present, perhaps because intrusive growth of imperforate tracheary elements is prominent. Some examples of storying may be seen in Figs. 20b and 21b.

Vessels. Vessels are in radial multiples or chains (Figs. 20a and 21a and e). In some species, vessels are dimorphic in diameter, markedly in *Apophyllum anomalum* (Fig. 21e and f), and less clearly so in *Cladostemon*, *Maerua*, *Niebuhreria*, *Quadrella* (Fig. 20d), *Ritchiea*, and *Stuebelia*. Mean number of vessels per group ranges upward from about 1.5 (*Capparis*, *Cladostemon*, *Quadrella*, Fig. 21e) but mostly lies below 2.5.

Perforation plates are simple. Lateral wall pits of vessels is circular and alternate. Pits appear to be non-vestured in *Maerua* (Fig. 22d), and minimally vestured in *Capparis spinosa* (Fig. 20f). Vestures that are prominent but confined to pit apertures were

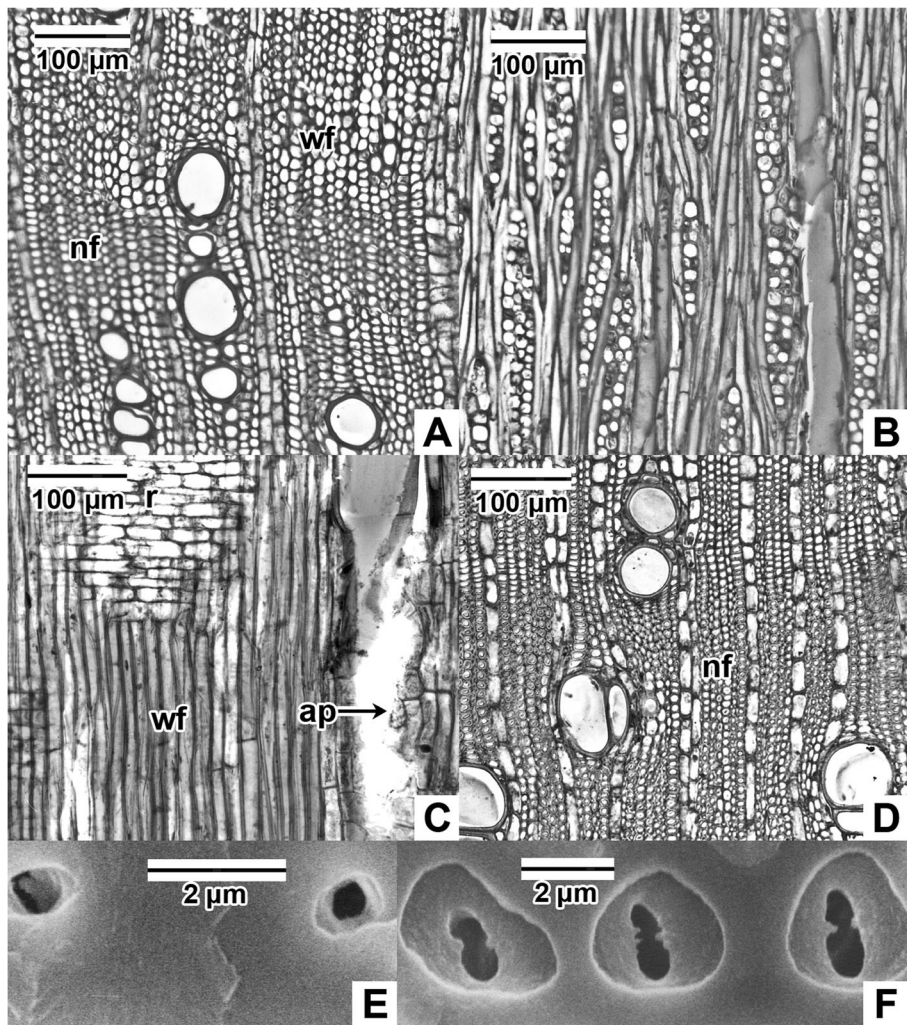


Fig. 20 Wood of Capparaceae, **a–c.** *Capparis flexuosa* L. Stem & Brizicky 236 Y. **a.** Transverse section; fiber dimorphism evident in narrow fibers (nf) compared to wide fibers (wf). **b.** Tangential section. Rays are Homogeneous Type I. **c.** Radial section. Axial parenchyma adjacent to vessel indicated by arrow; r = ray; wf = wider fibers. **d.** *Quadrella cynophallophora* (L.) Hitchc., Stem & Brizicky 262 Y. Transverse section. This area contains only narrow fibers, although elsewhere in the section, wide fibers are present. **e–f.** SEM micrograph of tracheary elements. **e.** fiber-tracheid with narrow pit borders. **f.** Three pits from vesicle; vestigial vestures on pit aperture

observed in *Atamisquea* (Fig. 22f) and *Apophyllum* (Fig. 22c). *Boscia hildebrandtii* has vestures limited to the central portion of the pit cavity; these can be seen from both the inside (Fig. 22e) and on the outer surface of vessels (Fig. 22f). More abundant vestures can be seen on vessels of *Crataeva* (Fig. 23a and b), *Quadrella* (Fig. 23c) and *Steriphoma* (Fig. 23d). *Crataeva* has very distinctive vestures as viewed from the insides of vessels (Fig. 23f). The vestures extend from one widened pit aperture to another in a kind of network; there is merging of the abundant vestures.

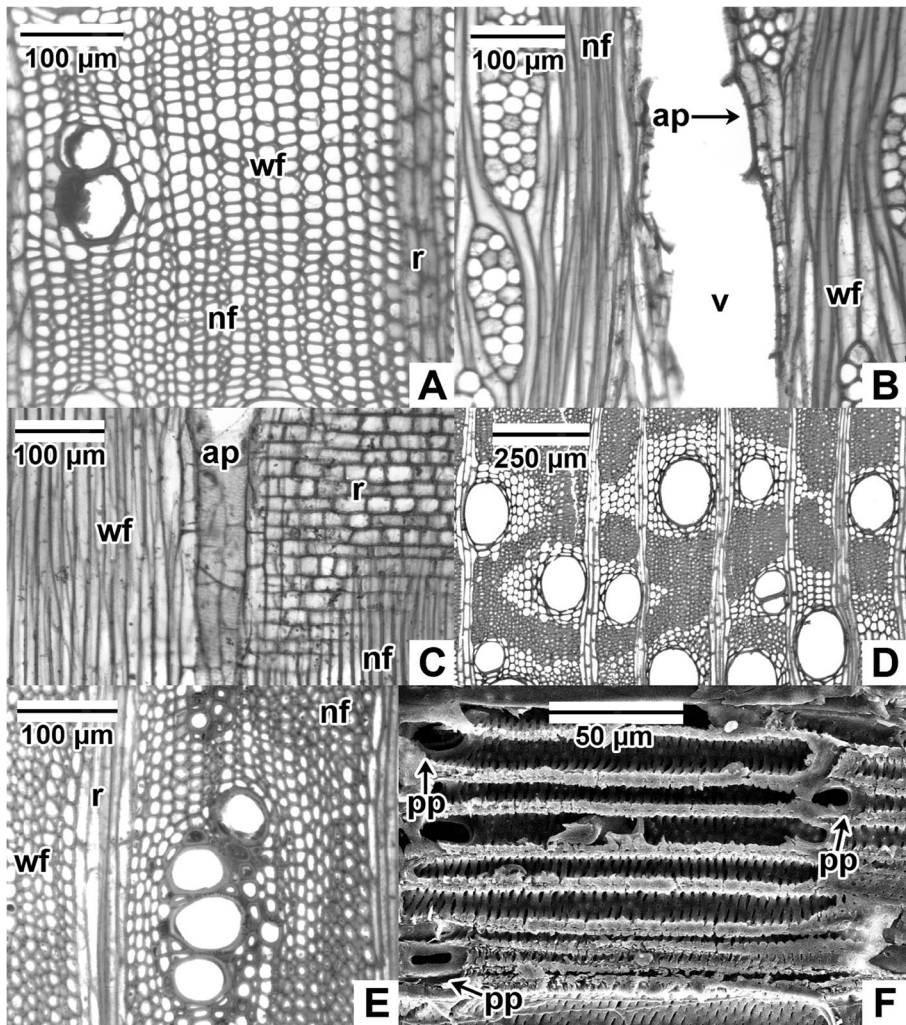


Fig. 21 Wood of Capparaceae. **a–c.** *Cladostemon kirkii* Pax & Gilg, Y-33950. **a.** Transverse section. Fiber dimorphism evident by virtue of narrow fibers (nf) and wide fibers (wf) r = ray. **b.** Tangential section; rays Heterogeneous Type IIB transitional to Homogeneous Type II; ap = axial parenchyma, nf = narrow fibers, v = vessel, wf = wide fibers. **c.** Radial section; axial parenchyma strands (ap) behind vessel; nf = narrow fibers, r = ray, wf = wide fibers. **d.** *Niebuhria linearis* DC, Imperial Forestry Institute, Oxford, 174. Transverse section showing aliform-confluent parenchyma around solitary vessels. **e–f.** *Apophyllum anomalum* F, Muell., Y-15906. **e.** Transverse section to show fiber dimorphism (nf = narrow fibers; wf = wide fibers), ray (r), and vessel dimorphism (narrow vessels are in dense dark patches). **f.** Radial section (SEM micrograph), showing a group of very narrow vessels (pp = perforation plates)

Prominent grooves interconnecting pit apertures were observed in narrower vessels of *Apophyllum* (Fig. 21f). Helical thickenings occur in vessels of *Atamisquea* (Fig. 22f).

Imperforate tracheary elements. Libriform fibers or fiber-tracheids with minimal borders (Fig. 20d) are present. A number of Capparaceae exhibit fiber dimorphism, a

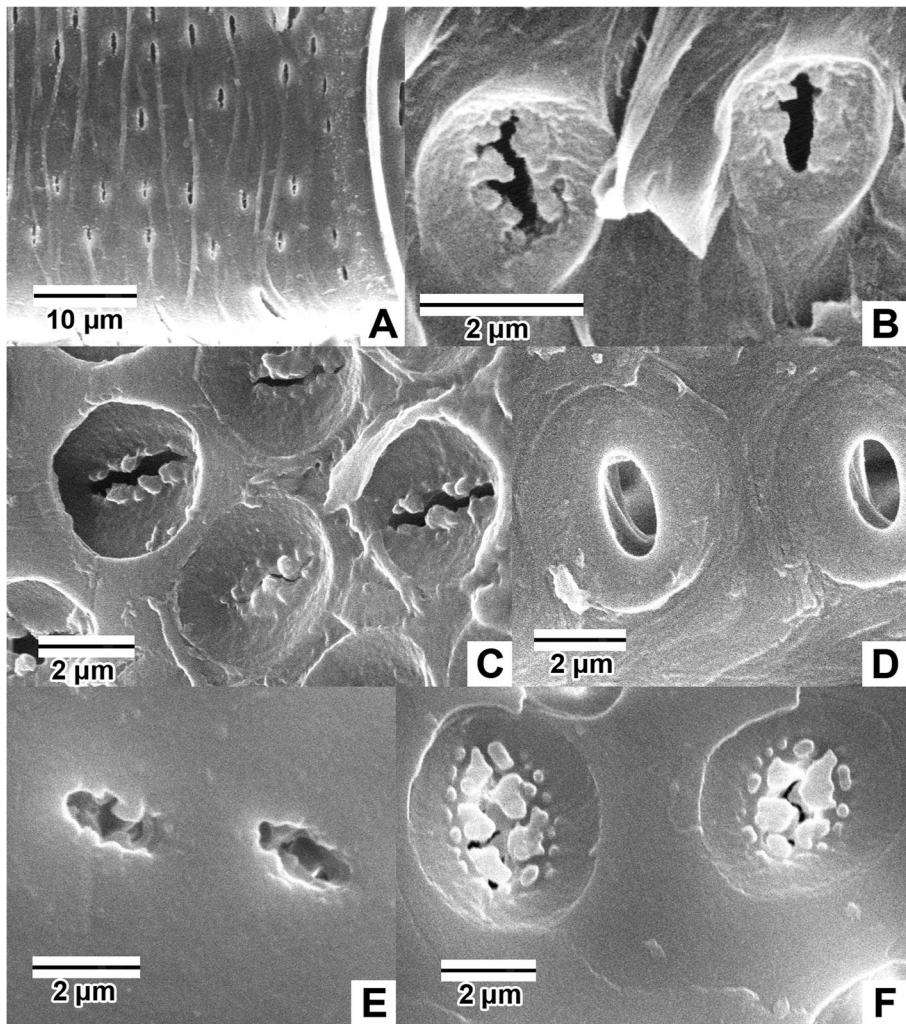


Fig. 22 Wood of Capparaceae. SEM micrographs of vessel details from radial sections. **a–b.** *Atamisquea emarginata*, Kw-628. **a.** Inner surface of vessel, showing fine helical thickenings. **b.** Two vestured pits from outside surface of vessel. **c.** *Apophyllum anomalum*, Forestry Commission of New South Wales SFCw-R658-8. Outer surface of vessel, several vestured pits. **d.** *Maerua angolensis* DC., USW-37704. Non-vestured pits seen from outer surface of vessel. **e–f.** *Boscia hildebrandtii* Gilg, USW-21795. **e.** Two vestured pits, inside surface of vessel. **f.** Two vestured pits, outer surface of vessel

phenomenon observed in Asteraceae (Carlquist 1958, 1961) and in a range of other angiosperm woods (Carlquist, 2014). This feature, which denotes alternating patches of wide (presumably living) fibers and narrow fibers, can be seen to advantage in *Capparis flexuosa* (Fig. 20a–c), *Quadrella cynophallophora* (Fig. 20d), *Cladostemon* (Fig. 21a–c), and *Apophyllum* (Fig. 21e). No fiber dimorphism was observed in *Maerua*, *Morisonia*, *Niebuhreria*, *Ritchiea*, and *Stuebelia*. Fibers are notably thin-walled and wide in *Ritchiea*.

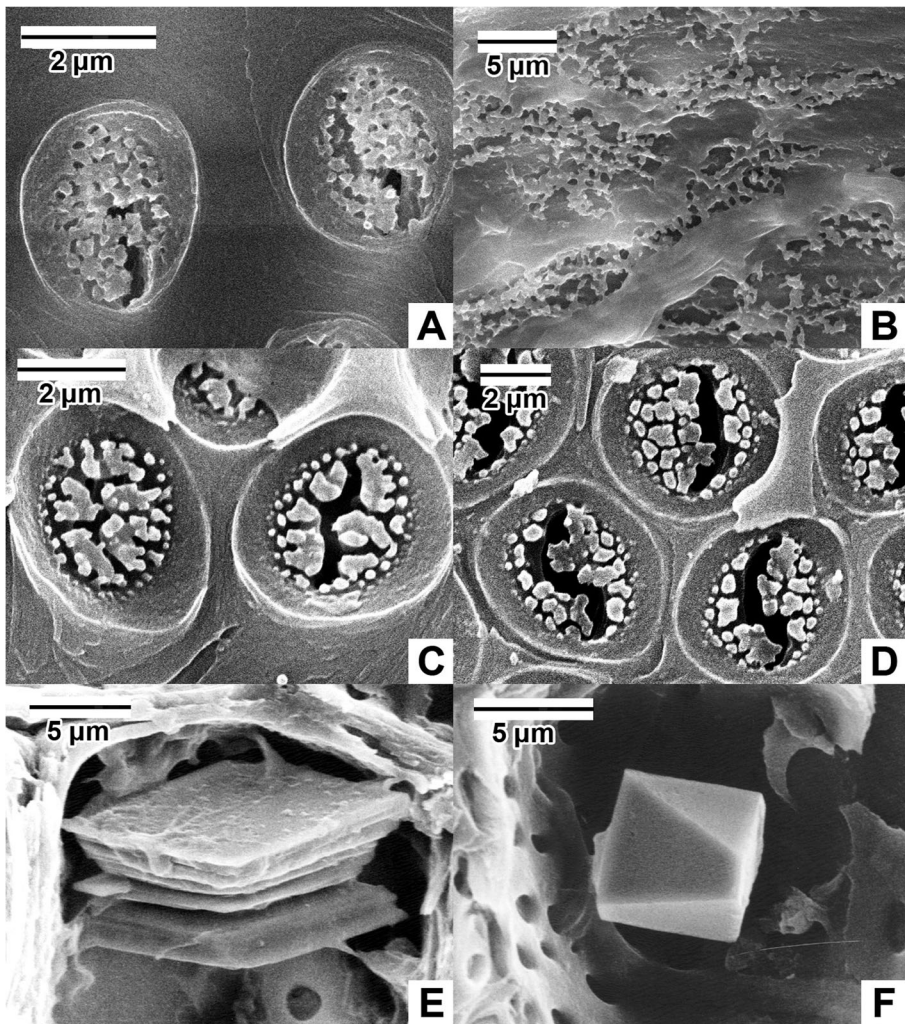


Fig. 23 Wood of Capparaceae. SEM micrographs. **a–b.** *Crataeva tapia* L., USw-9952. **a.** Two vestured pits from outside of vessel. **b.** Inner surface of vessel. Vestures extend beyond pit apertures and form a coalescing network. **c.** *Capparis scabrada* Kunth, Ittis & Ittis40058 WIS. Vestured pits seen from outside of vessel. **d.** *Steriphoma peruviana* Spruce ex Eichler, USw-10326, several vestured pits as seen on outer surface of a vessel. **e–f.** Crystals in ray cells. **e.** *Atamisquea emarginata*, Kw-628, lamellate rhomboidal crystal. **f.** *Cadaba farinosa* Forssk., MADw-43549, octahedral crystal

Axial parenchyma. Axial parenchyma is mostly vasicentric scanty, which is also the predominant type in Brassicales. More abundant paratracheal parenchyma (about two layers thick around vessels) can be seen in *Quadrella cynophallophora* (Fig. 20d). Aliform-confluent parenchyma characterizes *Niebuhreria linearis* (Fig. 21d). Axial parenchyma can be easily distinguished from instances of fiber dimorphism, and the two co-occur in species with fiber dimorphism. Axial parenchyma strands are composed of two to five cells (Fig. 21b).

Rays. Both multiseriate and uniseriate rays, corresponding to Heterogeneous Type IIB of Kribs (1935), occur in *Atamisquea*, *Maerua*, and *Niebuhrria*. Multiseriate plus uniseriate rays composed of procumbent cells only, the Homogeneous Type II of Kribs, occur in *Capparis* (Fig. 20b) and *Quadrella*. Rays that are multiseriate and consist wholly of procumbent cells, the Homogeneous Type II of Kribs, were observed in *Capparis flexuosa*, *Cladostemon kirkii* (Fig. 21b), and *Stuebelia*.

Crystals. Rhomboidal crystals were observed in ray cells in *Atamisquea* (Fig. 23e), *Cadaba* (Fig. 23f), *Capparis*, *Morisonia*, *Niebuhrria*, *Quadrella*, and *Steriphoma*. The crystals of *Atamisquea* are noteworthy for their lamellate structure. The crystals in *Cadaba* are apparently octahedral.

Comments. Wood features of Capparaceae are quite diverse, and include most of the character states found in other Brassicales, excluding presence of tracheids and interxylary phloem. Successive cambia, although reported in a few genera, are not characteristic of all species of those genera, may be infrequent, and need further investigation. Fiber dimorphism certainly occurs in several genera of Capparaceae characteristically, and has not been reported in any other family of Brassicales with the exception of Cleomaceae. Vestured pits are certainly well developed in the majority of Capparaceae and may represent the prime component of wood xeromorphy in the family. Vessel dimorphism, prominently present in *Apophyllum*, also occurs in Tropaeolaceae, and may represent another type of wood xeromorphy. There is a notable absence of juvenilistic features in wood of Capparaceae, suggesting that despite the abundance of herbaceous Cleomaceae and Brassicaceae, the three crown group families of Brassicales may have a woody ancestry.

18. Cleomaceae (data original and from Metcalfe & Chalk, 1950) (Fig. 24).

Because of the recent separation of Cleomaceae from Capparaceae (Iltis et al., 2011), there is no thoroughgoing account of wood of the family. Descriptions of wood of individual genera may be found in Solereder (1908) and Metcalfe & Chalk (1950) under Capparaceae.

Wood plan. Stems with a single cambium except for *Cleome droserifolia* Delile, which has successive cambia, illustrated by Fahn et al. (1986). The conjunctive tissue in that species contains a band of sclerenchyma. The remaining Cleomaceae studied have relatively wide rays (2—5 cells) separating fascicular areas with one or two radial strips of vessels (Fig 24a). The vessels are separated from rays by several layers of fibers, creating a “vessel restriction pattern” (Carlquist, 2009b).

Storying. Libriform fibers as well as vessels are storied in *Peritoma* (= *Isomeris*), which is a shrub with sufficient secondary xylem accumulation to show storying. Most species of Cleomaceae have less wood than *Peritoma*, hence, storying is not present in them.

Vessels. Vessels are in groups ranging from a mean of 1.4 vessels per group in *Cleome anomala* Kunth to 4.0 in *Peritoma arborea* (Fig. 24a). Vessels are not dimorphic in size, and are rather narrow (Fig. 24a and d) in them species studied.

Perforation plates absent. Vessel-to-vessel pits alternate and circular (Fig. 24c). Pits are prominently vestured (Fig. 24b). A wall interface between two vessels (Fig. 24c) show vestures on the wall seen from the lumen side (left), but vesture

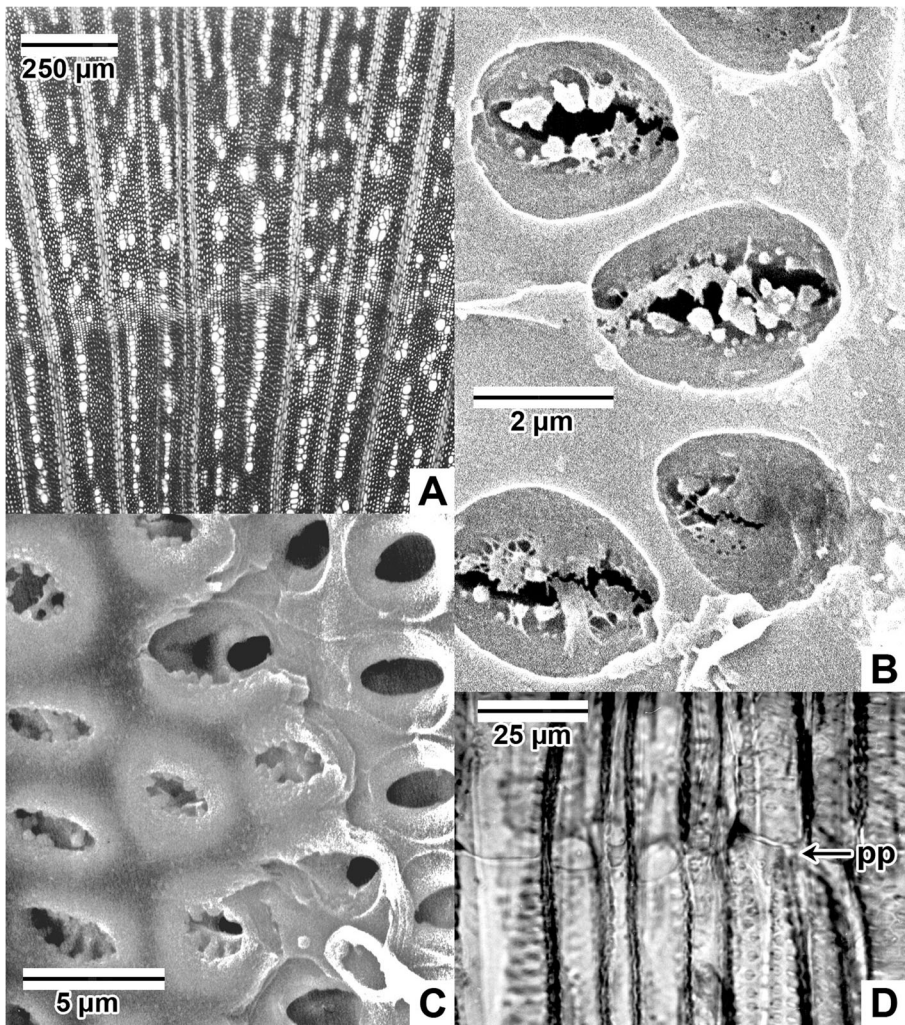


Fig. 24 Wood of *Peritoma arborea* (Nutt. ex Torr. & A. Gray) Iltis (*Isomeris arborea* of earlier authors). Cultivated at Santa Barbara Botanic Garden. **a.** Transverse section. Vessels are very narrow, in radial multiples and radial chains, and are separated from ray by fibers. **b–c.** SEM micrographs of vessel wall. **b.** Four vestured pits seen from outer surface of vessel. Two pits at bottom retain portions of pit membranes. **c.** Interface between two vessels, seen from inside a vessel; pits on vessel inner surface are vestured, whereas those on the outer surface of the adjacent vessel, at right, are not vestured. **d.** Radial section, showing a file of very narrow vessels (pp = perforation plates)

absence on the facing wall (right. This discrepancy may be due to handling, but it illustrates that not all vessel walls are alike when one searches to see whether vesturing is present.

Imperforate tracheary elements. Monomorphic libriform fibers are present in most species. Dimorphic fibers were observed in *Peritoma* (Fig. 24a, center).

Axial parenchyma. Vasicentric scanty, in strands of two cells, or undivided in *Peritoma*.

Rays. Rays are Heterogeneous Type IIB of Kribs (1935). Stems of *Cleomella oocarpa* A. Gray are rayless in earlier-formed secondary xylem (original observation).

Crystals. No crystals were observed in species studied.

Starch. Starch is abundant in the rays of *Peritoma arborea*.

Comments. The vessel restriction patterns of Cleomaceae are distinctive, and we have only preliminary explanations for this type of structure (Carlquist, 2009a) or several other non-random types of vessel distributions in woods. The high degree of vessel grouping in *Peritoma* is indicative of wood xeromorphy. Presence of dimorphic fibers in *P. arborea* is perhaps a link to Capparaceae and Brassicaceae. Although not all genera of these three families have dimorphic fibers, they are the only families in Brassicales in which they occur.

19. Brassicaceae (Metcalfe and Chalk, 1950; Carlquist, 1971; Schweingruber, 2006) (Figs. 26, 27 and 28).

References that describe wood of one or only a few species are listed by Gregory (1994). As the largest family (3710 species) of Brassicales, Brassicaceae is also minimally known with respect to wood anatomy. We do not even have thorough descriptions of ontogenetic stages and mature structure of secondary xylem of the cultivated species. To no small degree, this is related to the bias against less woody species where studies of secondary xylem is concerned. Absence of secondary xylem in herbaceous eudicots is, in fact, rare, and secondary xylem in herbaceous eudicots is worthy of investigation.

Wood plan. Most species have a single cambium; growth rings are inconspicuous (Carlquist, 1971). Latewood (as well as adjacent earlywood) bands of axial parenchyma that contain vessels are shown here for *Stanleya pinnata* (Fig. 25a, b and d). Such bands, which do not appear to be modifications of the basic axial parenchyma types (e.g., Kribs, 1937) also occur in *Alyssum spinosum* L., *Brassica fruticulosa* Cyrillo, and *Vella spinosa* Boiss. (Metcalfe and Chalk, 1950). The entire background tissue may consist of storage parenchyma, as in roots of *Raphanus sativus* L. or thick-walled parenchyma may surround vessels, with thinner-walled parenchyma more distal, as in *Armoracia* (Fig. 27b), or an entirely fibrous background consisting of libriform fibers may be present, as in most of the woody Brassicaceae studied by Carlquist (1971). Lateral meristem activity, which can add some vascular tissue as well as parenchyma, is evident in *Armoracia lapathifolia* (Fig. 27a–c) roots (“rhizomes” of some authors). Interxylary phloem strands are present in *Hirschfeldia incana* (= *Brassica nigra* and *B. aspera* of various authors), as shown in Fig. 26c. Such variations are probably more widespread in Brassicaceae, and would provide interesting material for further examination. The annual habit may be regarded as formation of a single growth ring. (Fig. 26a and b) and often features diminution in vessel diameter as secondary growth ceases. The 3710 estimated species of Brassicaceae are mostly annuals and thus this pattern is probably widely represented.

Storying. Various degrees of storying from inconspicuous (Fig. 25c) to prominent (*Crambe strigosa* L’Her.), but storying may be entirely lacking in many annual species.

Vessels. Compared to other families of Brassicales, Brassicaceae have relatively narrow vessels, with diameter means from 16 to 71 μm , with most species ranging from 30 to 50 μm in the species assemblage studied by Carlquist (1971). The widest vessels

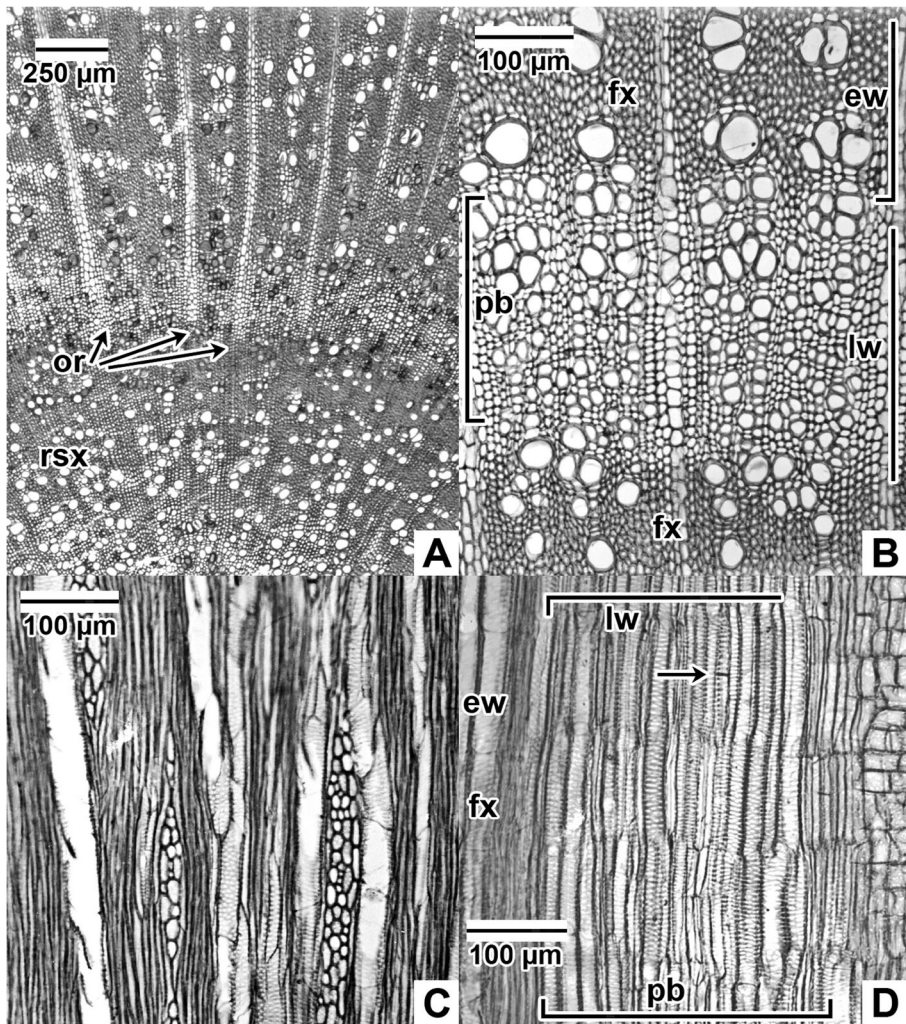


Fig. 25 Wood of *Stanleya pinnata* (Pursh) Britton (Brassicaceae), cultivated at Rancho Santa Ana Botanic Garden. **a–b.** Transections. **a.** Innermost secondary xylem, pith at bottom. The first-formed wood is rayless, followed by origin of rays as indicated (or, arrows). **b.** Parenchymatous band (pb) that includes vessels of various diameter, but latewood (lw) earlier along and some earlywood (ew); fx = fibrous xylem. **c.** Tangential section, rays Heterogeneous Type IIB. Storying evident in libriform fibers, upper left. **d.** Radial section of parenchymatous band (pb) and adjacent earlywood (ew) with fibrous xylem (fx). Latewood (lw) has extremely narrow vessels. Section corresponds to the zone covered in Fig. 25b

in that survey occurred in a laurel forest shrub from Madeira, *Cheiranthus mutabilis* L'Her. (54 µm, species mean); this species also has the longest vessel elements recorded for the family thus far, 100 µm. Number of vessels per group is highest in *Stanleya pinnata* (Fig. 25a and b) and lowest in *Cheiranthus mutabilis*. Vessel grouping most commonly takes the form of radial multiples or radial chains, or a mixture of those with non-radial multiples (Fig. 25a and b; Fig. 26a and b).

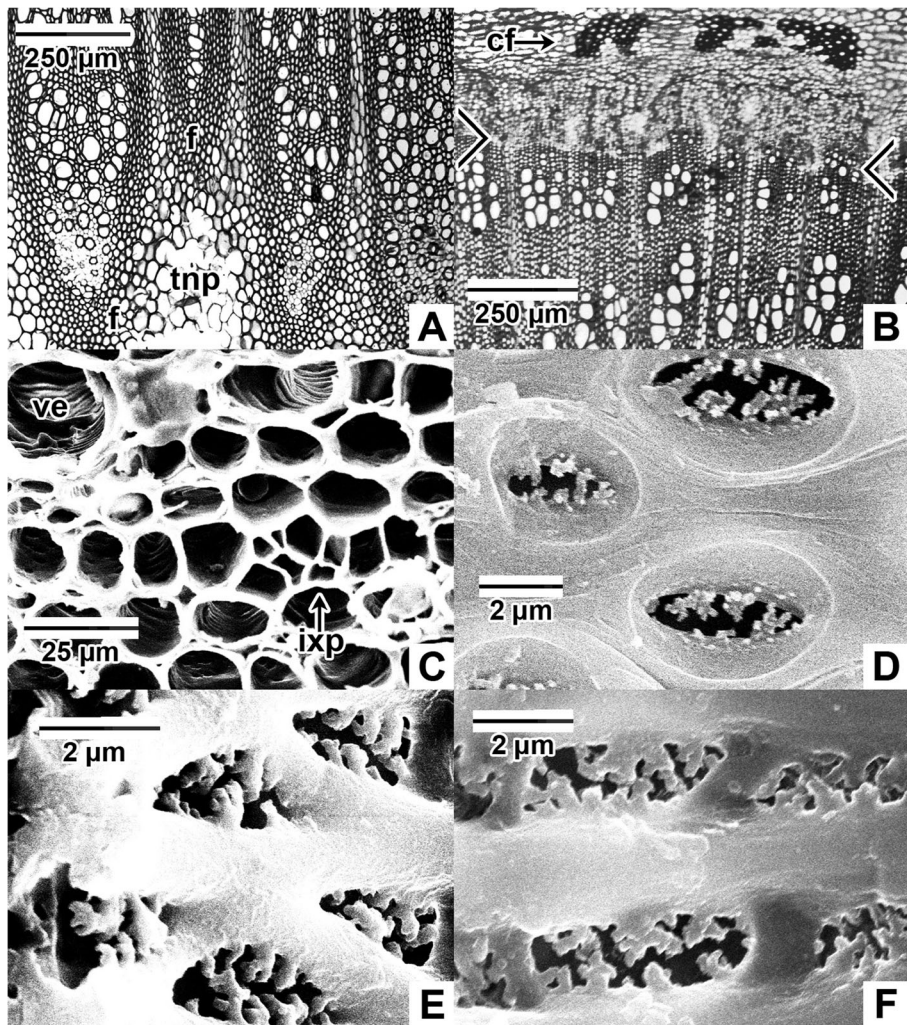


Fig. 26 Wood of *Hirschfeldia incana* (L.) Lagr.-Foss. (*Brassica nigra* (L) W. O. J. Koch and *B. strigosa* DC. of various authors), (Brassicaceae). **a–b.** Transections of secondary xylem. **a.** Section near pith; fascicular xylem sheathed in fibers; thin-walled pith (tnp) at bottom. **b.** Section including cambium (pointer); secondary phloem and cortical fibers (cf) at top. **c–f.** SEM micrographs. **c.** Transverse section, showing vessel (v) in a background of thin-walled fibers, and a strand of interxylary phloem (ixp). **d–f.** Views of vested pits from radial sections. **d.** View of pits from outer surface of vessel. **e–f.** Inner surface of vessels. **e.** Appearance of branched bases to vestures. **f.** pits in which vestures show some coalescence; bands of secondary wall subdivide two of the pits

Perforation plates are simple. Lateral wall pitting of vessels consists mainly of alternate circular bordered pits. The pit apertures facing the lumen may be interconnected by grooves (Figs. 26f and 27d) or such grooving may be absent, as in *Matthiola*. Degrees of prominence of the grooving as seen with SEM are shown in Fig. 26e, f and Fig. 27d. Pits are vested (Figs. 26d–f, 27d–f and 28a–f). Vestures are evident if vessels are viewed from outer surfaces (Figs. 26d, 27c and 28b–f) or inner surfaces (Figs. 26e–f, 27d, f and 28a). The appearances in these two views are different. As

viewed from the lumen side of a vessel wall, the vestures appear as branching coralloid outgrowths (Figs. 27f and 28a), which may appear to be in contact with the pit membrane—probably an artifact produced by drying. As seen from the outer surface, the vestures are seen as tips of the outgrowths (compare Figs. 27e with 27f; Fig. 28a with 28b). Thus far, there is no report of any Brassicaceae in which vested pits are absent from vessels.

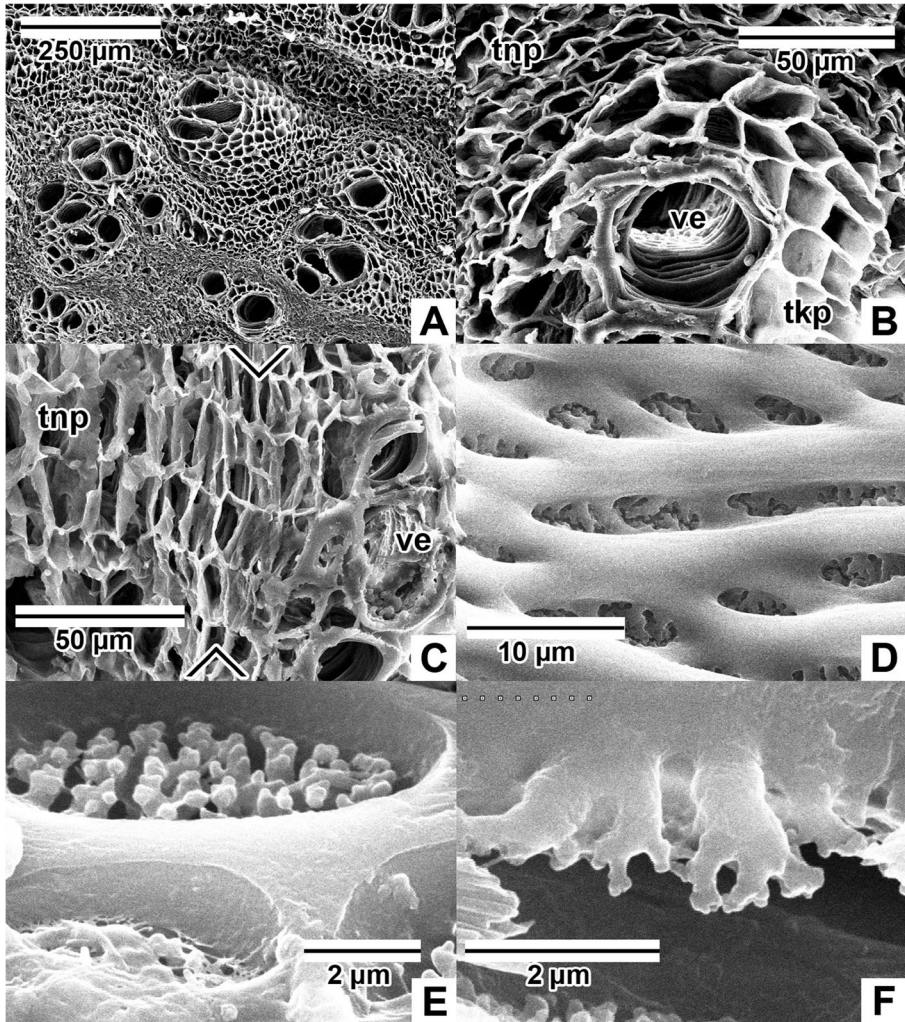


Fig. 27 Wood from *Armoracia lapathifolia* Gilib. (Brassicaceae) root; market source (horseradish). **a–c.** Transverse section. **a.** Irregular vascular increments in a parenchymatous background. **b.** Portion of vascular strand: ve = vessel element; thick walled parenchyma (tkp) is adjacent to vessel, thin-walled parenchyma (tnp) is distal. **c.** A site of cambial activity (pointers); a vessel (ve) has been produced to the right, and thin-walled parenchyma (tnp) to the left. **d–f.** Views of vested pits from radial section, SEM micrographs. **d.** View from inner surface of vessel, grooves interconnect pit apertures, bridges of secondary wall material subdivide some pits. **e.** Tips of vestures in a pit seen obliquely, on outer surface of a vessel. **f.** Portion of a pit aperture as seen from inner side of vessel, showing coralloid branching of vestures

Vestured pits have hitherto been figured only in secondary xylem, and are shown mostly in vessels, although tracheids may also have vesturing (e.g., Bataceae, Koeberliniaceae and Stixaceae). However, my material of *Raphanus raphanistrum* proved favorable for demonstrating vesturing in primary xylem. Vestures can occur on the helical thickenings of primary xylem of both stems (Fig. 28d) and roots (Fig. 28e–f). In vessel elements that are transitional between helical and reticulate, vestures may be restricted to or more common in the angles where adjacent helices fuse

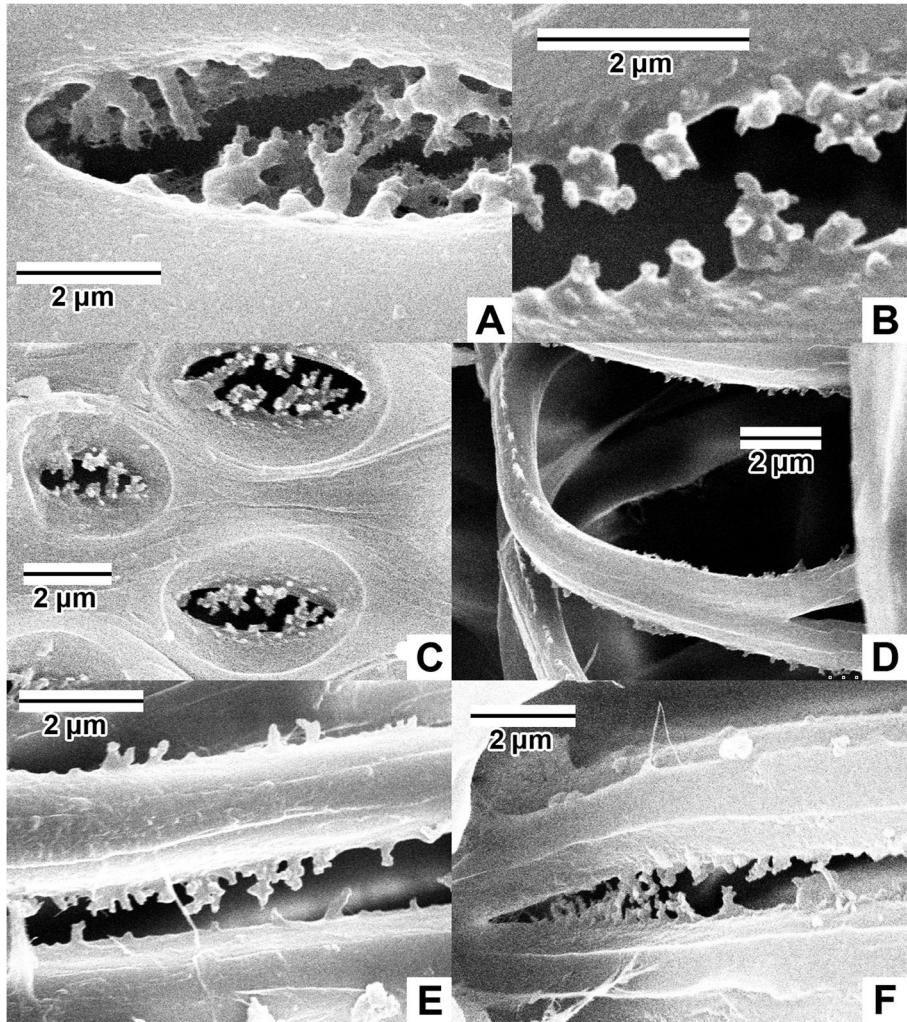


Fig. 28 Wood of *Raphanus raphanistrum* L. (Brassicaceae), adventive near Santa Barbara, CA. **a–c.** SEM micrographs of vestured pits from radial sections of root. **a.** Vestures in a pit seen from the inside of a vessel; coraloid vestures are in contact with pit membrane. **b.** Tips of branching vestures, seen from outside of a vessel. **c.** Three pits, seen from outer surface of a vessel. **d–f.** Vestures on protoxylem helices from radial section of stem base. **d.** Small vestures along helices, lower power. **e.** Vestures more abundant where helices are close. **f.** Vestures more abundant where two helices intersect

(Fig. 28f). Vestured pits of four species of Brassicaceae were illustrated by Carlquist & Miller (1999),

Imperforate tracheary elements. Metcalfe & Chalk (1950) indicate that pits of “fibers” of wood of Brassicaceae may have simple or minutely bordered pits, and thus in the terminology used here, be libriform fibers or fiber-tracheids, respectively. In a survey of fiber dimorphism (Carlquist, 2014), Brassicales were not included, but on the basis of the present study, fiber dimorphism occurs in some Capparaceae as well as some Brassicaceae. Fiber dimorphism is present in all of the Brassicaceae studied by Carlquist (1971) except for *Lepidium fremontii* S. Waston and *Parolinia ornata*. Is axial parenchyma different from the wide, thinner-walled fibers where fiber dimorphism is present? In Brassicaceae, axial parenchyma is scanty vasicentric, and only exceptionally are wide seasonal bands (e.g., Fig. 25b) present. Fiber dimorphism, on the contrary, tends to occur in irregular patches, as illustrated here for Capparaceae. In addition, axial parenchyma in Brassicaceae is often subdivided into strands of two cells (arrow in Fig. 25d), whereas subdivided wide fibers are unusual, a distinction evident in radial sections. In some species, the two cell categories appear to merge, as they do not in the instances described in Carlquist (2014), in which clear instances of fiber dimorphism were selected for illustration.

Axial parenchyma. In addition to the vasicentric scanty parenchyma mentioned in the preceding paragraph, paatracheal sheaths more than one cell in thickness have been observed in *Cheiranthus*, *Crambe*, *Descurainia*, *Lepidium*, *Sinapidendron*, and *Stanleya*. In the roots of *Armoracia lapathifolia*, vessels are sheathed in thick-walled axial parenchyma, whereas thin-walled axial parenchyma forms the background tissue of axial xylem (Fig. 27a–c).

Some Brassicaceae have parenchyma bands interpolated into fibrous secondary xylem, notably *Stanleya pinnata* (Fig. 25a and b). These bands occur as latewood events, but are much wider than the one or two layers common in terminal parenchyma, and can include some vessels from other portions of a growth ring. Schweingruber (2006) offers some additional examples: *Arabis alpina* L., *A. ciliata* Clairv., and *Ptilotrichium spinosum* Boiss.

Rays. Rays of Brassicaceae can mostly be classified as Heterogeneous Type IIB of Kribs (1935). Upright cells, however, are more common in rays of such species as *Cheiranthus mutabilis* and *Stanleya pinnata* (Fig. 25c), suggesting that in a number of species, ray histology is transitional to Paedomorphic Type I (Carlquist, 1988); this indicates some degree of protracted juvenilism in ray structure. In *Lepidium serra* H. Mann the rays are all multiseriate with upright cells very common, so that Paedomorphic Type II is present. In *Parolinia ornata* Webb, rays are Homogeneous Type I of Kribs (1935). Ray cells are notably thick walled in *Descurainia millefolia* and *Parolinia ornata*, and could be termed sclereids. Raylessness occurs in *Stanleya pinnata* (Fig. 25a), in which rayless secondary xylem is the first secondary xylem produced; rays originate soon. Raylessness is likely to be found in other Brassicaceae with habits similar to that of *Stanleya*.

Crystals. Small rhombic crystals are present in the libriform fibers of *Parolinia ornata* (Carlquist, 1971; Schweingruber, 2006). Rhomboidal crystals were reported for *Descurainia briquetii*.

Comments. Although wood anatomy of Brassicaceae is very similar to that of some Capparaceae and Cleomaceae, we as yet know relatively little about Brassicaceae in

comparison to the size of the family. Although a number of Capparaceae are clearly woody, woodiness is limited in Brassicaceae: the shrubby *Parolinia ornata* and woodier cultivars of *Brassica oleracea* L are among the woodiest representatives of the family. Although other less woody representatives of the family may appear less pertinent to study for those grounded in timber species, wood anatomy of less woody species very likely has much to offer.

Character Change and Adaptation in Wood of Brassicales

This section is designed to show how Brassicales exemplify ecophysiological principles in their wood character states. These concepts, however, should apply to woody angiosperms at large, differing most notably where ecology and growth form differ modally from those of the various brassicalean families. For each category of structure, an interpretive hypothesis is given, and evidence from histology of wood of Brassicales is cited in a series of statements that follow the hypothesis.

1. Imperforate tracheary elements. Bailey & Tupper (1918) offered a much-reproduced drawing schematizing the tracheid as a primitive type of tracheary element in angiosperms; from this, vessels (first with scalariform, ultimately simple perforation plates) were progressively derived on the one hand, whereas on the other hand, imperforate tracheary elements progressively lost pit borders and changed (by implication, gradually) from conductive to mechanical cells. Some data by Metcalfe & Chalk (1950, p. xlv) seem to support this as a generalization, but is it always true? Can, in fact, this “trend” run in the other direction in some instances, or even abruptly change from tracheid to libriform fiber with few if any intermediate stages? Brassicales are an ideal group for demonstrating phyletic events because we have good molecular data showing the probable clades of the order, and we have a number of instances of shift from one type of imperforate tracheary element to another.

Hypothesis: Although in angiosperms at large the tracheid may have been the ancestral imperforate tracheary element type, the phylogeny of Brassicales (Fig. 1, column ITE at right) suggests five instances in which tracheids have been evolved from fiber-tracheids (Tropaeolaceae, Koeberliniaceae, Emblingiaceae, Pentadiplandaceae, and Stixaceae). Almost all vascular plants have the genetic information for form bordered pits, as vessels demonstrate, so transferring this information to imperforate tracheary elements can result in formation of tracheids in a clade that ancestrally has fiber-tracheids. In a similar way, simple pits on imperforate tracheary elements (= libriform fibers) have developed in some clades (Salvadoraceae, Tovariaceae, Borthwickiaceae, Resedaceae, Capparaceae, Cleomaceae, Brassicaceae); angiosperms have the genetic information to form fibers with simple pits (e.g. phloem fibers), so this genetic information can be applied to formation of imperforate tracheary elements.

Evidence for the above can be found in the following:

- a. The early-departing branches of the Brassicales clade lack tracheids (in Tropaeolaceae, the tracheids are vasicentric tracheids, probably derived by vessel dimorphism). In Setchellanthaceae, Koeberliniaceae, Emblingiaceae, and

- Stixaceae, evolution of tracheids correlates with highly arid habitats. This evolution of “secondary tracheids” or “neotracheids” can be noted elsewhere in angiosperms, as in Krameriaceae, *Fabiana* of Solanaceae, *Rosmarinus* of Lamiaceae, etc. (Carlquist, 1985a, 1992a, 1992b, 2005; Carlquist and Hoekman, 1985).
- b. Tracheids confer safety to a conductive system because under conditions of negative pressure in water columns, the pit membranes prevent entry of air, whereas in vessels, and if a tracheid does embolize, emboli do not spread into other tracheids as they do in vessel elements, in which the simple perforation plates allow the cavitation to spread into much of a vessel (Zimmermann, 1983).
 - c. The families most likely to be ancestral to Brassicales (Fig. 1) are Malvales and Sapindales (Soltis et al., 2011), which most commonly have fiber-tracheids as an imperforate tracheary element type.
 - d. Libriform fibers in Brassicales may be considered, in agreement with the Bailey and Tupper (19i8) scheme, to be an imperforate tracheary element type in which borders have been lost from pits on fibriform cells. If libriform fibers are dead at maturity, they may be presumed to have a mechanical function rather than a storage function.
 - e. Libriform fibers may be living fibers (fibers with extended longevity); septate fibers may be considered the most frequently encountered type of living fiber. Living fibers (Akaniaceae, Moringaceae) are believed to be involved in storage of water and photosynthates, but may play, at the same time, a mechanical role (Wolking, 1969).
 - f. The parenchyma background of secondary xylem of Caricaceae, like roots in some Moringaceae, can be regarded as libriform fibers that have transitioned out of a mechanical function into a storage function entirely. The mechanical tissue of Caricaceae consists of abundant secondary phloem fibers, by way of compensation.

2. Fiber dimorphism has been often overlooked, but must be included in any understanding of wood physiology. Brassicales and related orders offer occurrences that are likely to offer such understanding.

Hypothesis: Fiber dimorphism is a mechanism for production of wide, thinner-walled libriform fibers together with narrower, thicker-walled libriform fibers, the two types co-occurring in patches that are not discretely defined as are parenchyma bands, and are usually not subdivided into strands (Carlquist 1958, 1961, 2014). This dimorphism offers enough differentiation between mechanical and storage cells to qualify the two types of resultant cells as functionally different. The occurrences of fiber dimorphism in Brassicales and related groups offer the following correlations:

- a. In species with fiber dimorphism, there is not a proportionate decrease in axial parenchyma volume, so the wide living fibers offer a net addition of storage tissue.
- b. As shown by Sauter (1966), the storage in wide fibers of *Acer* (formerly Aceraceae, now Sapindaceae) is in the form of starch which can be rapidly hydrolyzed into sugar that increases the sugar content of sap in vessels. This mechanism has not been studied in other species, but undoubtedly is operative, judging by the occurrence of wide fibers similar to those of *Acer*.

- c. Fiber dimorphism can be identified in some species of each of the three families Capparaceae, Cleomaceae, and Brassicaceae, which form the “crown group” of Brassicales. Similar distributions occur in Sapindales (Carlquist 2015a, 2015b). In all of these instances, fiber dimorphism is probably an apomorphy.

3. Vestured pits. Vestured pits occur in lateral wall pits of vessels of about half of the families of Brassicales that have been investigated with SEM (Fig. 1, column VES at right). Because all of the Brassicaceae investigated in this respect have proved to have vestured pits, the proportion of Brassicales with vestured pits may be much larger at the species level. Sampling the secondary xylem of thousands of species of Brassicaceae is not practicable, but sampling species selected according to ecological categories and systematic groupings would probably give good indicators of the presence of vestured pits in the family at large.

Hypothesis: Vestured pits offer a mechanism for prevention of, or repair of embolisms in vessels, or possibly both. Thus, vestured pits would be valuable in habitats where embolism risk is high: areas that are seasonally very dry, or where freezing of soil (which makes moisture unavailable to the plant). Potentially, vestured pits, vesturing on vessel walls, and similar structures (warts, which may not be separable from vesturing) are the most effective form of xeromorphy, but they may be complicated structures to evolve and therefore are not widespread. A very plausible functional explanation, that they are hydrophilic, has not been offered yet, but may prove to be operative.

- a. The work of McCully et al. (2014) shows that in *Zea*, walls of vessels are mostly hydrophilic, but with hydrophobic patches. These authors find that pit borders are hydrophobic, but turn hydrophilic when water touches them. We do not have any data yet on whether or not vesturing is hydrophilic or hydrophobic, because no species with vestured pits have been studied by the methods used by McCully et al. (2014). The results of such studies are likely to help us refine our ideas about the function of vesturing. Work on surface topography effects of vessel walls (Kohonen, 2006; Kohonen & Helland 2009) are very likely relevant in this respect. Angle of structures related to pits has been mentioned (e.g., Jansen et al., 2003), but this does not explain instances of vestured pits that occur as mere irregularities along the margins of pit apertures (e.g., *Batis*, *Setchellanthus*). The ideas offered to date on the mechanisms for functioning of vestures are unsatisfying or vague (see Jansen et al. 2003).
- b. Circumstantial evidence from geographic distribution of vestured pits has been offered by Jansen et al. (2004), who claim that vestured pits are most common in both mesic and dry tropical lowlands. Only a small number of eudicot families have vestured pits: 48 families in 11 orders (Jansen et al., 2001). These authors claim that the percentage of vestured pits drops to zero in boreal regions, but the large family Brassicaceae (3710 species) is largely boreal and cold temperate, and no species of that family were included in the survey by Jansen et al. (2004). To date, however, no species of Brassicaceae has been reported to lack vestured pits. The radiation of *Eucalyptus* and other Myrtaceae in Australia as well as the

- occurrence of vesturing in a number of Fabaceae, account for the idea that tropical and subtropical lowlands host the majority of examples reported to have vested pits, so systematics and speciation of particular groups is certainly involved here. The most speciose families of Brassicales, Brassicaceae and Capparaceae, may owe their success to having vested pits.
- c. Vesturing does not evolve readily (present in 48 eudicot families according to Jansen et al., 2001) whereas other forms of topographic relief on vessels (helical thickening, grooves) are apparently easily achieved morphogenetically (note that they probably parallel cyclosis in the last stages of wall formation). In fact, we may be amazed that vested pits have evolved as frequently as they have, considering their complexity. That they are as widespread as they are despite the precision and intricacy of their structure evidences the strong selective value that they have. Vestures may be lost, or suppressed, within particular families (e.g., Moringaceae). Vesturing is uncommon in plants with succulence (e.g., Caricaceae, Gyrostemonaceae and Moringaceae in Brassicales).
 - d. The examples of vesturing in Brassicales show that imaging vestures from the outer surfaces of vessels is insufficient. As one example, the rich assemblage of fused vestures on the insides of vessels of *Crataeva* (Capparaceae) would have been overlooked had only outer walls of vessels been studied. Vesturing in pits is three-dimensional and deserves study that can explore that. Sections through pits and TEM studies may also prove useful.
 - e. The imperforate tracheary elements of some Brassicales have vested pits, notably those of *Batis* (Bataceae), *Forchhammeria* and *Stixis* (Stixaceae), *Koeberlinia* (Koeberliniaceae), as well as *Pentadiplandra* (Pentadiplandraceae). Occurrence of vested pits on imperforate tracheary elements is always associated with fully bordered pits, and together, these two features are sufficient to declare such elements to be conductive cells, and therefore tracheids.
 - f. The demonstration of vestures on helices of primary xylem of *Raphanus* (Brassicaceae), a first report for angiosperms, is interesting in that there is not a clear connection between this occurrence and functioning of vestures, and it may represent a morphogenetic or developmental phenomenon in which complete exclusion of vestures from primary xylem in a species with vested pits in secondary xylem is not always achieved. The occurrence of the vesture-like papules on flanges paralleling perforation plates in *Azima* is also an unprecedented occurrence.

4. Helical sculpture in vessels. Brassicales contain all of the main kinds of helical sculpture in secondary xylem vessels: helical thickenings, grooves widening pit apertures or connecting helical series of pit apertures (coalesced pit apertures), and thickening bands adjacent to grooves. The varied contexts of these occurrences suggest several possible interpretations.

Hypothesis: The hydraulic significance of these structures is related to change in surface topography that controls wettability of surfaces. Kohonen (2006) suggests “engineering the roughness of the capillary [inner surface of a vessel or tracheid] walls to achieve complete wettability.” Enhanced wettability would permit recovery from cavitation, whether caused by drought or freezing.

- a. McCully et al. (2014) and Brodersen and McElrone (2013) have found patchy distribution of hydrophilic wall surfaces, although their work does not specifically address helical sculpture. Further studies in this regard are much needed.
- b. The occurrence of helical thickenings in the wet forest Akaniaceae may relate to cold conditions rather than to drought. Aridity seems related to helical thickenings in *Setchellanthus*.
- c. Grooves interconnecting pit apertures in vessels are almost universally present in the woody Brassicaceae studied by Carlquist (1971). These species experience wet winters but dry summers.
- d. Helical sculpture in vessels is highly correlated with xeric habitats in Asteraceae (Carlquist, 1966) and in the woody flora of southern California (Carlquist and Hoekman, 1985).

5. Vessel grouping. Although vessel grouping or lack of vessel grouping in woods has been figured beginning in the earliest works on plant anatomy, such as that of Grew (1682), Its physiological significance has been appreciated only recently (Carlquist, 1984).

Hypothesis: Vessel grouping is a form of xeromorphy that is present in angiosperm woods that do not have tracheids as an imperforate tracheary element type. Tracheids (sensu Bailey, 1936; Carlquist, 1988; IAWA Committee on Nomenclature, 1964; Sano et al., 2011) in angiosperm woods are more effective than vessel grouping as a xeromorphic feature, because vessels are mostly solitary in woods that have tracheids (or abundant vasicentric tracheids) as an imperforate tracheary element type. In woods with libriform fibers or fiber-tracheids, degree of grouping is proportional to habitat aridity. Vessel grouping represents a non-random form of vessel placement (Carlquist, 2009a). Solitary vessels randomly placed would be expected in angiosperm woods unless there is some factor promoting non-random placement. The effect of vessel grouping is one of redundancy, not of embolism reduction per se.

- a. The vessel grouping of some species of Brassicaceae is relatively elevated: 5.7 vessels per group in *Cheiranthus scoparius* Brouss., 5.9 in *Stanleya pinnata*; most Brassicaceae have values between 1.5 and 2.2 (Carlquist, 1971).
- b. Radial chains of vessels potentially represent not merely redundancy, but commissioning of new vessel elements so as to maintain the pattern of water columns in xylem as older vessels cease to conduct actively. This tendency is shown conspicuously in Cleomaceae.
- c. Minimal vessel grouping is shown in the brassiclean families Emblingiaceae, Koeberliniaceae, Pentadiplandraceae, and Stixaceae. These families all have tracheids as an imperforate tracheary element type. Tropaeolaceae, which have abundant vasicentric tracheids, also exemplify this phenomenon.

6. Vessel diameter. Vessel diameter was early recognized as exemplifying xeromorphy (Carlquist, 1966, 1975). The resistance of latewood vessels to cavitation was demonstrated by Hargrave et al. (1994). Wide vessels have conductive efficiency, as cited by Zimmermann (1983).

Hypothesis: there is a trade-off between the conductive ability of wide vessels and the conductive safety of narrow vessels. Both can be accommodated in a single wood, by means of growth rings in which latewood features narrowing of vessel diameter. Vessel diameter may increase in diameter in a stem if a plant is able to tap deeper and therefore moister levels of water availability, but in many groups, decrease in vessel diameter accompanies senescence. The more shallow the roots of a plant at maturity, the more likely it is to show narrowing of vessels with age. Particular growth forms (vines, lianas) show wider vessels than one would expect from shrubs or trees of the same stem diameter, but vessel dimorphism (few wide vessels, more numerous narrow vessels) is common in scandent species.

- a. Mesomorphy in woods of Brassicales is unusual, but is represented in Akaniaceae, as well as in species of Capparaceae which are native to moist forest areas. The wide vessels of Caricaceae and Moringaceae may relate to succulence more than to steady soil moisture (Olson & Carlquist, 2001).
- b. Scandent species are few in Brassicales, but the wide vessels of Tropaeolaceae exemplify the tendency for vining species to have wider vessels and vessel dimorphism.
- c. Brassicales that represent particular types of xeromorphy and grow in dry habitats include Koerberliniaceae (deserts with summer rainfall), perennial Resedaceae (arid scrub), and *Setchellanthus* (subtropical desert scrub). Bataceae grow in moist maritime habitats that are saline, which is a type of physiological drought. This is also true of *Cleome droserifolia* (Fahn et al., 1986).
- d. Annuals such as *Hirschfeldia incana* represent a single growth ring, with narrow vessels formed as the plant goes into flowering and fruiting. Vessel dimorphism occurs in non-scandent Brassicales. It is particularly common in Capparaceae (*Apophyllum*, *Atamisquea*, *Capparis*, *Maerua*, *Niebuhria*, *Quadrella*, *Ritchiea*, and *Stuebelia*). These are characteristically dry land shrubs in which narrower vessels could maintain conductive pathways even if wider vessels in a group cavitated.

7. Vessel density. Vessel density has been considered, along with vessel length and vessel diameter, a chief indicator of conductive characteristics. It should be roughly inverse to vessel diameter, based on packing considerations.

Hypothesis: deviations from the expected value (inverse of vessel diameter) do occur (Carlquist, 1975, p.183), and may tell us some unexpected information about the wood plans of particular species and their function. In particular, vining species have fewer vessels per sp. mm than expected because their conduction is related to the fourth power, not the square or cube, of the vessel diameter (the Hagen-Poiseuille equation). Succulents probably have less than peak flow because of lowered transpiration rate, and have lower vessel density accordingly. Plants with successive cambia have lower density because older vessels are still active in conduction, so the conductive area of a stem is greater than in species that retire vessels sooner.

- a. *Tropaeolum* does not have fewer vessels per sq. mm than the vessel diameter would dictate, but it does have fewer wide vessels. The narrower vessels confer conductive safety, countered by the high flow capacity of the larger vessels.
- b. More succulent portions of *Moringa* (Olson & Carlquist, 2001) have fewer vessels per sq. mm. than expected, probably because peak transpiration is lower during the warmest months, this in turn related to drought deciduousness. Caricaceae (Carlquist, 1998a) has only slightly less vessel density than expected based on vessel diameter, probably because its foliage has greater transpiration than does the foliage of a succulent with few or no leaves during the warmest months.
- c. The imperforate tracheary elements of *Forchhammeria* (Stixaceae) consist wholly of tracheids, which have great conductive safety (Carlquist et al., 2013). The foliage of *Forchhammeria* consists of narrow leaves with recurved margins and thick cuticle, likely to have transpiration characteristics lower than those of broad, thin leaves. More importantly, each of the successive vascular cambia continues to produce secondary phloem for long periods of time, suggesting that vessels function for longer, and thus a greater actual conductive area is achieved in a given stem without adding very many vessels in newer vascular increments. Some portions of *Forchhammeria* secondary xylem are devoid of any vessels.

8. Vessel element length follows trends in vessel diameter, vessel density, and vessel wall sculpture by being shorter in plants of xeric regions. The reasons for this may be multiple.

Hypothesis: Because air emboli tend to stop at perforation plates, regardless of whether perforation plates are scalariform or simple (Slatyer, 1967), shorter vessel length may be disabled in a plant with shorter vessel elements. In addition, shorter vessel elements, characteristic of plants of drier regions (Carlquist, 1975; Carlquist & Hoekman, 1985), are not likely to be concomitantly wide: deviations from a straight-line length/width ratio do occur, but are not extensive (Carlquist, 1975, page 183), for reasons of optimal cylindrical strength design.

- a. Vessel element length is notably short in Brassicaceae (Carlquist, 1971), which are mostly short-lived annuals or perennials.
- b. The only family of Brassicales with relatively long vessel elements is Akaniaceae, in which both monotypic genera are native to moist forests.
- c. Within Brassicaceae, the longest vessel element lengths occur in the species with the most mesic habitat, *Cheiranthus mutabilis*, from Madeiran laurel forest openings (Carlquist, 1971).

9. Axial parenchyma. Scanty vasicentric is the prevailing type of axial parenchyma present in Brassicales. The presence of other types of parenchyma in particular families invites interpretation based on ecophysiological and growth form characteristics primarily.

Hypothesis: A few cells adjacent to a vessel or a vessel group (vasicentric scanty parenchyma) suffice to control the conductive process in vessels. Additional thickness of parenchyma sheaths (paratracheal abundant) around vessels may serve for

intermediate steps in the conductive process, such as storage. Although some of this process remains hypothetical (Zwieniecki and Holbrook, 2009), the role of axial parenchyma and its exchange of solutes with vessel sap seem clear in terms of histological association (Carlquist, 2015a). Although mechanical tissue tends to be prevalent in first-formed wood of a stem in a suffruticose shrub, bands of parenchyma (which are especially common in latewood positions) may be added and have functions related to such functions as prevention of freezing damage. Conversion of part or all of the axial secondary xylem to parenchyma relates to succulence and storage.

- a. The parenchyma bands seen in *Arabis*, *Ptilotrichium*, *Stanleya* and other genera of Brassicaceae occur in genera and species that experience considerable drought and/or freezing exposure (Carlquist, 1971; Schweingruber, 2006).
- b. Instances of aliform-confluent axial parenchyma in Brassicales are scarce (*Niebuhrria*, Fig. 21d; *Sinapidendron*), and may relate to the balance between storage and mechanical strength and woodiness of these species.
- c. The water, starch, and sugar storage by parenchyma of widely known vegetables (e.g., *Brassica oleracea*) and condiments (*Armoracia lapathifolia*), although increased in extent in cultivars, is basic to storage for flower and seed production or for perennation.
- d. Succulence in Caricaceae and Moringaceae, achieved by pervasive axial parenchyma in secondary xylem, relates to survival of their arborescent and sarcorhizal growth forms in habitats that provide a seasonal water supply (Olson & Carlquist 2001).
- e. Bordered pits on cross-walls of axial parenchyma strands (*Akania* and *Bretschneidera*) are indicative of active axial flow of photosynthates in axial parenchyma,

10. Rays. The function of rays relates to histological feature: cell shape; types of pitting; and contents of ray cells. The predominant ray type in Brassicales is Heterogeneous Type II (which is also the most common type in angiosperm woods).

Hypothesis: Heterogeneous Type II rays feature multiseriates with a central group of procumbent cells, plus some upright sheathing cells, and few or no wings at tips of multiseriate rays. More abundant upright cells in such a ray qualifies it as Paedomorphic Type I, and relates to axial flow patterns in rays, whereas predominance of procumbent cells (Homogeneous Type I or II) indicates radial flow in rays. Active flow of photosynthates is indicated by presence of bordered pits, which are especially common in tangential walls of procumbent cells, indicating radial flow in procumbent cells. Raylessness is often exhibited in the form of first-formed secondary xylem that has only mechanical elements, and rays are initiated later; it is indicative of a trade-off between mechanical strength and radial flow of photosynthates in ray parenchyma in “woody herbs” that exhibit other features of secondary woodiness (Carlquist, 2015b).

- a. Raylessness occurs in a number of Brassicaceae in which there is branching of stems from the base of a shrub or subshrub, and in which each of these branches bears an appreciable weight in terms of flowers and fruits. Raylessness at the beginning of secondary growth, yielding to development of rays thereafter, was

- observed in the brassicaceous species *Cheiranthus mutabilis*, *Lepidium fremontii*, *Matthiola maderensis* Lowe, *Parolinia ornata*, *Sinapidendron* sp. (Carlquist 2760, RSA), and *Stanleya pinnata* (Fig. 25a) (new reports)
- b. Procumbent cells are most abundant, even present exclusively in some species, in Akaniaceae, Capparaceae, Koerberliniaceae, and Salvadoraceae. This is especially true in the larger wood samples.
 - c. Presence of upright ray cells predominantly or exclusively is seen in *Tropaeolum*, some Resedaceae, and numerous insular Brassicaceae (Carlquist, 1971; Lens et al. 2013). This ray feature is prominent in instances of secondary woodiness (Carlquist, 1962, 1969; Lens et al., 2013). This feature is present in species with protracted juvenilism (also termed paedomorphosis, which implies sexual reproduction while in a juvenile state of development).

Brassicales and Beyond: Basic Ideas in Wood Evolutionary Theory

Reversibility in Wood Characters: How Prevalent Is It?. In a thoughtful review of this topic, Baas and Wheeler introduced us to some elements of the question. Bailey (1944) asserted that trends associated with vessel element specialization are irreversible. He used vessel element length as a kind of key to advancement in wood features. Leaving aside fluctuation in character expression, was he right? Baas and Wheeler (1996) think that reversion can take place. However, reversibility/irreversibility is not a single concept, but several.

Hennigian character state methodology and its problems. The level of irreversibility that has received attention with respect to cladistics methodology is character state change as seen from morphological studies. For example, Koerberliniaceae, Pentadiplandraceae, and Stixaceae have tracheids, whereas fiber-tracheids with nearly simple pits appear basic to Brassicales. Are these families reverting to an ancient type imperforate tracheary element? Bailey & Tupper's (1918) diagram of evolution of progressively more specialized vessels on the one hand, and progressively more specialized imperforate tracheary elements on the other, beginning with a tracheid with fully bordered pits as the ancestral cell type. So have the three families of Brassicales just named reverted to an ancient cell type? Not really. The genetic information for formation of bordered pits has not been lost, as can be seen from bordered pits on vessels in all Brassicales, so it's merely a matter of applying that information to the formation of imperforate tracheary elements. Hence, we have "secondary tracheids" or "neotracheids" in Koerberliniaceae, Pentadiplandraceae, and Stixaceae. Baas and Wheeler might see this as a character reversion. We have been much influenced by the 0 and + designation of character states in the Hennigian mode. But character states are rated by human observation, whereas the plant may arrive at a character state in any of various ways. Similar instances could be cited elsewhere in angiosperms. *Fabiana* is exceptional in Solanaceae in having tracheids, *Rosmarinus* similarly exceptional in having vasicentric tracheids in Lamiaceae (Carlquist 1992a, 1992b). In fact, Baas & Wheeler (1996) do present a table citing reversions for the prevalent sequence from bordered to simple pits on imperforate tracheary elements in angiosperms. If Baas and Wheeler could have used the concept of "secondary tracheids," their table would be different.

Unfortunately, a table of Baas & Wheeler (1996) combines both the pit evolution and evolution of simple perforation plates from scalariform perforation plates, so we cannot

separate their reversion estimates for these two characters. In another table, they list families in which scalariform and simple plates are both present. However, the families cited have different stories. Dilleniaceae has long scalariform perforation plates in *Dillenia* from New Caledonian rain forest, but simple perforation plates in *Hibbertia* from dry Western Australian scrub. Araliaceae and Styracaceae, on the other hand, have scalariform and simple perforation plates in the same wood—the scalariform plates more common in latewood. And one can cite other modes of occurrence, showing that a simple 0 and + designation may be used in a data matrix, but may be misleading.

Ratchet Theory. Olson (2014) and Jansen & Nardini (2014) have attributed to me the idea of an evolutionary ratchet, which denotes evolutionary progression in a character and associated characters to the extent that reversion is not possible. Thus, change to a simple perforation plate in a clade may involve loss or silencing of information leading to the formation of bars on a perforation plate, and inevitably, other features in a wood (e.g., ray type) also change. This might explain the instances of perforation plates in Akaniaceae and Tropaeolaceae. In these families, perforation plates are simple, except for a very few plates that are imperfect versions of scalariform patterns. A scalariform perforation plate pattern will likely not return to such clades. We are not sure why the occasional malformed plates occur, although one can furnish ideas.

If scalariform perforation plates are associated with mesic conditions, simple plates in any given clade are not disadvantaged in mesic habitats, should there be a shift from more arid habitats where simple plates evolved in a clade, into some new mesic conditions. Virtually all of the plants on the summit of Mt. Waialeale, supposedly the wettest place on earth, have simple perforation plates.

Gene theory. If one looks at evolution at the gene level, one has a different perspective. Evolution is a forward progression. Even if a few genes (like those for variegation) can switch expression readily, most genes cannot, and a series of processes—silencing, modification, multiplication, loss, inversion of segments, pleiotropy—may be involved. Obtaining a picture of gene changes in wood anatomy for any given clade is so complex and perhaps impossible that we cannot readily think in these terms. But if we do use this perspective, we see that successive gene and gene combination changes are ongoing, they never return us to the ancestral DNA sequences.

Xeromorphy in Wood: Tiers of Effectiveness and Types of Action in Characters. In Asteraceae, number of vessels per group as seen in transverse section increases markedly with habitat aridity, as in *Olearia* (Carlquist 1960, 1966). Yet *Krameria*, *Prunus*, and some other desert genera never group vessels. Noting this, I reviewed the data on vessel grouping and for imperforate tracheary element type for each family in Metcalfe & Chalk (1950), and found a hitherto unappreciated correlation. Vessels also do not group in genera with abundant vasicentric tracheids (*Eucalyptus*, *Quercus*). Thus, the conductive safety provided by vessel grouping is subordinate to the safety provided by tracheids (Carlquist, 1984). The ultimate conclusion of this line of thinking is that no two xeromorphic adaptations have the same value, and the occurrence of one precludes the occurrence of others. Most characters, other than the pair just mentioned, are additive in nature.

Tier 1. Tracheids or abundant vasicentric tracheids; vested pits.

Tier 2. Vessel diameter; vessel density (number of vessels per sq. mm); vessel length; growth rings.

- Tier 3. Vessel grouping; vessel diameter dimorphism; helical thickenings in vessels.
 Tier 4. Vessel element length; water storage parenchyma in wood.

The assignment of relative values to each of these characters is tentative. Rankings may change if a family or genus lacks a particular character. For example, numerous genera of Brassicales have vestured pits; Asteraceae lack vestured pits and tracheids (except for *Loricaria*), so Asteraceae show more reliance on Tier 3 characters. Interestingly, Brassicaceae with wood or leaf succulence (Bataceae, Caricaceae, Gyrostemonaceae, Moringaceae, Resedaceae) have minimal vesturing or none at all.

Particular families show quite different priorities for characters that tend to insure conductive safety. In cacti, Mauseth (1993) cites wide-band tracheids (characteristic of no other family except for Anacampserotaceae) and water storage parenchyma, which may be placed variously with respect to mechanical tissue. The succulent nature of stems as a whole in cacti permit the volumetric change that extends to wide-band tracheids, which would be non-adaptive in a family with no volumetric change of stems in accordance with changing water content.

The co-occurrence of two Tier 1 features, vestured pits and tracheids or vasicentric tracheids, in Myrtaceae can be correlated with the amazing radiation of *Eucalyptus* and other myrtalean genera in Australia. In this respect, we should note that vestured pits are varied, and that they do not lend themselves to experimental work in wood physiology because of their minute size.

Heterochrony: An Extensive and Nuanced Source of Diversity in Angiosperm Woods. Heterochrony is the umbrella term for protracted juvenilism and accelerated adulthood. The terms paedomorphosis and progenesis have been used for those two phenomena, respectively, but the terms are zoological ones and are doubtfully applicable to organisms with an open system of growth. The wood of conifers shows no protracted juvenilism (except perhaps in rays of *Welwitschia*). However, angiosperms have a rich assemblage of juvenile features the expression of which may extend for various periods of time—perhaps for the entire vegetative history of a particular plant (Carlquist 1962, 2009a). One can say that all of these features relate in some way to cambial activity:

Juvenile	Adult
Cells upright only or predominantly in multiseriate rays	Multiseriate ray cells procumbent, upright ray cells on sides and tips of rays
Rays little changed during ontogeny	Rays subdividing, widening, or initiated more frequently as a stem or root grows.
Rayless	Rays present
Length-on-age curve for vessel element length descending	Length-on-age curve for vessel elements Once can say that all of these features relate ascending, then leveling
Storying absent or onset delayed	Storying onset earlier
Cambium in bundles absent (monocots)	Cambium present
Pseudoscalariform pitting on vessels	Rapid progression to circular or oval vessel pits.

The inclusion of absence of cambium in monocots and a few non-monocots (Nymphaeales), may seem unexpected, but absence of a cambium is the ultimate juvenile condition. One can see minimally active cambia in some Saururaceae, leading to cambial absence; Saururaceae are not ancestral to monocots (although they are not far from monocot origins), but the reduction of cambial activity in their stems very likely simulates the stage preceding monocot stem origin. Could one assume that short duration of cambial activity is equivalent to juvenilism, and that prolonged cambial activity equates to woodiness? As a generalization, this seems permissible. Annual herbs can certainly be considered a juvenile growth form, albeit with some special features, and wood of annuals contains some or most of the features listed above.

There is no one kind of descending length-on-age curve for vessel elements (Carlquist 1962, 2013). These remain to be explored.

Juvenilistic wood characters can and do occur independently of each other, and one may find only one, or several in a given wood.

Do Brassicales show protracted juvenilism? Certainly the many annuals in Brassicaceae qualify, as do the perennials (Lens et al., 2013). Olson (2007) shows how *Moringa* stem and root anatomy can be interpreted in terms of heterochrony, and similar considerations can be applied to Caricaceae. Tropaeolaceae can be considered a relatively non-woody derivative from ancestors like the arboreal family Akaniaceae. Tropaeolaceae have upright ray cells exclusively, as do some Resedaceae and some Gyrostemonaceae. The non-woody Cleomaceae have juvenilistic secondary xylem. With the exception of Brassicaceae, Brassicales are not exceptional among angiosperm orders in their degree of heterochrony.

Heterochrony in wood and in other respects of plant structure is basic to large territories of angiosperm evolution. The orders Asterales and Lamiales may be the richest assemblages of species representing juvenilistic structure modes, both in xylem and in other plant portions. A case has been made for considering basal angiosperms as stemming from less woody ancestors (Donoghue & Doyle, 1989; Taylor & Hickey 1996; Carlquist, 2009b), and we can no longer believe that trees are the basic growth form from which other angiosperm growth forms were derived. The full implications of heterochrony in angiosperms remain to be explored, but we can safely say that heterochrony, in addition to the simultaneity of embryo and endosperm development, was a major factor in the rise of angiosperms.

In Wood Anatomy, Which Characters are “Functional,” Which are “Taxonomic”? Solereder’s thesis (1885), *Über den systematischen Wert der Holzstrukture bei den Dicotyledonen*, set a precedent followed by his (1908) compilation “Systematic anatomy of the Dicotyledons,” and Metcalfe & Chalk’s (1950) “Anatomy of the Dicotyledons.” The arrangement of the data by families is not just for the convenience of accessing data on particular plants: it was developed from the conviction that wood features could be used as systematic tools for wood identification. The use of such data for wood identification continues to the present. However, any thoughtful viewer of these books will say, “why do particular woods have the assemblages of features that they do?”

One must remember that the great majority of woody species is in tropical latitudes, whereas academic and especially forest institutes where wood anatomy was studied

were in cold temperate regions. Thus, students of wood anatomy did not know the growth form and ecology of the woody species they studied. More importantly, with many different woods to compare, evolution of particular characters with respect to ecological and climatic factors was not highlighted. A situation of great simplicity was offered by the family Asteraceae, a family of about 26,000 species in which woods are essentially identical (owing to the recentness of radiation of the family) except for vessel characters that relate to ecology. By studying a large number of asteraceous woods from known habits, the patterns of vessel evolution were dramatically evident (Carlquist, 1966). The application of the knowledge gained in Asteraceae to angiosperms at large proceeded (Carlquist, 1975; Carlquist & Hoekman, 1985). Now, each family for which wood anatomy is reviewed, such as those of Brassicales, can be compared to our knowledge of ecological wood anatomy. Of course, physiological principles underlie the correlations, and students of wood physiology have been eager to discover those principles (e.g., Jansen & Nardini, 2014).

With the advent of molecular phylogeny, we no longer look to wood anatomy as a source of materials from which to build phylogenetic systems. More importantly, we realize that wood anatomy represents a series of character combinations, unlike from clade to clade, that offers various ways of satisfying the water economy needs of particular plants. We have now attributed ecophysiological functions to many wood characters. Not all wood characters relate to water economy. Mechanical functions of wood; carbohydrate and water storage; and herbivore deterrence are chief among the non-hydraulic features wood offers. If we are willing to interpret features like starch storage; crystals, silica bodies, gum or resin-like compounds; and various wood “fiber” configurations in terms of function, we have no residue of “purely taxonomic features.” Assemblages of functional wood character states in one genus or family can, however, be compared to those in another. Thus, in Brassicales, we see that rays of two distinct sizes, septate or living fibers, helical thickenings in vessels, and curious rare malformed scalariform perforation plates link *Akania* and *Bretschneidera*, previously widely separated in phylogenetic systems, into the family Akaniaceae. We also find resemblance between wood of Tropaeolaceae and that of Akaniaceae. Resemblance in functional systems between taxa that are shown to be closely related on the basis of molecular phylogeny is an expected outcome. There are some instances in which habit, wood anatomy, and habitat of one clade have veered markedly away from close relatives (e.g., Limnanthaceae, Koeberliniaceae, Stixaceae). Wood and other features of the family of annuals Limnanthaceae show few resemblances to the families closest to them in the system of Brassicales, Caricaceae, Moringaceae and Setchellanthaceae (Fig. 1).

Are there “relictual” features in angiosperm woods that persist even though they have little function? The scalariform perforation plate is an obvious example, because such perforation plates are common in basal woody angiosperms. The available evidence, inadequate though it is (see Jansen & Nardini, 2014), suggests that scalariform perforation plates may serve more than one function, including ones those authors do not list (e.g., Slatyer’s 1967 idea that embolisms tend to terminate at perforation plates and thus stop disabling of an entire vessel). Those who do experimental work are understandably uncertain about how to proceed in such situations. Baas & Wheeler (1996) cite Zimmermann’s (1978) suggestion that bars on perforation plates sieve out bubbles that form after frozen water in vessels thaws, but only half of boreal

angiosperms have scalariform perforation plates. Moreover, the majority of scalariform perforation plates are not located in areas that freeze, but rather in frost-free cloud forests of Malaysia, the Andes, equatorial Africa, and Indonesia. When confronted with multiple and uncertain scenarios like these, those interested in wood hydraulics understandably opt for some experimental procedure that does not involve scalariform perforation plates. Synthesis is the ultimate goal, but wood anatomy is such a varied terrain in angiosperms that both separating out individual functions and then synthesizing them across phylogenetic lines requires a level of knowledge that may not be possible, even in a group of well-chosen collaborators. Ironically, when we look at anatomical preparations of woods of the world, we can see the results of natural experiments, but the complexity of angiosperms at large is so great, and the limitations of any individual's expertise are so real that while individual wood features and individual species can be studied, the needed synthesis of information all too often remains elusive.

Acknowledgments Dr. Mark Olson kindly provided material of *Forchhammeria* and *Moringa*; his contribution to study of Brassicales extends well beyond those samples, however. Dr. David Boufford kindly provided materials of *Borthwickia* and *Stixis*. Wood samples from the xylarium of the Forest Products Laboratory of several families of Brassicales were provided through the kindness of Dr. Regis Miller and other staff members of that institution. Wood samples of Capparaceae were provided through the kindness of the curator of the wood collection of the U. S. National Museum of Natural History (Smithsonian Institution). Most of the SEM micrographs were obtained with the SEM at Santa Barbara Botanic Gardens, and the directors of that institution (Dr. Edward L. Schneider, Dr. Steven Windhager) deserve thanks for giving me access to that machine and keeping it in repair. Dr. Jocelyn Hall and the University of Wisconsin Herbarium furnished a twig of *Pentadiplandra arborea*. John Garvey applied scales, lettering, and symbols to the photographic figures.

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